

Prebid Queries

Vendor 1

Date :

30.11.2024

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Prebid Queries								
Name of Contractor : Vendor 1							Date :	29-11-2024
Sr. No. / Query No.	Bidding Document Reference(s) (Volume No / Section Number / Clause Number)		Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.					
1		A. CHILLER PLANT HIGHSIDE EQUIP	CHILLER PLANT ROOM HIGHSIDE EQUIPMENT	1	Supply, installing, testing and commissioning of WATER COOLED CENTRIFUGAL CHILLERS each complete with 1000 TR + 5 Nos Working and 1 No Standby actual capacity single compressor with motor, VFD starter panel, water cooled condenser, insulated evaporator. Refer to PEPL_MUM_PLE_ME_SCH_0001 for the chiller equipment schedule & Performance Characteristics.	1.Kindly confirm the requirement for auto tube cleaning system. 2. We understand that Chiller Vessels shall be with ASME Stamping kindly confirm? 3. Support structure for all chiller and equipment shall be provided by client free of cost please confirm? 4. Sound Level for the Equipment shall be as per OEM Manufacturing Standards.Please Confirm?	1. ATCS line added in the BOQ. 2.Yes, with stamping 3.Client/PMC to confirm 4.Follow the BOQ and Specification	
2		A. CHILLER PLANT HIGHSIDE EQUIP	CHILLER PLANT ROOM HIGHSIDE EQUIPMENT	2	CHILLER PLANT OPTIMIZER SYSTEM Supply, installation, testing & commissioning of chiller plant Optimizer as per chillers requirement. The plant optimizer shall be suitable to controll all the chillers, pumps & cooling towers being proposed and all suitable madules (I/O) shall have to be provided so individual Micro Processor controlled panels of machines can be synchronised for chillers and pumps operation and to enable sequential operation to achieve equal operating time and take care of load patterns 1. To control and operate the Primary & Variable secondary speed chilled water pumping system, Chillers and Cooling Towers. 2. To optimise the run time hours of all the pumps,Chillers and Cooling Towers. 3. To control and monitor the temperature of the chilled water and condenser water entering and leaving the chillers. 4. To control and operate the cooling towers and associated pumps 5. Chilled water energy usage for the different zones. 6. Interface with BMS system.	Kindly Share IO Summary for CPM	Refer the IO Summary in BMS tender	
3		A. CHILLER PLANT HIGHSIDE EQUIP	CHILLER PLANT ROOM HIGHSIDE EQUIPMENT	3	Supply, installation, testing and commissioning of COOLING TOWERS INDUCED DRAFT CROSS FLOW CTI approved for air-conditioning system. Each tower shall be complete with basin (with connection for drain,makeup water ,overflow, quick fill,float valve etc), casing, distribution system, filling, louvers, ladder, fan and motor. Motor shall be suitable for 415±10% volts, 50 cycles, 3 phase power supply with efficiency of class IE3, Isolator panel with MCCB with suitable enclosure for cooling tower and earthing, shall also be included. Cooling Tower for air-conditioning system shall be selected on the basis of leaving water temperature 31°C & entering water temperature 38° C, ambient wet bulb 28° C and nominal capacity shall be as per Equipment Schedule Ref : PEPL_MUM_PLE_ME_SCH_0006 . Cold water basin shall be a deep sump and made out of 5 mm thick FRP construction and UV stabilized on which cooling tower super structure shall be supported. Fillings shall be made of corrosion proof and rigid PVC film in honeycomb design and arranged in square / rectangular form. The cooling tower shall be quiet operating, and shall produce an overall sound no higher than 80 dB at a distance of 1.5 m.	1 Sound Levels Mentioned in the BOQ is Noise shall be less than 80db from 1.5 mtr Whereas in Schedule it is mentioned noise level shall not exceed 85dBA at 1,5m Please Confirm the Sound Levels for the Cooling Tower , Also Sound levels and effecency for the cooling tower shall be as per OEM Standard Selection. Please confirm? 2. Support structure for Cooling Tower and other equipment shall be provided by client free of cost.Please Confirm	1.Follow the BOQ and Spec. 2.Client/PMC to confirm	
4		A. CHILLER PLANT HIGHSIDE EQUIP	CHILLER PLANT ROOM HIGHSIDE EQUIPMENT	4	Supplying, installing, testing and commissioning of Chilled Water Pumps Vertical Inline Centrifugal Pumps consisting of adjustable frequency drive (VFD) for each pump, one dedicated microprocessor based pump controller per zone housed within the enclosure of one of the VFD with parallel pumping software duly down loaded and multiple differential pressure sensor / transmitters along with complete control cabling from transmitter to VFD Panel as per the site requirement. The entire system along with chilled water pumps as described above must be sourced from single manufacturer only, to ensure unit responsibility. Necessary factory test certificates for entire system shall be submitted along with the equipment. Sensor / Transmitters as described in specification. Complete assembly including pump, motor and drive shall be designed to limit THD to less than 5%. System shall be BMS compatible with open protocol communication port, so that complete data can be transported to remote IBMS console. (The pump motor shall be suitable for 415 + 10% volts, 50 cycles, 3 phase power supply. The motors of secondary chilled water pumps shall be suitable for operation on adjustable frequency drive. The motor selected shall be energy efficient having IE3 efficiency class rating. (The quoted price shall include cost of thermal insulation and cladding.) The system shall be complete in all respects and suitable for motor ratings as specified. Refer to PEPL_MUM_PLE_ME_SCH_0003,PEPL_MUM_PLE_ME_SCH_0004 & PEPL_MUM_PLE_ME_SCH_0005 for the pump schedule with performance characteristics.	1. Kindly Confirm the requirements for Harmonic Filter , if required Please Confirm whether Active or Passive. 2. Efficiency for Pump shall be as per OEM Standard Selection.Please Confirm. 3. As the pumps mentioned are vertical inline we shall not consider inertia base, please confirm?	1.Inbuilt harmonic filter shall be considered,refer the electrical section.Active filters to be considered 2.Follow the BOQ and Spec. 3.OK	
5		A1. CHW PUMPS & PHEX	CHILLED WATER PUMPS & HEAT EXCHANGERS	1	Supplying, installing, testing and commissioning of Variable Tertiary Chilled Water Pumps Vertical Inline Centrifugal Pumps consisting of adjustable frequency drive (VFD) for each pump, one dedicated microprocessor based pump controller per zone housed within the enclosure of one of the VFD with parallel pumping software duly down loaded and multiple differential pressure sensor / transmitters along with complete control cabling from transmitter to VFD Panel as per the site requirement. The entire system along with chilled water pumps as described above must be sourced from single manufacturer only, to ensure unit responsibility. Necessary factory test certificates for entire system shall be submitted along with the equipment. Sensor / Transmitters as described in specification. Complete assembly including pump, motor and drive shall be designed to limit THD to less than 5%. System shall be BMS compatible with open protocol communication port, so that complete data can be transported to remote IBMS console. (The pump motor shall be suitable for 415 + 10% volts, 50 cycles, 3 phase power supply. The motors of secondary chilled water pumps shall be suitable for operation on adjustable frequency drive. The motor selected shall be energy efficient having IE3 efficiency class rating. (The quoted price shall include cost of thermal insulation and cladding.) The system shall be complete in all respects and suitable for motor ratings as specified. Refer to PEPL_MUM_PLA_ME_SCH_0005 for the pump schedule with performance characteristics Common base plate from Hot dip galvanized iron for required size as per manufacturer recommendation with heavy duty anti vibration pads and inertia bases. Supply Installation Testing and commissioning of Imported portion of VFD, Pump Controller & Sensor Transmitters.Refer to the chilled water schematic for the Location and number of Differential pressure sensor. Cost to include cabling from panel to pump.	1. Kindly Confirm the requirements for Harmonic Filter , if required Please Confirm whether Active or Passive. 2. Efficiency for Pump shall be as per OEM Standard Selection.Please Confirm. 3. As the pumps mentioned are vertical inline we shall not consider inertia base, isolator , please confirm?	1.Inbuilt harmonic filter shall be considered,refer the electrical section.Active filters to be considered 2.Follow the BOQ and Spec. 3.OK	

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6		B TFA	TREATED FRESH AIR UNITS	1	Supplying, installing, testing and commissioning of TREATED FRESH AIR UNIT WITH Heat Recovery Wheel (HRW) with 50mm thick double skin construction with Thermal Break Profile as per specification, The unit shall comprise of filter section, fresh air inlet / exhaust air outlet, fan section with EC plug type fan with motor and drive package including the variable frequency drive housed in the control panel. The unit shall be have MERV-15 + PCO + UV LAMP. Prefilter should be MERV-8.Refer to the equipment schedule PEPL MUM PLA ME SCH 0004 and specification for air filtration requirements. Min 6 Row Deep water coil of copper tube & aluminium fins construction and base frame, condensate drain connections with U trap. Coil size shall be selected for a maximum face velocity of 500 feet/minute, Double skinned panels shall be made of galvanized steel, pressure injected with foam insulation 50 mm thickness having density 40 kg/m3	1. Can we consider pre filter and fine filter on common base frame kindly confirm? Sound level and efficiency shall be as oer OEM standard selection please confirm?	1.Yes consider. 2.Follow the BOQ and Spec.		
7		C AHU	AIR HANDLING UNITS	1	Supplying, installing, testing and commissioning of double skin construction draw thru type AIR HANDLING UNITS (CHW AHUs) with Thermal Break Profile as per specification, each complete with motor and drive package. The unit shall be have MERV - 8 + PCO + UV LAMP. Refer to the equipment schedule and specification for air filtration requirements. Double skinned panels shall be made of galvanized steel, pressure injected with foam insulation 50 mm thickness having density 40 kg/m³ Min 6 Row Deep water coil of copper tube & aluminium fins construction and base frame, condensate drain connections with U trap. coil size shall be selected for a maximum face velocity of 500 feet/minute	Sound level and efficiency shall be as oer OEM standard selection please confirm?	Follow the BOQ and Spec.		
8		E.BTU METER & THERMOSTAT	BTU METER & THERMOSTAT	2	Supply, installation, testing and commissioning of BTU Meters with necessary temperature measuring Instruments (2 per BTU Meter) (No need to quantify these temp sensors separately), test ports controls (2 per BTU Meter) (No need to quantify these temp sensors separately) and arrangement as shown in the schematic. The Cooling/BTU meter shall be MID approved and comply with EN 1434 standard. The MID approval must be displayed on the meter. (MID - Measuring Instruments Directive)Minimum Pressure Rating: 16 bar Shall be Ultra sonic - Side Wall mounted, with RTD Sensor and consisting of M-bus protocol which would support BMS. Selection The flow meter shall be selected such that the expected flow shall be below the meter's Nominal flow rate. The Minimum flow (Qi) measurable shall be 1/100 of the meter's Nominal flow (Qp). The flow meter shall be selected such that the below criteria are fulfilled; Threaded meters: The outer connection on the meter shall be of the same size as the pipe size. In no condition meter shall be selected that outer connection on meter is one size lower than the pipe size. This is because in threaded meter actual measuring is done in one size lower chamber inside the meter. This will ensure longer life of meters as no cavitations happen in the meter. Flanged meters: The meter shall be of same size as pipe size or maximum one size lower than pipe size. In no condition meter shall be selected two sizes lower than pipe size. This will ensure longer life of meters as no cavitations happen in the meter. The Minimum flow measurable by the meter shall be 1% of the Nominal flow of the meter. Qi:Qp = 1:100 (Qi - Minimum flow, Qp - Maximum flow) CHW ranning for each tenant space to have BTU meter	1.Please confirm the type of BTU Meter?	Ultrasonic type BTU meter		
9		F.PIPING	PIPING	1	CHILLED WATER PIPING Providing & fixing the following MS ERW Class C pipes with specified wall thickness as per IS 3589, cut to required lengths and supporting arrangements with all welded joints, providing and fixing in position necessary fittings like bends, elbows, tees, reducers, mating flanges & sockets for BAS, fabricated to size and shape in workshop or brought out ready made where available of acceptable quality. Shoe fittings in pipe header to be factory fabricated. Pressure testing of pipes to be done. The rate shall be inclusive of Rubber support inserts having steel reinforcement, with overall material density of 1400 Kg/m³ for the pipes where the Supports would have contact with the chilled water pipes. All hangers, clamps, brackets etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns as per the approved makes unless specified otherwise and then supply, installation of the same shall also be included for rates under this head. Supplying, installing, testing and commissioning of Seismic Supports for vertical riser piping to take care of wind , building settlement and seismic effect (Refer schematic for seismic support detail) The contractor shall bring on board the seismic contractor & submit all the required calculations & methods.	1)We will consider welded joints for piping kindly confirm? 2)We will consider socket welded fittings for pipe dia upto 50mm and ERW butt welded fittings for pipe dia above 50mm kindly confirm	Follow the BOQ and Specification		
10		I.VRV	VARIABLE REFRIGERANT VOLUME (VRV) UNITS		Include for the Supply, complete installation, Testing & Commissioning as detailed in the specifications and on the Drawings and to the complete satisfaction and approval of the Engineer and all concerned Statutory Authorities. (refer equipment schedule: PEPL MUM PLA ME SCH 0007) Air Conditioning units including remote, thermostat, Sight glass, drier, filter,flare nut, refrigerant gas charging & top-up etc. complete Ductable VRV/VRF Indoor units, shall be provided with fire retardant flexible connections constructed of flexible double canvas sleeve as per the approved shop drawings. Supply,installation in position ,testing of Variable refrigerant flow /volume outdoor condensing unit complete with compressors, air cooled condensers, motors, with blygold coating on coils,starters, gauges, necessary valves and accessories Along with internal and external refrigerent piping connected between FCUs and out door unit. controls in microprocessor based control panels (starter panel), necessary hook up with indoor units / IR sensors with control cables. R410 a refrigerant, Refrigerant used shall be Ozone friendly and should not contain HCFCs and Halons. Power peak control unit and all other standard accessories whether specifically mentioned or not including mounting on a rigid steel base frame. Spring isolators of 25mm static deflection as detailed in specification and drawings and complying to necessary requirement with COP at AHU conditions. Unit control panel shall have the necessary provision for compatibility with HA / IBMS system and in accordance with specification, drawings and equipment schedule. Allow for all protective and decorative painting and cleaning of piping, fittings and equipments. Allow for all identification labels, plates, charts and identification of plant, equipment's etc. Allow for training and presentation of employer's personnel on all aspects of the operation of system and equipments Allow for part and integrated inspection, testing and commissioning of the HVAC installation as stated in the contract documents. Allow for inspection, testing and commissioning of HVAC system installation by the OEM. Indoor units to be with condensate drain pump. Allow for providing comprehensive equipment documentation including operating instructions and maintenance manuals, schedules of manufacturers recommended tools and spare parts for each piece of equipment. record documents, warranties, guarantees, drawings and the like.	1. we will quote on TR basis CFM shall ber as per OEM standard selection please confirm? 2. we will not consider any type of deration kindly confirm?	1.Consider optimal selection based on both TR & CFM criteria 2.Follow BOQ & specification		
11		I.VRV	GERANT VOLU	1.03.5	Supply,Installation and testing of sealing of upto 100 mm dia PVC sleeves for all refrigerant pipes with expandable PU foam	Please share the quantity for the same as it is given in LOT	Consider in LOT		

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12		J.FANS	FANS	1	Supply, installation, testing and commissioning of TUBE AXIAL FANS . The motor shall be suitable for 415 V ± 10% volts three phase 50cycle. The fan shall be equipped with mounting bracket inlet cone & outlet cone required at discharge. Fans shall be selected for minimum 65% efficiency. Fan shall be with flexible fire proof connection at inlet and outlet. The motor selected shall be energy efficient having IE3 efficiency class rating. The unit shall be suspended or floor mounted through spring isolators and threaded rod as per the standard installation details and as mentioned in specifications. All hangers, clamps, brackets, anchor/fixd support,guides,rollers etc. shall be shall be modular hot dip galvanized having galvanizing thickness 75 microns as per approved makes unless specified otherwise and then supply and installation of the same shall also be included for rates under this head. The fan shall have low sound power level not exceeding 75 dBA at three meter distance at both ends. Fans to be AMCA air & sound certified. Fans to have double canvas flexible connections. Allow for all protective and decorative painting and cleaning of piping, fittings and equipments. Allow for all identification labels, plates, charts and identification of plant, equipment's etc. Allow for training and presentation of employer's personnel on all aspects of the operation of system and equipments Allow for part and integrated inspection, testing and commissioning of the HVAC installation as stated in the contract documents. Allow for inspection, testing and commissioning of HVAC system installation by the OEM. Allow for providing comprehensive equipment documentation including operating instructions and maintenance manuals, schedules of manufacturers recommended tools and spare parts for each piece of equipment. record documents, warranties, guarantees, drawings and the like. Refer to PEPL_MUM_PLA_ME_SCH_0001 & PEPL_MUM_PLA_ME_SCH_0008 for the fan performance characteristics Refer Technical Specification Ref : 4.15 Fans	1. The motor mounting plate thickness shall be as per OEM standards please confirm? 2. Sound Levels & Efficiency for the Unit shall be as per OEM Standard selction please confirm?	Follow BOQ & specification		
13		J.FANS	FANS		Supplying, installing, testing and commissioning of SMOKE SPILL FANS FOR SMOKE EXHAUST as per specifications.Fan shall be with direct driven tube axial fan.Fan and casing shall be suitable for smoke exhaust application.Motor and all components shall be thermally rated for 250 Deg. C for 2 hours as per [BS-12101 Part-3 : 2002]. Motors shall be suitable for 415±10% volts, 50 cycles , 3 phase AC supply provided with class H insulation and static pressure shall be as indicated. Fans shall be activated from the Car park ventilation PLC panel. Fan shall be with flexible fire proof connection at inlet and outlet. Fans shall be selected for minimum 65% efficiency. Fan shall be with flexible fire proof connection at inlet and outlet. The motor selected shall be energy efficient having IE3 efficiency class rating. The unit shall be suspended or floor mounted through spring isolators and threaded rod as per the standard installation details and as mentioned in specifications. All hangers, clamps, brackets, anchor/fixd support,guides,rollers etc. shall be shall be modular hot dip galvanized having galvanizing thickness 75 microns as per approved makes unless specified otherwise and then supply and installation of the same shall also be included for rates under this head.	Sound Levels & Efficiency for the Unit shall be as per OEM Standard selction please confirm?	Follow BOQ & specification		
14		J.FANS	FANS		Supplying, installing, testing and commissioning of INLINE CENTRIFUGAL DUCT FANS suitable for installing in any position in vertical or horizontal ducts. The casing shall be single skin, internally acoustically lined and constructed of galvanised steel. The motor shall be suitable for 220±10% volt single phase 50 cycles AC supply. All units shall be complete with duct flexible connector double canvas sleeve with sandwiched 25 mm thick fibreglass insulation and volume control damper. The fan shall have low sound level exceeding not more than 40 DBA at three metre distance. The unit shall be suspended or floor mounted through spring isolators and threaded rod as per the standard installation details and as mentioned in specifications. All hangers, clamps, brackets, anchor/fixd support,guides,rollers etc. shall be shall be modular hot dip galvanized having galvanizing thickness 75 microns as per approved makes unless specified otherwise and then supply and installation of the same shall also be included for rates under this head.	Sound Levels & Efficiency for the Unit shall be as per OEM Standard selction please confirm?	Follow BOQ & specification		
15		K.DUCTING & ACCESSORIES	DUCTING & ACCESSORIES	1	General Ventilation system Supply, fabrication and erection of factory fabricated Rectangular GSS Ducting in complete with all fittings such as tee, bends, special off chutes, turning vanes, splitter, dampers inspection doors, transformation pieces,ball mouth, wire mesh etc. as required and as per SMACNA Standards including suspension and supporting arrangement plenum, duct conforming to specifications and in accordance with the approved shop drawings.Duct price to include all required scaffolding/fixing arrangements for installation the ducts The below mentioned accessories also shall be a part of the duct system i)Turning vanes. ii) Hot dip galvanised modular support systems for hanging/supporting, suspending, floor mounting etc.,. iii) Closed cell neoprene gasket. iv) with 180 GSM zinc coating (including both sides) v) All hangers, clamps, brackets, anchor/fixd support,guides,rollers etc. shall be shall be modular hot dip galvanized having galvanizing thickness 75 microns as per approved makes unless specified otherwise and then supply and installation of the same shall also be included for rates under this head. vi) Duct sleeve to be with fire sealant. Refer Technical Specification Ref : 8 Air Distribution 8.2, 8.3, 8.4	1. Any Civil work shall be in Client Scope please confirm? 2. we will not consider any type of Modular or Siesmic Support for Duct.Please Confirm	1.Yes 2.Follow the BOQ		
16		K.DUCTING & ACCESSORIES	DUCTING & ACCESSORIES	10	Supply, installation, testing and commissioning of Motorised fire damper tested as per UL555 suitable for 230V, complete with control panel, inter connecting wiring at locations with 16 G GI steel outer frame and 16G GI steel blades, with 18G sleeves of minimum 450 mm length, accesres doors at locations as shown in approved shop drawings and as per specifications complete with actuators and control wiring. Refer Technical Sncedification Ref : 8.8 Fire & Smoke Dampers	can we consider CBRI approved fire damper but the construction shall be as per UL	OK		
17		Make List			Request for Additional Make Approval				
18		Make List			VRF	Bluestar			
19		Make List			Expansion tank & Air Sperator	KD Agencies, Anergy, Emerald			
20		Make List			AHU	Waves			
21		Make List			FCU	Midea			
22		Make List			Chemcail Dosing	Inchem, Ion Exchange	Follow makelist		
23		Make List			Fans	Airflow, Elta, Nicotra			

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1	PHE BOQ	A.Sanitary Fittings & Acc	A		SANITARY FITTINGS & ACCESSORIES (ONLY INSTALLATION, TESTING & COMMISSIONING)	We will consider installation of Sanitaryware only, all CP & Sanitaryware will be free issue by Client for site.	Yes				
2	PHE BOQ	E. Pump Room Equipment	E, Item No. 10		Providing ,fixing, testing and commissioning of Cast Iron Motorized Butterfly valve interconnected with CAPACITIVE TYPE level sensor of sizes as indicated on drawings and schematics. Control logic to be referred from the technical specifications.	Please confirm PN Rating for Valve.	PN 10				
3	FPS BOQ	A. Sprinkler	A, Item No. 04		Supplying, installing, testing and commissioning of 15mm CP Brass quartzoid bulb type screw end connected Sprinkler Head , UL & FM Approved with approved chrome finish including joints to steel pipe work. The sprinkler shall be either pendent or upright type/side wall as specified on drawings and shall not be universal style. All concealed pendent sprinkler will be provided with rosette plate. Sprinkler shall be quick response type for all areas except car park and mechanical areas (i.e. Refer technical specs for sprinkler discharge coefficient table. (Sprinkler K-Value = K-80 Contractor shall supply 2% spare sprinkler heads and one spanner neatly installed in a wooden cabinet with glass shutters as approved by the Employer. This is to be included in the cost. Fittings to be UL & FM Approved. Technical Specification Ref : 3.4 Sprinkler Heads	Can we consider standard coverage sprinklers.	Follow BOQ & specification				
4	FPS BOQ	A. Sprinkler	A, Item No. 11		Pressure Reducing Valve Sprinkler System only Supplying, installing, testing and commissioning of Diaphragm type Ductile iron/Cast Iron PRV station with pressure (Pressure dependent valve) reducing valve as per the specification,schedule,drawings and as described here under. The same shall be with Ductile iron body & cover and internal bypass line, "Y" strainer, isolation valves and pressure gauges before and after PRV with bypass arrangement shall be provided as indicated on schematic and drawings and suitable for upstream operating pressure of 16 bar . Refer to typical detail no in drawing PEPL-MUM-XXX-FP-D-5001 for the accessories which are to be part of the PRV Assembly. Technical Specification Ref : 3.9 Zone Control Valve & Inspection and Test Valve Assembly	Please specify MOC of PRV (Ductile iron / Cast Iron)	Follow specification,page no 31, Sub-section no 3.9				
5	FPS BOQ	B. Hydrant	B, Item No. 04		Supplying, installing, testing and commissioning of indoor type Lockable Fire Hose Cabinet (Concealed Type). The entire cabinet will be masonry construction except for the door which shall be two door and fabricated from 16 gauge MS sheet with half front glass door and locking arrangement, suitable to accommodate the below listed fittings/accessories/equipment's. Technical Specification Ref : 2.9 Fire Hose Cabinets (FHC) Concealed type	As Cabinet is Concealed masonry construction ,we have to consider only Doors for same. - Please confirm	Doors will be in architectural scope, this item is RO				
6	FPS BOQ	E. Pump Room Equipment (2)	E, Item No. 12		Installation Control Valve Providing,Fixing,testing and commissioning of Installation Control Valve with turbine type automatic Alarm Gong to be connected with control valve and all other necessary components as per manufacturer's specifications complete as required. Color No. 536 of IS:5 for valves. Technical Specification Ref : 3.5 Installation Control Valve	Please specify MOC of Installation Control Valve.	Ductile iron, Follow BOQ				
7	FPS BOQ	E. Pump Room Equipment (2)	E, Item No. 16		Providing ,fixing, testing and commissioning of Solenoid valve interconnected with electronic level indicator	Please specify MOC of Installation Control Valve.	Follow specification				
8	FPS BOQ	F. Fire Extinguishers	Section F		Fire Extinguisher	Cartarage type or Stored Pressure type Fire extinguisher to be consider - Please confirtn.	Cartridge type, Follow BOQ & specification				

Prebid Queries

Name of Contractor : Vendor 2

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09.12.2024

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	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.						
1	BOQ_MEP_TOV	PHE BOQ		B. Water Supply	Supply of Odent Mepla Multilayer pipe (to be considered as optional WOC along with CPVC) for supply system in the dimensions d 16–75 mm(Operating Temperature: 0–70 °C(up to 95 °C intermittent peak temperature) and Operating pressure:10 bar) with system approval mark DVGW for potable water installation systems in accordance with EN 806, DIN 1988 Mepla multilayer pipe d 16–75 mm, medium-transporting pipe made of PE-RT II, supporting pipe made of aluminum, butt-welded, exterior protective coating made of PE-RT II, lengthways non-positive connection through direct pressing of pipe onto fitting without sleeve, pipe ends hygienically sealed, pipe length 5 m and 50 m.Mepla Pipe need to supply along with necessary fittings such as Polyvinylidene fluoride fittings (PVDF)-Tees,Elbows,Couplers,Reducers etc,Brass or gunmetal fittings for fittings with threaded connection, with hygienic cap. The seal ring made of EPDM fulfils all hygiene requirements such as BGA, KTW recommendation and DVGW W 270. Processing and laying must be carried out as per EN 806, EN 1717 and national collateral standards of the series DIN 1988, the manufacturer specific regulations as well as subject to the relevant standards. Manufacturer specific characteristics are to be confirmed with evidence. with clamps/Hot dip galvanised modular supports for vertical stack structural metal supports as required/directed at site cut to required length laid and clamped in dry wall chase, ceiling and floor including jointing, making connection to fixtures like Shower Head, etc along with threaded elbow, piping with purpose made connectors and along with necessary fixing,nuts,bolts,gaskets,washers,plugs,supports, metal to plastic adopter if required , and arrangement as per specification and drawings. Pressure rating as per schematic and specification. The following points to be considered in the cost and not limited to : Chasing in brick & RCC Wall making good the same to its original conditions & complete in all aspects is in contractor's scope. All hangers, clamps, brackets, anchor/fixed support, guides etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns as per approved makes unless specified otherwise and then supply and installation of the same shall also be included for	Kindly confirm if we need to consider both Mepla & Mapress pipe for Water Supply.	MEPLA to be considered in cost		
2	BOQ_MEP_TOV	PHE BOQ		B. Water Supply	Supply of Odent Mepla Multilayer pipe (to be considered as optional WOC along with CPVC) for supply system in the dimensions d 16–75 mm(Operating Temperature: 0–70 °C(up to 95 °C intermittent peak temperature) and Operating pressure:10 bar) with system approval mark DVGW for potable water installation systems in accordance with EN 806, DIN 1988 Mepla multilayer pipe d 16–75 mm, medium-transporting pipe made of PE-RT II, supporting pipe made of aluminum, butt-welded, exterior protective coating made of PE-RT II, lengthways non-positive connection through direct pressing of pipe onto fitting without sleeve, pipe ends hygienically sealed, pipe length 5 m and 50 m.Mepla Pipe need to supply along with necessary fittings such as Polyvinylidene fluoride fittings (PVDF)-Tees,Elbows,Couplers,Reducers etc,Brass or gunmetal fittings for fittings with threaded connection, with hygienic cap. The seal ring made of EPDM fulfils all hygiene requirements such as BGA, KTW recommendation and DVGW W 270. Processing and laying must be carried out as per EN 806, EN 1717 and national collateral standards of the series DIN 1988, the manufacturer specific regulations as well as subject to the relevant standards. Manufacturer specific characteristics are to be confirmed with evidence. with clamps/Hot dip galvanised modular supports for vertical stack structural metal supports as required/directed at site cut to required length laid and clamped in dry wall chase, ceiling and floor including jointing, making connection to fixtures like Shower Head, etc along with threaded elbow, piping with purpose made connectors and along with necessary fixing,nuts,bolts,gaskets,washers,plugs,supports, metal to plastic adopter if required , and arrangement as per specification and drawings. Pressure rating as per schematic and specification. The following points to be considered in the cost and not limited to : Chasing in brick & RCC Wall making good the same to its original conditions & complete in all aspects is in contractor's scope. All hangers, clamps, brackets, anchor/fixed support, guides etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns as per approved makes unless specified otherwise and then supply and installation of the same shall also be included for	Kindly provide layout in AutoCAD format instead of Pdf format.	CAD layout is already included in tender		
3	BOQ_MEP_TOV	PHE BOQ		C. Drainage	Drainage Piping Inside Office and Toilet (Single Stack System)	Kindly provide layout in AutoCAD format instead of Pdf format		Has been shared.	

Sr. No. / Query No.	Bidding Document Reference(s) (Volume No / Section Number / Clause Number)			Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.						
4	BOQ_MEP_TOV	PHE BOQ		C. Drainage	Supply, installation , testing commissioning of HDPE PE 63 PN6 pipe work and fittings as per EN 12056 cut to required lengths, with Electrofusion joints (as per manufacturer recommendations) & making proper connections including all necessary fittings and specials such as bends, tees, unions, reducers, expansion joints,Y joints,Rodding Eye, flanges, supports cowl and plugs etc. Fixing at wall/ ceiling level supported by clamps, thrust blocks, velocity breakers. Hot dip galvanized modular supports hangers etc. as per the specification and drawings. Chasing in brick & making good the same to its original conditions & complete in all aspects is in contractor's scope. Electrofusion jointing process to be followed as per code / Manufacturer recommendations. All hangers, clamps, brackets, anchor/fixed support, guides, rollers etc. to be modular hot dip galvanized having galvanizing thickness 75 microns as per approved makes unless specified otherwise and then supply and installation of the same shall also be included for rates under this head. Welding of any kind on the galvanized support / hanger shall not be permitted . All pipes shall be suitable for PN6 rating. The following points to be considered in the cost and not limited to : Cleanouts & Rodding Eyes will be quoted with Pipes. Covers for capped off connections to be considered. Electrofusion welding & Butt welding machines to be considered in the cost. OEM Reports to be prepared on a fortnightly basis. Refer enlarged detail of toilets showing connection details, types of fittings etc. Allow for all protective and decorative painting and cleaning of pipes, fittings and equipment's. Refer Technical Specification Ref : 3.4 HDPE Pipes & Fittings	1) Kindly provide rainfall intensity. 2) Kindly provide the layout in Auto CAD format instead of pdf format.	1.Consider 150mm/hr rainfall intensity 2.CAD layout is already included in tender		
5	BOQ_MEP_TOV	PHE BOQ		D. Car Park & External SWD	Supply & Installation of CI Flat Grating -150mm x 150 mm size with pipe connection 200 mm length with accessories for car park drain, Plant rooms at basement levels. Refer Technical Specification Ref : 3.12 Floor Drains for Car Park Areas	1) Kindly provide load capacity of CI grating.	Follow specification, Grade B-125		
6	BOQ_MEP_TOV	PHE BOQ		E. Pump Room Equipment	Supply, Installation , testing, commissioning and handing over of below mentioned pumps along with motor, base frame, mounting arrangement, protection drive sets pressure with isolation cock. Mechanical seal, suitable vibration elimination inertia bases of approved design, drain pipe with valve (25 dia) for the pump and all other necessary accessories as complete package fully complying to specification B,C, and D and tender drawings. The pump shall be suitable for 415±10% volts 3 phase AC supply. (Pump room valves & accessories are quantified separately below)	1) Kindly provide the schedule the Pumps PEPL_MUM_PLA_PL_SCH_0001.	Already provided in the tender	Has been shared.	
7	BOQ_MEP_TOV	PHE BOQ		F. Water Tank Fittings	A solution of chlorine shall be applied directly to the surfaces of all parts of the storage facility that would be in contact with water when the storage facility is full to the overflow elevation as specified.	Kindly confirm if we need to consider the scope only tank disinfection or both tank cleaning & tank disinfection.	Consider both		
8	BOQ_MEP_TOV	PHE BOQ		H. Electrical	Design, Unloading, Safe storage , handing, Installation , testing, commissioning and handing over of free standing, front operated, fully compartmentalized cubical type, extendable on both sides , Panel Boards suitable for indoor installation & for use on 415Volts 3-phase, 4wire, 50Hz system having (with-stand) fault current carrng capacity as specified on the SLD, housed with incoming breakers, suitable manual interlock arrangement wherever applicable, outgoing feeders, busbar & necessary metering with interconnections etc. Complete as per schematic drawing, relevant standards & including all accessories as per site conditions to make the system fully operational. Notes: 1) Sheet steel - 2mm thick for frames and equipment mounting plates, 1.6mm thick for partitions. Panels shall be powder coated. 2) Form of protection as shown in SLD. 3) Welded construction with transport sections bolted together. All such joints to be gasketed. 4) The cubicles shall be totally dust and vermin proof. 5) All doors to be of hinged type except busbar chamber covers, which shall be of bolted type. 6) The panels shall be of flush front design and hence shall be designed for maintenance with back access only. 7) The construction shall be such as to facilitate easy extension at both ends.	Kindly confirm the scope of Panels , as per BOQ description supply is not in our scope.	Panel supply is in PHE contractor scope		
9	BOQ_MEP_TOV	BMS BOQ		General	Kindly provide the IO suumary for BMS system.		Refer the IO Summary in BMS tender		

Name of Contractor : Vendor 2

Date :

09.12.2024

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	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.						
10	BOQ_MEP_TOW	FPS BOQ		A. Sprinkler	Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239/IS-3589 (heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall / ceiling with clamps, hangers (using anchor fasteners) as per specification.(For horizontal piping - Grooved flexible & rigid coupling type shall be used for all pipe sizes.) Fittings to be UL & FM Approved. G.I. pipe sleeve of suitable higher size shall be provided wherever the pipes are crossing the walls/floors and sealing the sleeves with glass wool in between & fire sealant compound at either end all as per Project Manager's / Consultants requirements including cutting holes and chases in brick, R.C.C work and making good the same to original conditions complete in all respects. Allow for builders work/sleeves associated with fire protection installation work. All hangers, clamps, brackets etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns as per the approved makes unless specified otherwise and then supply, installation of the same shall also be included for rates under this head. Welding of any kind on the galvanized support / hanger shall not be permitted. For pipe pressure rating refer specification and schematics. Pipes shall be coated with the two coats of Zinc chromate (Yellow colour) and painted with two coats of synthetic enamel paint of approved make and shade No. 536 of IS:5 as per specifications and final approval of consultant.Allow for all protective and decorative painting and cleaning of pipes, fittings and equipment's. Technical Specification Ref : 2.1 Pipes & Fittings	Kindly confirm if we need to consider schedule 40 seamless butt welded fitting for complete piping or we have to consider grooved fitting with coupling.	Grooved fittings to be considered		
11	BOQ_MEP_TOW	FPS BOQ			General	Kindly provide AutoCAD layout for FPS package.	CAD layout is already included in tender		
12	BOQ_MEP_TOW	FPS BOQ		G. Electrical	Design, fabrication, assembling, wiring, supply, installation, testing and commissioning of fire fighting switchgear as noted below and on Contract Document. Cable gland plates shall be provided on top of the panels. Panels shall be treated with all anticorrosive process before painting as per specifications with 2 coats of red oxide primer and final approved shade of Pot office red powder coated paint. 2 Nos. earthing terminals shall be provided for 3 phase, 4 wire, 50 Hz supply system per panel. Lifting hooks shall also be provided in case of large panels. Approval shall be taken for each panel before fabrication. Cadmium Plated hardware shall be used in fabrication of panels.	Kindly confirm the scope of supply for Panels for FPS package.	Panel supply is in FPS contractor scope		
13	BOQ_MEP_TOW	HVAC BOQ		M. ELECTRICAL	Supplying, unloading at site, shifting into the store, Laying, Testing & Commissioning of 1100V grade, Fire Survival FS/PH120, XLPE Insulated, Steel Wire Armour, Copper conductor Cables to be laid with necessary supports on cable trays/ on ceiling / wall complete in all respect as per site requirement, technical specifications, and as per approval from client / consultant. To be read in conjunction with contract documents. Armour to be used as CPC.	Kindly confirm if we need to quote FS cables as per IS code or BS code.	Please adhere to specs and quote as per BS code		
14	BOQ_MEP_TOW	HVAC BOQ		M. ELECTRICAL	AHUs to be with Starter/ control panel which will house the VFD. Control Cabling from AHU Starter Panel to the respective equipment located on piping and ducting shall be included. VFDs shall be designed to limit THDi to less than 5%. All required control cabling from VFD, Control panel shall be included in the scope. Panel shall have provision for providing 230 V & 24 V supply for respective instruments and spare supply provisions	1) We understand that the scope of starter panel for AHU is already considered in Electrical sheet in HVAC BOQ,hence we are only considering the cost of AHU only, Kindly confirm. 2) We are considering VFD with AHF for AHU starter panel mentioned in the electrical sheet , Kindly confirm.	1) Please adhere to boq description and quote with starter panel 2) AHU is with EC plug fan hence AHF not required		
ELECTRICAL									

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	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.						
15	BOQ_MEP_TO WER_ABCE	A - 1, HT & XMER	1.1 to 1.7			We are considering HT Panel from OEM approved system house / approved channel partner. Please confirm. Approved make for the HT panels are Siemens ,ABB, Schneider,Lauritz Knudsen (earlier L&T). Please confirm we have to consider the panel directly from OEM or from OEM approved system house / approved channel partner. Also please confirm IP indoor or outdoor requirement for the panels.	OEM approved panels are ok . This is IP42 panel,indoor		
16	BOQ_MEP_TO WER_ABCE	A -1.2, Battery and charger	1.2.1			we understand charger rating shall be (3 Phase) 110V-40A Float cum boost charger with 110V-200AH VRLA Battery.	Charger rating is 3phase 110V-40A float cum boost charger		
17	BOQ_MEP_TO WER_ABCE	Busduct			Delta on HT Side & Star on LT Side with neutral terminal brought out for solid earthing in addition to neutral terminal in the L.V. cable box. Provide a 3000/5A CL.1 5 VA NCT for Stand by Earth Fault protection.	We have considered Bus duct flange for LV termination. However Please re-confirm LV Terminal arrangement whether it is cable box or Bus duct.	The transformer LV is with busduct		
18	BOQ_MEP_TO WER_ABCE	LV Electrical panel board				we are considering authorised channel partner for LT panel please confirm ?	OEM channel partners are ok. Please share IEC 61439 certificate for all MLTP and SMDB . Please share sample discrimination report of project of similar scale .		
20	BOQ_MEP_TO WER_ABCE	Point Wiring			Supply, Installation, Testing and Commissioning of below said point wiring with pre laid HMS PVC conduits concealed in wall/Slab/wooden partitioning/ and GI conduit and flexibles for below false ceiling fixtures, fixed with spacers, GI saddles etc. with bends, sockets, couplings, junction boxes etc. to make piping rigid with chasing in wall if required and making good, including concealed metal box (Suitable for modular type switch boards) with ceiling	As per BOQ description, it is mentioned that the conduit (PVC / GI) is already laid in the slab, please confirm do we have to consider the conduit or not.	Conduits are pre laid in slab. Dont consider them in point wiring item.		
21	BOQ_MEP_TO WER_ABCE	Point Wiring			PIR, BLANKING PLATES & PUSH BUTTONS : Supply,installation and testing of high performance PIR sensors for Parking bays including 3runs 1.5sqmm HFFR wiring to 2 lights & toilets	Please share high performance PIR sensors Cat No. or please confirm can we consider normal standard occupancy sensor.	Please consider Occupancy sensor		
22	BOQ_MEP_TO WER_ABCE	Point Wiring			Conduit GI/MS	We are considering standard thickness for 1.4 to 1.6mm for GI/MS conduit upto 25mm and 32mm & above 2mm.	GI conduit thickness min 1.6mm thick upto 25mm dia and 2mm thick for 32mm dia and above		
23	BOQ_MEP_TO WER_ABCE	Common area			DB Thickness	We are considering thickness 1.2mm for the standard DB's.	Please adhere to the specifications ,provide 1.6mm thick		
24	BOQ_MEP_TO WER_ABCE				Civil Work	We have considered chipping work and cleaning the debris from the site. Filling the chasing area, making cutting area good with cement and final finishing with plaster and paint will be in client scope. Please confirm.	Chipping and cleaning the debris from the site and making good is in Electrical contractor scope. Final finish and paint will be in client scope		
25		General			Tax BOCW	Please confirm BOCW tax to be consider in our scope or it will be paid by client.	PMC to address	Contractor can mention the BOCW cess separately in the summary sheet of the Bid submitted by them	
26		General			Construction power	We understand that Construction power is available at site and we will get a single point supply further distribution in our scope.	PMC to address	Please refer the special conditions of contract and responsibility matrix point no. 17 and 19 carefully for detailed responsibility of the MEP contractor. The cost of creating site infra for own use and other contractors use, employer will ask MEP contractor to provide the water and power and will be in scope of MEP contractor. All the meter and sub meter provisions are in the scope of MEP contractor. MEP contractor to include all such costs in their quoted rates.	
27		General			CEIG	We are considering only CEIG coordination and High rise approval.	Please adhere to boq description		
28		General			Transformer, DG set	We are not considering any civil work or foundation for transformer.	ok	Okay, please refer the responsibility matrix carefully for further details	

Name of Contractor : Vendor 3

Date : 09.12.2024

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		Plumbing Queries							
1		Plumbing _ B Water Supply	B3 _ Water Supply Valves & Accessories _ 3.PR.V station in mumbai		Supply, installation , testing commissioning of Forged Brass PRV station with balanced seat, standard pattern, set point scale, high quality synthetic material valve insert, SS Wire mesh, NBR diaphragm, built in filter, NBR seats, pressure gauges, Y-strainer, isolation valves before and after PRV, with isolation valve in by pass line and necessary arrangement , mating flanges to suit the specified piping material, as specified and complete as required in water supply drop pipes. on-line type as shown in drawings/ schematics and in accordance with technical specification To suit pressure rating as specified on schematics and specification. For inlet and outlet pressure rating refer PRV schedule and schematics. PRV shall be complete with station with isolation valve, pressure gauge before after PRV along with bypass line with stand by PRV and isolation valves, pressure gauge before and after PRV. Each station consists of 1 PRV assembly. 1 PRV Assembly Set consists of 2 Nos of PRVs, 4 isolation valves and 2 nos of pressure gauges. Refer typical details & schematic. PRV shall be diaphragm type only. Note: (1) Up to 50mm dia threaded forged brass type and above 50mm dia-CI flange type. (PN 10 Operating Pressure) The pressure gauge shall be constructed of die cast aluminum and stove enamelled. It shall be weather proof with an IP 55 enclosure. It shall be a stainless steel Bourdon tube type pressure gauge with a scale range from 0 to 20 Kg / cm square and shall be constructed as per IS-3524. Each pressure gauge shall have a siphon tube connection. The shut off arrangement shall be by Ball Valve. Dial gauges shall have adequate response for the pressures encountered within the specified (Range 0-20 Kg/sq.cm). Pressure gauges shall be of Bourdon tube and spring type accurate within 1% with minimum 115 mm dia. dial and with recalibrating screws. Refer Technical Specification Ref : 2.11 Pressure Reducing Valve	Please share SLD	SLD is shared in the tender		
2		Plumbing _ B water Supply	B3 _ Water Supply Valves & Accessories _ 4. Bib tap		Refer Technical Specification Ref : 2.11 Pressure Reducing Valve Supply, Unloading, Safe storage , handing, Installation , testing, commissioning and handing over of Brass material hose Bib tap with hose connector and lock type arrangement in accordance with technical specifications.	We are only considering Installation please confirm	To be included in ITC		
3		Plumbing _ B water Supply	B3 _ Water Supply Valves & Accessories _ 5. Automatic Air vent		Refer Technical Specification Ref : 2.12 Brass Bib Cocks & Stop Cocks Supply, installation, testing & commissioning of Auto Air Vent with bronze body and non ferrous float for supply risers, with isolation valve, with unions and accessories with necessary mating flanges to suit the specified piping material and in accordance with specification. To suit pressure rating as specified on schematics and specification. Refer Technical Specification Ref : 2.20 Automatic Air Release Valves		Query missing		
4		Plumbing _ C Drainage	c.3 _ Sanitary and Waste Vent pipes within the building _ 1. Oil interceptor & Sump pit- Vent Stacks		Supply, installation, testing commissioning of Vent pipe of 1. Indian UPVC SWR Type - B as per IS 13592 for pipes 80 mm and Above. 2. Indian UPVC Class-3 as per IS 4985 for pipes Below 80 mm. Solvent cement joints with necessary fittings such as making proper connections to equipment's including all necessary fittings and specials such as bends, tees, reducers, nuts, bolts, gaskets, supports and plugs etc. Fixing at wall/ ceiling level supported by clamps, expansion joints branches, hangers etc. as per the specification, drawings and schematics. Chasing in brick & making good the same to its original conditions & complete in all aspects is in contractor's scope. All hangers, clamps, brackets, anchor/fixed support, guides, rollers etc. to be modular hot dip galvanized having galvanizing thickness 75 microns as per approved makes unless specified otherwise and then supply and installation of the same shall also be included for rates under this head. Welding of any kind on the galvanized support / hanger shall not be permitted. Note:- Cleanouts & Rodding Eyes will be quoted with Pipes. Allow for all protective and decorative painting and cleaning of pipes, fittings and equipment's. Refer Technical Specification Ref : 3.3 Vent Pipes	Please confirm Chasing is required or not for this pipes as these are vent pipes.	Follow BOQ line		
5		Plumbing _ C Drainage	c.3 _ Sanitary and Waste Vent pipes within the building _ 3.1 Drainage water accessories		Supply, installation , testing commissioning of UPVC Single Flow Trap / drains with provision for fixing MS gratings with epoxy paint of 150 x 150 mm with one nos. of inlet socket and one no. out let socket with diameters as indicated on drawings and schematics with P/S trap of 50mm and to suit the embedded installations along with making proper connection with solvent joints, cutting chase / hole in floors / slabs and bringing the same in proper condition and shape after placing the trap in right position and in accordance with specification. Refer Technical Specification Ref : 3.9 Traps	Please provide the size of the trap and drawing	Size is given in BOQ line item		
6		Plumbing _ C Drainage	c.3 _ Sanitary and Waste Vent pipes within the building _4 Utrap		Upvc type ' U' Traps for waste pipe before connection to main sewer header	As per the market standered U trap is only available in size of 110mm. So, Kindly suggest on how to proceed	This item is RO		
7		Plumbing _ C Drainage	c.3 _ Sanitary and Waste Vent pipes within the building _5 vent cowl		Supply, installation , testing commissioning of Roof vent cowl (PVC as per pipe MOC). Installation as per standard installation details.	Please provide the clear details about the type of pipe is it for SWR pipe or Low noise SWR Pipe Please confirm on the given Diameters as 100 and 80mm dia cowls not as per the market standered. So, Please check and confirm on how to proceed	BH will revert		
8		Plumbing _ C Drainage	c.3 _ Sanitary and Waste Vent pipes within the building _6 Scupper Rain water head		Supply, installation, testing commissioning of Scupper type Rain water head (for Refuge areas and peri metered Terrace floors) of epoxy coated CI body with SS screw. Include required length with puddle flange to fix head in RCC slab.	Please provide the make along with model and please confirm we need to add the Puddle flange in the price or not	Follow makelist, consider ancillary costs.		
9		Plumbing _ D. Car Park & External SWD	D.1 _Car Park Drainage @ Basement-04 Level		Supply & Installation of CI Flat Grating -150mm x 150 mm size with pipe connection 200 mm length with accessories for car park drain, Plant rooms at basement levels. Refer Technical Specification Ref : 3.12 Floor Drains for Car Park Areas	Please provide the pipe details type of the pipes and size of the pipe And please share any reference drawing or image	Follow specification, refer layout		
10		Plumbing _E. Pump Room Equipment	E. Pump Room Equipment		E. Pump Room Equipment	For all the pumps please provide the separate drawings along with schedules. Due to heavy PDF we are unable find the same. Kindly provide the Capacities of pumps in the BOQ only.	CAD layout was given in tender, refer pump schedule		
11		Plumbing _E. Pump Room Equipment	9.Electronic Type Water Level Indicator		Providing ,fixing, testing and commissioning electronic type Water level indicator with LED panel mounted, with necessary control wiring from control panel to level indicators and motorized valves, complete with probes, & all control wiring etc. (Level indicators in Terrace level tanks & their cabling , Level indicators in Basement level tanks & their cabling-shall be part of this Tender.) Each set includes low & high level sensors along with all necessary accessories. Control logic to be referred from the technical specifications. Refer Technical Specification Ref : 2.15 Level Transmitters for Water Tanks 2.16 Level Switch, 2.17 Level Sensors	Please provide the size of level indicator and tank details	Refer tank layout		
12		Plumbing _E. Pump Room Equipment	11. Water meter		Supply, installation , testing commissioning of testing and commissioning of Electronic ultrasonic water meter with BMS Compatible & direct reading dial in KI with all integral parts of gun metal or brass with necessary fitting such as threaded pieces, unions pressure gauge and isolation cock, flanges piece for future removal, flanges / unions complete with all necessary testing charges and obtaining test certificates from municipal authorities, on following size pipe lines (1 no. strainer, 1 no. of NRV & 2 nos. pressure gauges shall be provided at inlet & outlet and cost shall be inclusive for the same). (PN10 Operating Pressure) Refer Technical Specification Ref : 2.25 Ultrasonic Water Meter	Please provide the SLD	BH will revert (water meter typical detail)		

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13		Plumbing _F_ Water Tank Fitments	4. vent with mesh		50mm Dia vent with mosquito proof wire mesh	Please material for the vent an please confirm is the vent cowl or different	SS wire mesh		
		M.Electrical BOQ							
1		Under M.Electrical BOQ - Sec: M -HVAC ELECTRICAL INSTALLATIONS	Sec: M -HVAC ELECTRICAL INSTALLATIONS:Excel S .NO :8		Include for the Design, Manufacture, Loading, Supply, Transport, Unloading at site, Erection on proper place, Installation, Testing and Commissioning of below listed panels the complete in all respects as per technical specifications, relevant drawings & to the satisfaction of Client and Consultant and concern statutory Authorities under following sections with associated protective devices, relays, meters, terminations, isolators, supports, accessories etc.; To be Read in conjunction with ---- for Details.	Please confirm, In main discription they mentioned Installation only,in sub description they mentioned Supply also,Kindly confirm Shall we consider both supply and installation or only installation Kindly provide the separate SLD	SITC will be in contractor's scope		
2		Under M.Electrical BOQ - Sec: 2 -CABLES AND TERMINATIONS	Sec: 2 -CABLES AND TERMINATIONS:Excel S.No :92		Supplying, unloading at site, shifting into the store, Laying, Testing & Commissioning of 1100V grade Fire Retardant Low Smoke Zero Haolgen (FR-LSZH) , XLPE Inslated, Steel Wire Armour, Aluminium conductor Cables to be laid with necessary supports on cable trays/ on ceiling / wall complete in all respect as per site requirement, technical specifications, and as per approval from client / consultant. To be read in conjunction with contract documents. Armour to be used as CPC.	Please confirm, Shall we consider the excavation for LT Cables or not	Please don't consider excavation for LT cables		
		Electrical Queries							
3		Under A. HT & XMERI BOQ - Section A :HT PANEL	Section A : HT PANEL : Excel S NO : 7		Include for the Supply, Installation, Testing & Commissioning, Unloading, Handling of free standing, front operated, fully compartmentalized cubical type, HT Panel suitable for indoor installation with IP Rating and Form of construction as noted on SLD PEPL_MUM_PL_C_EL_D_7001 & PEPL_MUM_PL_C_EL_D_7007 , including all accessories, Readymade hot dipped galvanised supports as per direction of Engineer incharge site conditions to make the system fully operational. Notes: 1) Sheet steel - 2mm thick for frames and equipment mounting plates 1.6mm thick for partitions and powder coated with RAL 7032. 2) Modular construction with transport sections bolted together. All such joints to be gasketted with Neoprene Gaskette 3) The cubicles shall be totally dust and vermin proof 4) All doors to be of hinged type except busbar chamber covers, which shall be of bolted type with Earth Bar Link 5) The panels shall be of flush front design, mounted against the wall and hence shall be designed for maintenance with front access only. 6) The construction shall be such as to facilitate easy extension at both ends. 7) SMC supports and heat shrinkablePVC sleeves shall be provided for busbars. 8) Panel to be provided with Aluminium Basbar (3 Pole) 9) A Aluminium earth strip/bus of adequate size/section shall be run throughout the length of the Panel to which all earth terminals of the various components shall be connected. Provide one terminal for each outgoing circuit. Connect directly to earthing terminal without dependence on exposed conductive parts of enclosure. 10) Panel should be minimum IP42 rated and to be equipped with fire suppresion system. 11) HT Panel switchgear to be IEC 62271 Complied.	Please provide the, Auto cad drawings to route for HT panels	Files have been shared		
4		Under A. HT & XMERI BOQ - Section C :TRANSFORMERS (As per Technical specification & Schedules)	Section C :TRANSFORMERS (As per Technical specification & Schedules) Excel S .NO :71		Include for Supply, loading at factory, unloading at site, storage, moving it to respective location, erection,testing and commissioning of 11kV primary and 415V secondary, 3phase, 50Hz, step down, dry type off load tap links $\pm 7.5\%$ in steps of 2.5% transformer with Busbar box on LT side and cable box H.T. Side as per the drawings and as specified in the technical specifications and as directed. The rate shall include the lifting of transformer to the necessary height, necessary hardware, grouting, with all allied civil works etc, as required as complete. (Cable and busbar entry/Exit-Top). Transformer design should be as per IS: 2026. Also refer PEPL_MUM_PL_C_EL_D_7001 & PEPL_MUM_PL_C_EL_D_7007 SLD. Factory Visit for witnessing performance and adherence to match design parameters or surpass them. 4 Persons to be considered for Inspection, Testing of Equipment. Travels & Stay shall be included in the quoted rates. 2500kVA transformer to have only positive tolerance for impedance ie impedance will be strictly in the range of 6.25% to 6.87% . Refer Technical Specification Ref : 2.1.6 Transformer	Kindly provide the detail description of transformers and also prvide the SLD	Please refer specifications and Schematic		
5		Under A. HT & XMERI BOQ - Section E :CABLE CONTAINMENT, TRUNKINGS AND RACEWAYS	Section E :CABLE CONTAINMENT, TRUNKINGS AND RACEWAYS : S .NO 137		Supply & Laying of hot dipped, galvanized Iron, Ladder having 2mm thickness & 75mm height as per specifications, of following sizes, suspended rom ceiling or fixed vertically on wall / column including necessary readymade hot dipped galvanized supports like anchor bolts, hangers, brackets etc. and accessories like Elbows, Bends, Tee, requisite support/suspenders, hardware etc. as per site requirement. All hangers, clamps, brackets etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns as per the approved makes unless specified otherwise and then supply, installation of the same shall also be included for rates under this head. Supplying, installing, testing and commissioning of Seismic Supports for vertical riser to take care of wind, buidinging settlement and seismic effect. The contractor shall bring on board the seismic contractor & submit all the required calculations & methods. The cable trays support shall be fixed at every 1 meter spacing. Ladder type cable trays shall be made from 2mm thick sheet formed in 'C' section of 75mm height and inward flanges of 15mm as side runners and 30mm wide x 10mm high rungs ('C' shaped) from a 1.5mm thick sheet. Perforations as mentioned above shall be provided in the width of the rungs. Pitch of the rungs shall not exceed 250 mm centre to centre. Rungs shall be tack welded to the side members. Perforated trays shall be of the formed channel shape. The trays will be joined with fish plates,nuts,bolts and NOT welded . For all cable trays $\geq 600\text{mm}$ web height should be 75mm and all containment < 600mm web height should be 50mm . The zinc coating thickness for cable trays shall be minimum 460 gms/sq. meter as per IS 4759 (1996). Allow for all identification labels, plates, charts and identification of plant, equipment's etc as per section B.4-MEP Identification & Labelling-PROJ-BHE-ZZ-ZZ-SP-Z-1004.	Kindly confirm, shall we consider the supports in line item wise or seperately measured in kg's. Also provide the MS angle size for cable tray.	Please adhere to boq		
6		Under B. TENANT BOQ - Section G :ATS	Section G :ATS :S.No : 6		Supply, Installation, Testing and Commissioning of below listed ATS equivalent to 7000 Series of Asco, in RAL 7032 IP 21 metal enclosure & with all necessary accessories, supporting arrangement, gland plate etc; . For ATS,one cable entry from top and second is busbar and exit is busbar All entry/exit from top. Please refer SLD sheet: PEPL_MUM_PL_C_EL_D_7001 to PEPL_MUM_PL_C_EL_D_7010 Refer Technical Specification Ref : 2.6 Cable Containment Refer Technical Specification Ref : 3.19 Auto Transformer Switch (ATS)	kindly provide the, Auto cad drawings	Drawings have been shared		

Name of Contractor : Vendor 3

Date : 09.12.2024

Sr. No. / Query No.	Volume No.	Bidding Document Reference(s) [Volume No / Main Section No. / Sub Section or Appendix number]	Section Number / Clause Number	Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [LL / PEPL]	Remarks
7		Under B. TENANT BOQ - Section I : BUSDUCT AND RISING MAINS AND TAP OFF UNITS	Section I : BUSDUCT AND RISING MAINS AND TAP OFF UNITS S. No :61		Supply, installation, testing, Transport, Loading, unloading on proper place at site as directed by Engineer In Charge with Supervision & Commissioning of compact sandwiched Type, Aluminium Busduct / Rising Mains with Class B insulation in GI extruded enclosure with 2 Nos Integral GI strip earth strip to run along with rising mains with all accessories like bends, supports,copper flexibles, phase changeovers,flange end, end piece, plug in every floor for each riser, necessary supporting arrangements, hangers etc. as below complete in all respects as per approval of Client / Consultant / PMCs. The Rising Mains shall be suitable for IP54 class of protection with provisions for tap off boxes at every floor with additional 1nos plug in point every floors . The rising mains / busducts shall be approved by CPRI, Local electricity board, PWD Inspector, & any other governing authority. Contractor to furnish all the required certifications, documents for clients / consultants / PMCs review & approval purpose prior to start manufacturing. Supports to be powder coated with same finish as the busduct. Busduct quantification to be inclusive of all joints, terminations and accesories.The busduct enclosure also should be able to carry the fault level kA for 1 sec. All hangers, clamps, brackets etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns as per the approved makes unless specified otherwise and then supply, installation of the same shall also be included for rates under this head. Supplying, installing, testing and commissioning of Seismic Supports for vertical riser to take care of wind, buidinging settlement and seismic effect. The contractor shall bring on board the seismic contractor & submit all the required calculations & methods.Allow for all identification labels, plates, charts and identification of plant, equipment's etc as per section 8.4-MEP Identification & Labelling -PROJ-BHE-ZZ-ZZ-SP-Z 1004. Refer Technical Specification Ref : 2.3 LV BUSDUCT Factory Visit for Busduct for witnessing performance and adherence to match design parameters or surpass them. 4 Persons to be	Kindly confirm, shall we consider the supports in line item wise or seperately measured in kg's.	Please adhere to boq		
8		Under B. TENANT BOQ - Section j -CABLES AND TERMINATIONS	Section j -CABLES AND TERMINATIONS S. NO : 83		Supplying, unloading at site, shifting into the store, Laying, Testing & Commissioning of 1100V grade, Fire Retardant Low Smoke Haolgen (FRLSH), XLPE Insulated, Flat Steel Wire Armour, Aluminium conductor Cables to be laid with necessary supports on cable trays/ on ceiling / wall complete in all respect as per site requirement, technical specifications, and as per approval from client / consultant,(This cabling comprises of cables from metering panel to individual office Isolators)/MCCB To be read in conjunction with contract documents. Armour to be used as CPC. The cable from tap off box to ATS and ATS to metering Panel is in Utility scope . Refer Technical Specification Ref : 2.4 L.V. Power & Control Cables, IT/Telecom & TV Cables.	Please confirm, Shall we consider the excavation for LT Cables or not	answered above		
9		Under C. Common Area BOQ - Section N -LV ELECTRICAL PANEL BOARDS, DISTRIBUTION BOARDS AND ISOLATORS	Section N -LV ELECTRICAL PANEL BOARDS, DISTRIBUTION BOARDS AND ISOLATORS S. No : 7		Include for the Design, Manufacture, Loading, Supply, Transport, Unloading at site, Erection on proper place, Installation, Testing and Commissioning of below listed panels complete in all respects as per technical specifications, relevant drawings & to the satisfaction of Client and Consultant and concern statutory Authorities under following sections with associated protective devices, relays, meters, terminations, isolators, readymade hot dipped galvanised supports, accessories etc; Panels to be IEC 61439 compliant. To be Read in conjunction with Specifications No. TPM-BHE-ZZ-ZZ-SP-E-1000 and Drawing No.PEPL_MUM_PL_C_EL_D_7001 To PEPL_MUM_PL_C_EL_D_7010. Refer Technical Specification Ref : 2.2 L.V. Panels, Enclosed switchgear, Distribution Boards, VFD's, Control Panels & Starters Notes: 1) Sheet steel - 2mm thick for frames and equipment mounting plates, 1.6mm thick for partitions and powder coated with RAL 7032. 2) Welded construction with transport sections bolted together. All such joints to be gasketted 3) The cubicles shall be totally dust and vermin proof 4) All doors to be of hinged type except busbar chamber covers, which shall be of bolted type. 5) Wherever necessary, if the switch boards are to be mounted against a wall, the layout of the components and interconnections shall be designed to facilitate maintenance from the front. 6) The construction shall be such as to facilitate easy extension at both ends. 7) SMC supports and heat shrinkable PVC sleeves shall be provided for busbars. 8) The design shall be such as to have individual feeders for MCCBs ACBs in separate segregated compartments, with proper barriers between adjacent feeders, busbar chamber and cable termination chambers. However groups of MCB's can be housed in a compartment.	kindly provide the Auto cad drawings to quote for panels and distribution boards Kindly confirm Supply is client scope or contractor Scope		Files have been shared	
10		Under C. Common Area BOQ - Section 1.3 - Distribution Boards (DB) (As per Technical Specification, Schedules & SLD) For Lighting and Power	Section 1.3 -Distribution Boards (DB) (As per Technical Specification, Schedules & SLD) For Lighting and Power S. No : 148		Supply, Installation, Testing & Commissioning of Double Door Distribution Boards including MCB/ELCB/RCBO, copper busbar, neutral link. The Distribution Board shall be mounted on wall/ partitions with necessary supports, making zaris, chipping & again back filling by using cement mortar etc; complete in all respect as per drawings, technical specification, load details and as specified and as directed. Supports to be Hot dipped galvanized. To be Read in conjunction with Specifications and SLD no. PEPL_MUM_PL_C_EL_D_7001 To PEPL_MUM_PL_C_EL_D_7010. The contractor to quote in accordance with the DB schedule. The rate shall be inclusive of allied civil works. Distribution boards shall be powder coated with RAL 7032. Allow any other item or equipment not mentioned above but required by the OEM / Contractor to provide a fully functional system including but not restricted to connectors leads, jumpers, cabling, nuts, screws, bolts, mounting plates, cover plates, back boxes, Cable End Boxes and 1.6mm thick 1000mm GI trunking of suitable size and length for termination of large cables/wires and accessories as required to make the system complete and operational in all respects. Refer Technical Specification Ref : 2.2.17 Distribution Boards	Please confirm what is the thickness of DBs,Please confirm that we have to provide Timer and contactor for every DB, DBS: Distribution boards are not found in the drawings list,Kindly provide the separate drawing list for DBS	1. for thickness of DB ,adhere to boq and specifications .Please quote as per boq with timer and contactor . Please refer DB schedules in CAD drawing format		
11		Under C. Common Area BOQ - Section 1.40 - Distribution Boards	Section 1.40 -Distribution Boards : S. NO :319		Supplying and fixing metal double door DB. Distribution board with IP-42 protection and IK09, concealed in wall after cutting the wall & mending good the damages to original finish incl. Inter connection with suitable size of copper wire and neutral link & provision for earthing attachment. PDB-FCU-TA-L15-01 (32A MCB incommmer and 18 nos. of 6A TPN MCB Outgoings)	Kindly confirm,Shall we consider the DB price or not,and also provide the drawings	Consider DB price and refer DB schedule		

Name of Contractor : Vendor 3

Date : 09.12.2024

Sr. No. / Query No.		Bidding Document Reference(s) (Volume No. / Section Number / Clause Number)		Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [LL / PEPL]	Remarks
12		Under C. Common Area BOQ - Section O -STATIC INVERTERS AND UPS	Section O -STATIC INVERTERS AND UPS S. No :392		Include for the Supply Installation, Testing and Commissioning of the complete installation as detailed in the Specifications and on the Drawings and to the complete satisfaction and approval of the Employers Representative and all concerned Statutory Authorities and under the following sections including accessories complete with supports including communications cable etc. Item includes for equipment, emergency battery, accessories, sub circuit monitor, supports, etc., required to complete the scope of works complete as described. Allow any other item or equipment not mentioned above but required by the OEM / Contractor to provide a fully functional system including but not restricted to connectors leads, jumpers, cabling, nuts, screws, bolts, mounting plates, cover plates, back boxes etc and as required to make the system complete and operational in all respects 3Phase In / 3 Phase Out, Battery back time: 15 minutes Note: 1) Rate includes for Inverter with battery charger combined with control and changeover module and battery packs plus and require intercabling between the various components. Interwiring to be 2hr fire rated. Rate shall also include GI battery rack suitable to house appropriate no. of batteries corresponding to below mentioned ratings.Earthing for the UPS and battery system also to be considered. TO BE READ IN CONJUNCTION WITH TENDER PACKAGE & DETAILED NOTES SHEETS FOR THESE SYSTEMS REF DRAWINGS: PEPL_MUM_PLC_EL_D_7001 To PEPL_MUM_PLC_EL_D_7010	Kindly confirm,shall we consider 1Ph in / 1 Ph out or 3Ph in /3 Ph out and also confirm back up of Ups	Consider 3phase in and 3 phase out		
13		Under D.Point wiring BOQ : Section V POINT WIRING	Section V POINT WIRING : S. No :7		Refer Technical Specification Ref : 2.5 Point Wiring Supply, Installation, Testing and Commissioning of below said point wiring with pre laid HMS PVC conduits concealed in wall/Slab/wooden partitioning/ and GI conduit and flexibles for below false ceiling fixtures, fixed with spacers, GI saddles etc. with bends, sockets, couplings, junction boxes etc. to make piping rigid with chasing in wall if required and making good, including concealed metal box (Suitable for modular type switch boards) with ceiling rose,switches,sockets,face plates etc for all exposed light points on ceiling & with concealed round boxes for wall light points, complete as per approval of Client/Consultant. Refer Technical Specification Ref : 2.5 Point Wiring.	Kindly provide the measurements for point wiring to exact quote.	Please adhere to boq and refer drawings in CAD		
14		Under D.Point wiring BOQ : Section V POINT WIRING	Section V POINT WIRING : S. No :7		Supply, Installation, Testing and Commissioning of below said point wiring with pre laid HMS PVC conduits concealed in wall/Slab/wooden partitioning/ and GI conduit and flexibles for below false ceiling fixtures, fixed with spacers, GI saddles etc. with bends, sockets, couplings, junction boxes etc. to make piping rigid with chasing in wall if required and making good, including concealed metal box (Suitable for modular type switch boards) with ceiling rose,switches,sockets,face plates etc for all exposed light points on ceiling & with concealed round boxes for wall light points, complete as per approval of Client/Consultant. Refer Technical Specification Ref : 2.5 Point Wiring.	Kindly Confirm Shall we exclude the conduit and metal box price	Please exclude conduit but include metal box as in the boq description		
15		General			Capital Equipments	The capital equipments supply shall we consider or only installation. Since most of the clients remove this capital equipments like Panels, transformer,bus duct , etc., Kindly confirm.		Please quote for SITC works. As we have given separate columns for the supply and installation of the BOQ item. However, Client will have the liberty to descope these items if the rates quoted by Vendor found on higher side.	
16		General			Technical specifications	Kindly provide all technical data sheet will be separate parts.		As discussed during the pre-bid meeting, the technical data sheets can be readily extracted from the tender document.	
		General				Please provide the below details and documents		Kindly review the tender document, as all the required data is included within.	
						Type of Project			
						Type of construction			
						Towers			
						Floors			
						Total no of flats			
						Area Salable			
						Area Buildup			
						Commencement date			
						Time for Completion			
						Price Validity			
						Mobilization Period			
						PBG			
						Labour Cess			
						Electricity			
						Water			
						Labour accomodation			
						Mobilization Advance			
						Billing Cycle			
						Defect Liability Period			
						Insurance & Policies			
						Satutory compliance			
						Escalation			
						Free Issue/ Client supply materials			
						Basic rate by client			
						Site office store			
						Lifting and shifting of materials Material			
						hoist			
						Liquidated Damage			
						Scaffolding and staging			
						Payment terms			

Pre-Bid Queries for MEP Works for The Prestige, Sales Tower (ABCE) at Mahalaxmi, Mumbai									
Name of Contractor: Vendor 4						Date: 29.11.2024			
Sr. No. / Query No.	Bidding Document Reference(s) (Volume			Tender PDF File Page No.	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
	Volume No.	Main Section No. Sub-Section or Appendix No.	Clause No.						
	ELECTRICAL								
1	III / Specs		1.6.4	1584	As and when required by the Employers, the contractor shall arrange to test the materials and/or portions of works at his own cost to prove their soundness and efficiency.	Tests shall be conducted as per requirement mentioned in BOQ & specification. Any additional test shall be conducted at extra cost.	ok		
2	IV / BOQ	HT Panel Note-10			Panel should be minimum IP42 rated and to be equipped with fire suppression system	Please confirm the requirement of fire suppression system in HT panel as the same is not mentioned in BOQ.	Fire suppression to be considered for HT Panel		
3	IV / BOQ	HT Panel / TF / LT Panel Busduct			Factory Visit for Panels / TF/LT Panel / Busduct for witnessing performance and adherence to match design parameters or surpass them. 4 Persons to be considered for Inspection, Testing of Equipment. Travels & Stay shall be included in the quoted rates.	We request you to considered cost towards travel & stay in your scope.	Please adhere to BOQ description		
4	IV / BOQ III / Specs	HT & XMER Note-8	1.1.2-J-a	1589	BOQ: Panel to be provided with Aluminium Basbar (3 Pole) Specs: Main Bus: Copper for full length of switchgear.	We understand that as mentioned in BOQ, HT panels are with Aluminium busbar.	yes		
5	III / Specs					Please confirm about the requirement of earthing truck / earth switch in HT panel.	Provide Earth switch and it should be type tested design		
6	III / Specs		2.1.5-2.1.6.5, 2.1.6.16, 2.1.7.3-j	1591-1594, 1596, 1603	Blurred text	Texts under clause no. 2.1.5 to 2.1.6.5, 2.1.6.16 & 2.1.7.3-j are unclear. Please provide the clear texts.	will share again		
7	IV / BOQ	HT & XMER	Subhead-B		Note:- Contractor to include minor civil work like ..., excavation and back filling in ground for cable laying, ... and also fire proof sealing the extra wall/floor openings which are not already considered under the specific fire sealing item in this boq	i) We understand that excavation / back filling is not required for HT cable laying. ii) Please incorporate item of Fire Sealant in the BOQ	i) Excavation /back filling is not required for HT cable laying 2) Fire sealant will be part of separate tender	Refer the responsibility Matrix for the same where responsibilities are clearly defined	
8	IV / BOQ	HT & XMER	Subhead-C		2500kVA transformer to have only positive tolerance for impedance ie impedance will be strictly in the range of 6.25% to 6.87% .	Please accept impedance of 6.25% with IS tolerance	Please adhere to the boq description		
9	III / Specs	Transformer	2.1.6.11-i	1595	Temperature rise by no load plus short circuit method.	We understand that temperature rise test shall be conducted on one transformer	yes		
10	BOQ/Specs	Transformer				Please provide losses for transformer	they will be as per ECBC 2017		
11	IV / BOQ	HT & XMER	Subhead-D, item no. 4		Providing and laying of CPRI tested ,UL listed ,3m long 17.2mm dia Copper rod....	Please accept copper bonded steel rod in place of copper rod	Please adhere to boq description		
12	IV / BOQ	HT & XMER	Item no. 1.9		Charges for submission and obtaining approval before start of work. Survey & approval of Installation location. Inspection and approval of the work under scope of this tender (H.T. & L.T.) Co-ordination with Supply Company & Electrical Inspector for commissioning of the installation. Any other activity / approval related to the scope of this tender. Submission of required Test Reports to E. I. / Supply Authority. (This includes the statutory fees of liaison, incidental expenses and any other charges paid by the contractor).	i) We understand that scope of this item is CEIG approval of our installations. ii) The same item is captured in SH-M of TENANT BOQ with total 324 qty. Please clarify the requirement.	i) yes ii) The tenant meters installation have to be PWD approved		

Name of Contractor: Vendor 4							Date: 29.11.2024		
Sr. No. / Query No.	Bidding Document Reference(s) (Volume			Tender PDF File Page No.	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
	Volume No.	Main Section No. Sub-Section or Appendix No.	Clause No.						
13	IV / BOQ	TENANT BOQ	SH-G		ATS equivalent to 7000 Series of Asco...	We understand that ATS of other makes (ex, socomec) are also acceptable	Please adhere to the approved makes list		
14	IV / BOQ	Common Area BOQ / LT Panel	SH-N		Panels to be IEC 61439 compliant	We understand that LT panels are PTTA type. Please confirm	Please adhere to boq description		
15	II / SLD III / Specs	SLD Specs	Point no. 15 CI no. B, 2.2.9.1	478 1617, 1620	SLD: MCCB & ACBs status to be hard wired to BMS terminal block within the panel. All the meters shall have RS 485 modbus cimmunication port for BMS interface. Specs: The breaker should be equipped with micro-controller based, communicable type release with RS 485 port for communication...	We understand that as mentioned in the SLD switchgears shall not be provided with RS 485 communication port.	All incomers ,ACB and MCCB have to have RS485 except the tenant metering panels and control panels		
16	III / Specs	Specs	2.1.9.1.2, part-6	1621	All MCCB with microprocessor based release unit, the protection shall be adjustable Overload, Short circuit and instantaneous earth fault protection with time delay(Provide only where noted on SLD).	Please accept MCCBs up to 250A with thermal magnetic releases and above 250A with microprocessor based releases.	Please adhere to specification description		
17	IV / BOQ II / SLD	General Notes-B/17 Notes		478	BOQ:The Load Manager to be provided in the incomer of Main LT Panel . All the outgoing of Main LT panel shall have MFMM and other SMDB or Control Panel CP shall have Ammeter, voltmeter, selector switch and indication lamps etc. SLD: MFMM in incomer only	We understand that meters shall be provided as shown in SLD.	Please follow the SLD for meters		
18	II / SLD	Note 17		478	All panels to be provided with 20% spare capacity	LT panels are shown in SLD with spare feeders. Hence we shall not consider 20% additional capacity.	ok		
19	IV / BOQ	Common Area BOQ / SH-N	Item no. 1.3		Distribution board with IP-42 protection	Distribution board shall be with 18 gauge thickness	Please adhere to specification description		
20	III / Specs	BBT / Specs	2.3.5-F	1646	Ratings - Three phase with Neutral Bus Bars shall be designed to carry the rated full load current under normal operating site conditions without exceeding a 55 degree C rise above a 40 degree C ambient There shall be no derating of busbar upto 55 degC.	Our offered busduct shall comply to the requirement of "...without exceeding a 55 degree C rise above a 40 degree C ambient....". Hence the clause "There shall be no derating of busbar upto 55 degC" shall not be applicable.	Please adhere to the specifications		
21	IV / BOQ	Common Area BOQ / SH-P			Busduct	Please incorporate items for busduct accessories (bends, copper flexible, flange ends, etc.). If not, please provide busduct layout drawing to calculate the same.	Busduct layout is shared		
22	IV / BOQ III / Specs	Common Area BOQ / SH-Q Specs	Item no. 1,2,7 CI no. 2.4.3.1.1	1652	BOQ: Fire Retardant Low Smoke Haolgen (FRLSH), XLPE Insulated, Flat Steel Wire Armour, Aluminium conductor Cables Specs: Inner Sheathing : Flame Retardant, Type ST2, Low Smoke, Low Halogen (LSZH) PVC. Over Sheathing : LSZH PVC,TYPE ST2.	We understand that as per BOQ LT cables (for normal service) covered under item no. 1,2,7 (& elsewhere) are with FRLS in outersheath only.	Please consider normal cables as FRLSZH and not FRLS		

Name of Contractor: Vendor 4							Date: 29.11.2024		
Sr. No. / Query No.	Bidding Document Reference(s) (Volume			Tender PDF File Page No.	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
	Volume No.	Main Section No. Sub-Section or Appendix No.	Clause No.						
23	IV / BOQ	Tary BOQ / SH-E, K, R			Supply, installation, testing and commissioning of Seismic Supports for vertical risers, to take care of wind, building settlement and seismic effect. The contractor shall bring on board the seismic contractor & submit all the required calculations & methods. All supporting structure & hardware including nuts & bolts etc. shall be of hot dipped galvanized iron as approved. The cable trays support shall be fixed at every 1 meter spacing.	Please confirm the requirement of seismic supports in cable tray. If required, please incorporate separate BOQ items for the same.	Please adhere to the boq description		
24	IV / BOQ III / Specs	Tary BOQ / SH-E, K, R Specs	Cl no. 2.6.2-j	1666	BOQ: The cable trays support shall be fixed at every 1 meter spacing Specs: Support interval- up to 600 mm- 1.5 meter; above 600 mm - 1 meter	We understand that, as mentioned in specification, supports shall be required at every 1.5 meter for size up to 600 meter and at every 1 meter for above 600 mm trays.	Please adhere to the specifications		
25	IV / BOQ II / SLD	HT & XMER BOQ			1.4: HT-DG-TB-PWR-B01-01 AIR INSULATED (VCB) 11 kV SWITCHGEAR 1IC 2OG 1.5: HTP-TC-L00-01 AIR INSULATED (VCB) 11 kV SWITCHGEAR 2IC 3OG 1.7: HTP-TE-L01-01 AIR INSULATED (VCB) 11 kV SWITCHGEAR 1IC 3OG	SLDs for HT panels covered under BOQ item no. 1.4, 1.5, & 1.7 are not provided. Please provide the same.	Will be provided in addendum		
26	IV / BOQ II / SLD	Common Area BOQ			LT Panels				
		1.1			Main Distribution Boards (MDB/MLTP) (As per Technical Specification, Schedules & SLD)	1. SLDs of these LT panels are not provided. Please provide the same. 2. SLDs for tower B are not available. Please provide the same.			
					Tower B				
		1.1.9			MLTP-TB-PWR-L00-01 (FIRE SUPPRESSION SYSTEM) 50kA fault level				
		1.1.10			MLTP-TB-PWR-LS-L00-02 (FIRE SUPPRESSION SYSTEM) 50kA fault level				
		1.1.11			MLTP-TB-PWR-L02-01 (FIRE SUPPRESSION SYSTEM) 50kA fault level				
		1.1.12			MDB-TA-EVSE-B1-01 (FIRE SUPPRESSION SYSTEM) 50kA fault level				
					Tower E				
		1.1.25			BB-TE-PWR-L01-01 (FIRE SUPPRESSION SYSTEM) 65kA fault level				
		1.2.2			DG Backed SMDB (Essential Supply)				
					Tower A				
		1.2.2.8			SMDB-TA-VENT-B4-01				
		1.2.2.9			SMDB-FCU-TA-B1-01				
		1.2.2.10			SMDB-TA-VENT-B1-02				
					Tower B				
		1.2.2.12			SMDB-PL-TB-L00-01				
		1.2.2.13			SMDB-TB-LIFT-L00-01				
		1.2.2.14			SMDB-TB-LIFT-L00-02				
		1.2.2.15			SMDB-TB-PWR-L+P-L00-01				
		1.2.2.16			SMDB-TB-VENT-N-L00-01				
		1.2.2.17			SMDB-TB-EVSE-B1-01				
		1.2.2.18			SMDB-TB-PARK-B3-01				
		1.2.2.19			SMDB-TB-F&L-L00-01				
		1.2.2.20			SMDB-TB-ST-B4-01				

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		1.2.2.21			SMDB-TB-ST-B3-01				
		1.2.2.22			SMDB-TB-ST-B1-01				
		1.2.3			Life Safety SMDB (DG Backed) (FIRE SUPPRESSION SYSTEM)				
					Tower B				
		1.2.3.6			SMDB-TB-VENT-LS-L00-01				
		1.2.3.7			SMDB-TB-PWR-LS-B1-01				
27	II / SLD				Panels in SLD	These panels are shown in SLD but not available in BOQ. Please confirm the requirement.			
					MHTP-TE-HTDG-L02-01		Its in HT DG boq tab	Refer the BOQ shared recently	
					MHTP-TE-HTDG-L02-03				
					MHTP-TE-HTDG-L02-05				
					MHTP-TE-HTDG-L01-02				
					MHTP-TE-HTDG-L01-04				
28	IV / BOQ II / SLD	Common Area BOQ			Panels captured in both Electrical & HVAC BOQ				
		1.2.2.26			SMDB-TC-VENT-L48-01	These panels are captured in both Electrical & HVAC BOQ. Please advise where to quote.	These are part of HVAC boq,don't consider in Elec boq		
		1.2.2.28			SMDB-TC-VENT-L68-01				
		1.2.2.35			SMDB-TE-VENT-L01-01				
		1.2.2.36			SMDB-TE-WCH-B01-01				
		1.2.2.37			SMDB-TE-CT-L01-01				
		1.2.2.39			SMDB-TE-WCH-B01-02				
		1.2.2.40			SMDB-TE-CT-L01-02				
29	II/Drawings					Please provide drawings in AutoCAD format.			
30		Quantity mismatch				Some of quantities mentioned under "Total Qty Column" is not matching with total of individual quantity. Please provided corrected BOQ.			
31	III / Specs	HT Panel Makes		1362	Siemens ABB Schneider Lauritz Knudsen (earlier L&T)	Please accept HT panels from Authorized Channel Partners of approved OEMs	ok		
32	III / Specs	Cubicle LV Panels Main Distribution Boards (MDB), Sub-Main Distribution Boards (SMDB)		1362	Schneider ABB Siemens Lauritz Knudsen (earlier L&T)	We understand that Authorized Channel Partners of approved OEMs / CPRI approved panel builders shall be acceptable for LT Panels.	Authorized channel partner with the IEC 61439 and discrimination report of similar sclae project to be shared in Technical bid		
33		Sensor make			PIR Sensors	Please provide makes for PIR sensors	Please consider Occupancy sensor .Schneider/Honeywell/Legr and		

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34	III / Specs	Makes		1362 - 1368		Please accept following additional makes			
		11kV/22kV/33kV HV Cables			Polycab Finolex KEI RR Kabel	Avocab	Please adhere to Approved makes list		
		LT Cable / FS Cable			Finolex Polycab RR Kabel KEI	Avocab Bonton			
		Transformer (Dry Type)			Kirloskar C&G Voltamp RPG Raychem	Sudhir Ames Impex Rishabh			
		ELCB, MCB, RCCB & DBs			Schneider Electric ABB Legrand Lauritz Knudsen (earlier L&T)	Hager / Indo Asian			
		Capacitors with Harmonic Filters			Siemens- EPCOS ABB DUCATI Lauritz Knudsen (earlier L&T)	Neptune			
		Busducts			Siemens Zucchini- Legrand Schneider Electric Lauritz Knudsen (earlier L&T)	C&S Godrej Alfaduct			
		Cable Tays / Trunking			Profab Indiana Asian Globe	MEM BEC RM Con			
		Battery Charger			Hitachi Amara Raja Masstech Control	Mahamai Engineers Chhabi			
		UPS			Vertiv (Emerson) APC Toshiba Corporation Socomec Consul Neowatt	Numeric Eaton			
	ELV (FAPA, BMS)								
1	BOQ-FAPA	Fire Detection & Alarm System	A - Notes -(1)		All the components of the system and the complete system shall be UL listed and FM approved	As per the details mentioned in the notes section, all the components of Fire Alarm System shall be UL & FM approved. However, required approval for each item is also mentioned in the each individual item. We shall quote as per the approval mentioned in the individual item.	ok		
2	BOQ-FAPA	Fire Detection & Alarm System	A - Notes -(2)		The wiring core, size of conductor, type of wire shall be verified by the OEM suitable for loop length. The wire size shall not be less than 1.5sqmm for initiating devices and notification devices.	We shall consider size of conductor as per BOQ i.e. 1.5 sq. mm.	ok		

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3	BOQ-FAPA	Fire Detection & Alarm System	A1		Microprocessor based Analog Addressable MAIN network Fire Alarm Control panel with in-built battery back up for 24 hours under normal operation and 4 hours under alarm operation . The panel shall come with an integrated Fire Mans Telephone and Digital Voice Command Centre (DVCC) with touch screen display/QWERTY keypad for programming. The entire control panel, battery charger etc, shall be UL listed. The panel shall have 10 loop capacity, with Network card (RJ45) supporting BACnet protocol and Digital Alarm Communicator Transmitter (DACT) . Each loop should have 20% spare capacity . Main FAVA panel shal have built-in 2 way fire fighting communication system in built(microphone) Paging system. The panel shall have built-in ethernet/USB port , webserver & backnet/Modbus support for integration with BMS. Selection of Panel to be done based on length of Loop. Location: Fire Command Center at L00	1. We shall consider back up for 24 hours under normal operation and 30 mins under alarm operation as standard practice. 2. Type of networking required for panel is not mentioned in the BOQ. We understand networking shall be required between panel. We shall consider copper networking card for panel. Please accept. 3. We shall consider single BMS card for each tower. Please accept. 4. Please clarify the requirement for webserver.	1. ok 2. Suitable network card to be considered by checking the schematic and network requirement 3. Single BMS gateway is okay but it all shall cover all addressable points like detectors, modules and system events like fire and trouble 4. Webserver not required		
4	BOQ-FAPA	Fire Detection & Alarm System	A8		SITC of Addressable Multi Criteria (Smoke and Heat Detector) with surface & or recessed mounted standard base and fixing accessories. Listed spacing shall be as per OEM.	We shall consider detectors qty. as per BOQ. Any change in the quantity due to space modification shall be in client's scope.	ok		
5	BOQ-FAPA	Fire Detection & Alarm System	A12		SITC of Addressable Smoke Detector (void/above false ceiling) with standard base and fixing accessories & including Response Indicator. Listed spacing as per Tender Documents.	As per BOQ, response indicator has been asked with smoke detector. Since separate item for response indicator is already available in the BOQ (A14) . Request to measure the response indicator from the same item.	Please adhere to the boq		
		Fire Detection & Alarm System	A13		SITC of Addressable Smoke and Heat Detector (void/above false ceiling) with standard base and fixing accessories & including Response Indicator. Listed spacing as per Tender Documents.				
		Fire Detection & Alarm System	A14		SITC of Response Indicator (For closed Plant Rooms, Shafts)	We shall consider Agni/Ravel make response indicator. It shall be Non-UL listed. Please accept.	ok		
6	BOQ-FAPA	Fire Detection & Alarm System	A15		SITC of Beam Detector and receiver with standard base and fixing accessories .	We shall consider conventional beam detector along with module. Please accept.	Please consider UL FM approved		
7	BOQ-FAPA	Fire Detection & Alarm System	A16		SITC of Addressable Gas Detector with in built audible alarm and supervised power supply in built to be interface with fire alarm though monitoring module detector shall be complete with 2 relay outputs for interface including standard base and fixing accessories. It shall be UL listed.	1. As per BOQ, addressable Gas Detector has been asked. We understand it is LPG gas detector. Kindly confirm. 2. Make is not mentioned for Gas detector in the approved make list. We shall consider "Bosean" make. Please accept.	1. This is LPG gas detector for commercial kitchen application 2. Drager/ Msa/Detronics		
8	BOQ-FAPA	Fire Detection & Alarm System	A24		Supply,installation,testing and commissioning of Fire rated terminal box for termination of FS armoured cable and FRLS wire	As per BOQ, Fire rated terminal box has been asked. We understand it is for installation of detectors. We shall consider MS back box. Please accept.	Please consider GI back box and not MS ,refer standard details for same		

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9	BOQ-FAPA	Fire Detection & Alarm System	A26		Supply, Installation, Testing and commissioning of Fire alarm red revolving Beacon light with required accessories .Lamp should be LED type. Details of control panel and beacon Lamp and interface scheme with cabling details with Fire control module shall be submitted .equivalent to model no AVIL HD S	There are three makes available for Beacon in the approved make list. Cooper / Tomar Electronics / Fedral Signal Corporation Request to add " Kheraj" make for the beacon. Also, the model no mentioned in the BOQ description is of Kheraj Make.	ok		
10	BOQ-FAPA	Cables and Wires for Fire Alarm and Detection System	SI No. B1-i		2Cx1.5sqmm Fire Survival (FS) multi stranded, twisted pair copper cable.	Armoured/unarmoured is not mentioned in the given item description. Please clarify.	unarmoured		
11	BOQ-FAPA	Cables and Wires for Fire Alarm and Detection System	SI No. B2-i SI No. E 2 i		2Cx1.5sqmm FR-LSH multi stranded, twisted pair copper cable and FS cable for vertical exposed loop run	1. Armoured/unarmoured is not mentioned in the given item description. Please clarify. 2. FR-LSH and FS both types are mentioned in the description. We understand it is FR-LSH cable. Please confirm.	The detectors which are mounted on the true ceiling will be wired with FRLS wires pulled in pre liad conduits and the detectors which are on the false ceiling will be wired with FS cable		
12	BOQ-FAPA	Containment/Trunking/Conduits	SI No. C1		Supply & Laying of hot dipped, galvanized Iron, Perforated type, Cable trays having 1.6mm/2mm thickness & 50/75mm height as per specifications, of following sizes, suspended from ceiling or fixed vertically on wall / column including necessary supports like anchor bolts, hangers, brackets etc. and accessories like Elbows, Bends, Tee, requisite support/suspenders, hardware etc. as per site requirement. All supporting structure & hardware including nuts & bolts etc. shall be of galvanized iron as approved. The cable trays support shall be fixed at every 1 meter spacing. For Fabrication work in MS, the rate shall include the cost for painting with 3 Coats of enamelled / metal applied after rust treatment of two coats of red oxide. The colour of the paint & cable trays shall be as approved by Architect/Client. To be in	1. As per BOQ, thickness and height for cable tray is not clear. Request to share the same for all cable trays in the BOQ which has width from 50mm to 750 mm. 2. We shall consider MS angle support for horizontal support and threaded rod for vertical support. Support shall be provided at every 1.5 meter.	1. Its mentioned in boq 2. Please adhere to boq description only		
13	BOQ-FAPA	Containment/Trunking/Conduits	SI No. C2 (i)		25mm PVC HMS Conduit	We shall consider conduit having 1.6 mm thickness. Please accept.	Please adhere to boq description		
14	BOQ-FAPA	Containment/Trunking/Conduits	SI No. C2 (i)		32mm GI Conduit	Thickness is not mentioned for GI conduit. We shall consider 1.6 mm thickness for all sizes of GI conduit.	ok		
	BOQ-FAPA	Voice Annunciation System	D4		Supply, installing, testing and commissioning of 200W Power Amplifiers (PAmp) with required accessories. The units shall be UL listed and FM approved. To include for terminating speaker circuits into device and fixing. Provide two audio in channels and four outgoing channels for broadcasting messages. The units must allow for monioring. Refer Technical Specification Ref : 3.17.6 Power Amplifier (Power AMP)				

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15	BOQ-FAPA	Voice Annunciation System	D5		Supply, installing, testing and commissioning of 480W Power Amplifiers (PAmp) with required accessories. The units shall be UL listed and FM approved. To include for terminaitng speaker circuits into device and fixing. Provide two audio in channels and four outgoing channels for broadcasting messages. The units must allow for monioring. Refer Technical Specification Ref : 3.17.6 Power Amplifier (Power AMP)	As per BOQ, all the power amplifier has been asked with UL & FM approved. Since, power amplifiers are not available with UL and FM approved, we shall confirm the available approval for power amplifier with the OEM.	Please consider the Power amplifiers of wattage specified in boq .They should be at least CE approved		
	BOQ-FAPA	Voice Annunciation System	D6		Supply, installing, testing and commissioning of 600W Power Amplifiers (PAmp) with required accessories. The units shall be UL listed and FM approved. To include for terminaitng speaker circuits into device and fixing. Provide two audio in channels and four outgoing channels for broadcasting messages. The units must allow for monioring. Refer Technical Specification Ref : 3.17.6 Power Amplifier (Power AMP)				
	BOQ-FAPA	Voice Annunciation System	D7		Supply, installing, testing and commissioning of 800W Power Amplifiers (PAmp) with required accessories. The units shall be UL listed and FM approved. To include for terminaitng speaker circuits into device and fixing. Provide two audio in channels and four outgoing channels for broadcasting messages. The units must allow for monioring. Refer Technical Specification Ref : 3.17.6 Power Amplifier (Power AMP)				
16	2 watt speaker	Voice Annunciation System	D7		Supply, installing, testing and commissioning of 2 Watt type speakers (with suitable tappings as per OEM) with required accessories . The Units shall be UL listed and FM approved. To include for terminaitng circuit cabling into device and backbox.	As per BOQ, all the speakers has been asked with UL & FM approved. We shall consider 2 watt UL approved speaker. Please accept.	2W UL approved speakers are ok		
17	2 watt speaker	Voice Annunciation System	D8		Supply, installing, testing and commissioning of 2 Watt false celing mounted type speakers (with suitable tappings as per OEM) with required accessories . The Units shall be UL listed and FM approved. To include for terminaitng circuit cabling				
18	20W speaker	Voice Annunciation System	D 10		Supply, installing, testing and commissioning of 20W, Surface Mounted, horn speaker with required accessories. The Units shall be UL listed and FM approved. To include for terminaitng circuit cabling into device and backbox.	1. We shall consider CE complied speaker, since for 20 watt speaker, UL and FM approved speakers are not available with the OEM. 2. As per BOQ description of D11 item , , 20 Watt Horn with strobe has been asked. Wattage is not applicable for Horn. We	1. ok		
	20W speaker	Voice Annunciation System	D 11		Supply, installing, testing and commissioning of 20W, Surface Mounted, horn with Strobe & required accessories. The Units shall be UL listed and FM approved. To include for terminaitng circuit cabling into device and backbox.		Standard hooter strobe to be considered .Strobe shall have minimum 110cd output .It shall UL listed .The hooter output shall be 75db at 10 feet		

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	20W speaker	Voice Annunciation System	D 12		Supply, installing, testing and commissioning of 20W, Weatherproof, Surface mounted horn with Strobe & required accessories. The Units shall be UL listed and FM approved. To include for terminaitng circuit cabling into device and backbox.	wattage is not applicable for Horn. we understand speaker is being asked in this item. We shall consider 20 watt speaker for this item. Please accept.	Standard weatherproof hooter strobe to be considered .Strobe shall have minimum 110cd output .It shall UL listed .The hooter output shall be 75db at 10 feet		
19	BMS BOQ	BMS				IO Summary is not available in the tender document. Request to share the same.	IO summary is given in BMS tender		
20	BMS BOQ	CONTROLLERS & SYSTEM INTERFACES	SI No. 2.1	Page No. 3860	1. The above shall be housed in a vandal proof, lockable & secure MS powder coated Cabinets to be supplied along with the Controllers. 2. Must support BACnet communication protocol. 3. All panels shall be provided with a power failure/auto restart capability. 4. 10% spare capacity for each point type (i.e. AI, AO, DI, DO) should be allowed inside each DDC controller/panel rack for future expansion. The DDC should support hot swappable IO.	As per BOQ, 10 % spare points has been asked. However, 25 % spare points has been asked as per technical specifications . We shall consider spare points as per BOQ.	OK		
21	BMS BOQ	CABLES AND ACCESSORIES	SI NO. 4.01		Supplying, installing, testing and commissioning of 2 core x 1.0 mm ² twisted pair cable, with 1mm Annealed tinned copper as per specification. Cables as Tagged under "A" More than two cables should not be saddled to wall /ceiling. Perforated Cable tray /trunking of aproprate size should be used more than two cables to be layed thru ceiling /walls.	1. As per technical specifications, armoured cable has been asked. Nothing is mentioned in the BOQ with respect to armouring. We shall consider armoured cable as per technical specifications. 2. Required cable tray shall be measured from the respective line item of cable tray.	1.Follow BOQ 2.OK		
22	BMS BOQ	CABLES AND ACCESSORIES	SI NO. 4.02		Supplying, installing, testing and commissioning of 2 core x 1.5 mm ² twisted pair cable, with 1mm Annealed tinned copper as per specification, for BTU meter daisy chain wiring More than two cables should not be saddled to wall /ceiling. Perforated Cable tray /trunking of aproprate size should be used more than two cables to be layed thru ceiling /walls.	1. As per BOQ, 2 core x 1.5 sq. mm cable has been asked. However, as per technical specifications, 2 pair cable has been asked. We shall consider 2 core x 1.5 sq. mm cable as per BOQ. 2. As per technical specifications, armoured cable has been asked. Nothing is mentioned in the BOQ with respect to armouring. We shall consider armoured cable as per technical specifications. 3. Required cable tray shall be measured from the respective line item of cable tray.	1.Follow BOQ 2.OK 3.OK		
23	BMS BOQ	CABLES AND ACCESSORIES	SI NO. 4.03		Fibre Optic Cable The cable length as measured using the schematicBHE-XX-ZZ-DR-BM-7000 to 7003, is 2225metre. Vendor/OEM to confirm the length of the fibre optic cable as per equipment specific wiring. The wiring unit of measure mentioned over here is based on point wiring, considering the number of network controller	Please clarify the type of Fiber optic cable required.	BH to revert		

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24	BMS BOQ	CABLES AND ACCESSORIES	SI NO. 4.04		CAT-6 cable, tagged as "C" Supplying, installing, testing and commissioning of CAT-6 cable, Cables as Tagged under "C" More than two cables should not be saddled to wall /ceiling. Perforated Cable tray /trunking of aproprate size should be used more than two cables to be layed thru ceiling /walls. The cable length as measured using the schematic BHE-PL-ZZ-DR-BM-7000 to 7003, is 27560metre. Vendor/OEM to confirm the length of the fibre optic cable as per equipment specific wiring The wiring unit of measure mentioned over here is based on point wiring, considering the number of ASC (Application specific ctrlr) / DDC & Chiller plant optimiser	1. We shall consider flexible cat-6 cable. 2. Required cable tray shall be measured from the respective line item of cable tray.	1.BH to revert 2.OK		
25	BMS BOQ	CABLES AND ACCESSORIES	SI NO. 4.05		Supplying, installing, testing and commissioning of 2 mm thick GI (Hot deep galvanized) trunking with cover of the following sizes suspended from ceiling or fixed vertically on wall / column including necessary supports like anchor bolts, hangers, brackets etc. and accessories like Elbows, Bends, Tee, requisite support/suspenders, hardware etc. as per site requirement. All supporting structure shall be Hot dip galvanised (having galvanizing thickness 75 microns) modular support systems for hanging/supporting, suspending, floor mounting etc	We shall consider MS angle support for horizontal support and threaded rod for vertical support. Support shall be provided at every 1.5 meter.	Follow BOQ		
26	BMS BOQ	Sundries	SI NO. 6.09		One year comprehensive system operation & maintenance after successful handing over of the system	We understand only DLP shall be in our scope. Operation and maintenance shall not be in our scope. Please confirm.		Vendor to quote for Operations separately. However maintenance within DLP is in vendors scope.	
27	BMS BOQ	Sundries	SI NO. 6.13		Allow for providing all tools, spare parts and consumables .	Please confirm the requirement for spares.	Spare is required, follow BOQ		
28	BMS BOQ	Sundries	SI NO. 6.17		Contractor shall list here any other items not included above but necessary for the complete and satisfactory completion of each section of the work. Such items shall be deemed to be included	We understand our scope shall be limited to BOQ only. Any extra item required, shall be in client's scope.	Follow the line item, extra item have to be notified		
	HVAC								
1	BOQ_MEP_TOWER_ABCE	A. CHILLER PLANT HIGHSIDE EQUIP	Sr. No - 1	Chiller	BOQ calls for "Sound Pressure level for chillers shall not exceed 75 dBA at 1m distance from chiller."	Noise level shall be 85 dBA at 1.5 Mtr from Skid	Follow the BOQ and Specification		
2	BOQ_MEP_TOWER_ABCE	A. CHILLER PLANT HIGHSIDE EQUIP	Sr. No - 1	Chiller	BOQ calls for "Brush cleaning system on each chiller for tube cleaning and improved heat transfer."	We understand this indicates Auto Tube cleaning System.We request you to make line item for ATCS.	ATCS revised specification will be shared.		
3	BOQ_MEP_TOWER_ABCE	A. CHILLER PLANT HIGHSIDE EQUIP	Sr. No - 1	Chiller	BOQ calls for "Supply imported portion of above chillers CIF (Mumbai) Indicate for all chillers port charges, port handling charges, other local duties, loading, transportation/insurance upto site, octroi charges and all other taxes and incidentals complete till delivered at site." Customs duty and countervailing duty for the above imported equipment.	Applicable as per Chiller Manufacturer as chiller can b sourced from India	Applicable for imported chillers only		

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4		A. CHILLER PLANT HIGHSIDE EQUIP		Chiller		1. The access road should be made available by client till tower / store / our working space 2. Foundation near tower should be able to withstand load of crane	Client/PMC to confirm.	Refer the responsibility Matrix for the same where responsibilities are clearly defined. Discuss during technical round meeting no. 01 as well	
5	BOQ_MEP_T OWER_ABCE	A. CHILLER PLANT HIGHSIDE EQUIP	Sr. No - 2	CPM	Sensor and control wirings not BOQ item	We request you to make sensor and control wiring as a separate item for CPO	Please refer, H.FIELD DEVICES BOQ		
6	BOQ_MEP_T OWER_ABCE	A. CHILLER PLANT HIGHSIDE EQUIP	Sr. No - 3	Cooling Tower	BOQ calls for "INDUCED DRAFT CROSS FLOW"	We request you to accept Cross/Counter flow type as per OEM Standard.	Follow the BOQ		
7	BOQ_MEP_T OWER_ABCE	A. CHILLER PLANT HIGHSIDE EQUIP	Sr. No - 3	Cooling Tower	BOQ calls for "The cooling tower shall be quiet operating, and shall produce an overall sound no higher than 80 dB at a distance of 1.5 m."	Noise level shall be 80 dB at a distance of 3 MTR	Follow the BOQ		
8	BOQ_MEP_T OWER_ABCE	A. CHILLER PLANT HIGHSIDE EQUIP A1. CHW PUMPS & PHEX	Sr. No - 4	Pumps,TF A,AHU	Inspection and Factory Test Prior to Shipment, Pumps shall be subjected to witness Factory Test by owners representative (4 Persons) to ascertain the design requirements	1. We shall consider testing of one pump and one AHU 2. Please confirm whether cost of travel, food and lodging of client representative is in our scope or client will bear for pump , AHU, TFA etc.	1) OK 2) Factory test will be in vendor's scope		
9	BOQ_MEP_T OWER_ABCE	A. CHILLER PLANT HIGHSIDE EQUIP A1. CHW PUMPS & PHEX	Sr. No - 3	Tank	BOQ calls for "Expansion Tank shall have third party certificate (TUV / lloyds registered).Vessel 100-1000 liters :in accordance with EN13831/ 1200-8000 liters : in accordance with AD2000."	Expansion tank shall be as per IS standard and construction feature shall be OEM standard.	Follow the BOQ and Specification		
10	BOQ_MEP_T OWER_ABCE	B.TFA C.AHU	Sr. No - 1,2 Sr. No. - 1	TREATED FRESH AIR UNITS/AH U	BOQ Calls for "TREATED FRESH AIR UNIT WITH Heat Recovery Wheel (HRW) with 50mm thick double skin construction"	We request you to accept 43 mm PUF panel	Follow the BOQ		
11	BOQ_MEP_T OWER_ABCE	D.FCUI	Sr. No - 1	FCU	BOQ calls for "Fan coil units shall be selected for the lowest operating noise level having standard sound level rating of NC 30 at low speed and NC 35 at medium/high speed."	Noise level shall be as per OEM Standard	Follow the BOQ		
12	BOQ_MEP_T OWER_ABCE	D.FCUI	Sr. No - 1	FCU	BOQ calls for "PN 16 and PN 20 rated FCUI"	We understand that these PN ratings are for valve package	This is for complete AHU package.		
13	BOQ_MEP_T OWER_ABCE	D.FCUI	Sr. No - 1	FCU	BOQ calls for "Fan Coil units to be selected at Medium Speed."	FCU can be selected at high speed	Follow the BOQ		
14	BOQ_MEP_T OWER_ABCE	E.BTU METER & THERMOSTAT	Sr. No - 2	BTU	BOQ calls for "The Cooling/BTU meter shall be MID approved and comply with EN 1434 standard. "	MID is not applicable for BTU so request you to accept as per OEM Standard	Follow the BOQ		
15	BOQ_MEP_T OWER_ABCE	F.PIPING	Sr. No - 1,2,3	PIPING	BOQ calls for "All hangers, clamps, brackets etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns"	We request you to accept site fabricated MS painted support	Follow the BOQ		

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16	BOQ_MEP_T OWER_ABCE	F.PIPING	Sr. No - 1,2,3	PIPING	BOQ calls for "Supplying, installing, testing and commissioning of Seismic Supports for vertical riser piping to take care of wind , building settlement and seismic effect (Refer schematic for seismic support detail) The contractor shall bring on board the seismic contractor & submit all the required calculations & methods.)	We request you to make seismic support as a separate BOQ item	Refer the schematic to identify the riser qty.		
17	BOQ_MEP_T OWER_ABCE	F.PIPING	Sr. No - 1,2,3	PIPING	BOQ calls for " fittings like bends, elbows, tees, reducers, mating flanges & sockets for BAS, fabricated to size and shape in workshop or brought out ready made where available of acceptable quality."	Fitting for size 200 mm above shall be fabricated at site from same dia pipe.	Follow the BOQ		
18	BOQ_MEP_T OWER_ABCE	F.PIPING	Sr. No - 4	CONDENS ATE DRAIN PIPING	BOQ calls for "Providing and fixing in position the following UPVC Class-3 (IS 4985)"	UPVC shall be as per Such 40	Follow the BOQ and Specification		
19	BOQ_MEP_T OWER_ABCE	G.VALVES	Sr. No - 6	Cast Iron Non- Return valve	PN rating is not given in BOQ or Specs	We understand Non - Return Valve are PN 16 Ratings	PN rating shall be as per pump head, Minimum PN 16		
20	BOQ_MEP_T OWER_ABCE	H.FIELD DEVICES	Sr. No - 1 -1.2	Auto Air vent	BOQ calls for "Pipes greater than 100mm diameter – kinetic air release valve as per IS-14845"	We have considered 25mm diameter automatic air vent.	Follow the BOQ, 25 dia is acceptable.		
21	BOQ_MEP_T OWER_ABCE	J.FANS	Sr. No - 1,2	Axial Fans	The fan shall be equipped with mounting bracket inlet cone & outlet cone required at discharge.	We request you to inlet cone / outlet cone under duct as a sqmt item.	Included in ducting quantity.		
22	BOQ_MEP_T OWER_ABCE	J.FANS	Sr. No - 1,2,3,6,7,8	Axial/Cabin et/Inline Fans/Scrub ber Units	BOQ calls for "All hangers, clamps, brackets, anchor/fixed support, guides, rollers etc. shall be shall be modular hot dip galvanized having galvanizing thickness 75 microns "	We request you to accept site fabricated MS painted support	Follow the BOQ and Specification		
23	BOQ_MEP_T OWER_ABCE	J.FANS	Sr. No - 8	Scrubber Unit	BOQ calls for "Supply, installation, testing and commissioning of Kitchen Exhaust air fan with Scrubber Unit (Electro static precipitator & Activated carbon filter bank units or Electro static precipitator unit only)"	Please confirm whether we have to consider Activated Carbon filter.	Activated carbon filter to be installed after scrubber.		
24	BOQ_MEP_T OWER_ABCE	K.DUCTING & ACCESSORIES	Sr. No - 1 to 4	Duct	BOQ calls for "All hangers, clamps, brackets, anchor/fixed support, guides, rollers etc. shall be shall be modular hot dip galvanized having galvanizing thickness 75 microns"	We request you to consider wire rope hanger for duct	Follow the BOQ and Specification		
25	BOQ_MEP_T OWER_ABCE	K.DUCTING & ACCESSORIES	Sr. No - 1 to 4	Duct	iii) Closed cell neoprene gasket. Iv) Duct sleeve to be with fire sealant."	1. Gasket shall be Neoprene equivalent 2. We request you to make fire sealant as a separate item	Follow the BOQ and Specification		
26	BOQ_MEP_T OWER_ABCE	J.FANS	Sr. No - 3	Jet Fans	Fans to be AMCA air & sound certified.	Jet fans shall not be AMCA certified	Jet fan shall be AMCA certified.		

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27	BOQ_MEP_T OWER_ABCE	K.DUCTING & ACCESSORIES	Sr. No - 5	Curtain Type Fire damper	BOQ calls for " Curtain Type Fire damper"	We have considered blade type fire damper and fire damper shall be as UL 555 only	Follow the BOQ and Specification, Fire damper shall be as UL 555.		
28	BOQ_MEP_T OWER_ABCE	K.DUCTING & ACCESSORIES	Sr. No - 7.01	Actuators	BOQ calls for "Supply, installation, testing and commissioning of Actuators for the above motorised damper suitable for 24V, complete with inter connecting wiring at locations as shown in drawings and as per specifications."	We request you to pay connecting wiring under electrical works	Follow the BOQ		
29	BOQ_MEP_T OWER_ABCE	K.DUCTING & ACCESSORIES	Sr. No - 19	Sound Attenuato rs		Mode of measurement of sound attenuator shall be based on surface area like duct	Yes		
30	BOQ_MEP_T OWER_ABCE	I.VRV	Sr. No - 1	VRV	BOQ calls for "with blygold coating on coils,starters, gauges, necessary valves and accessories"	Coating shall be as per OEM standard	Follow the BOQ		
31	BOQ_MEP_T OWER_ABCE	I.VRV	Sr. No - 1	Outdoor VRV Condensin g Units including ODU refnets.		We understand the VRV outdoor unit shall be Cooling only type	Yes		
32	TPM - Tower ABCE - MEP Works Tender Docuement 20241122	Tube Axial Flow Fan	Page No - 1168 of 4237	Axial Fans	Motors shall be quiet operation and motor speed shall not exceed 1440 rpm.	Motor RPM shall be as per OEM selections	Follow the BOQ and Specification		
33	TPM - Tower ABCE - MEP Works Tender Docuement 20241122	PUMPS – General criteria for all types of pumps	Page No - 1119 of 4237	Pumps	Driver motor shall be rated at least 20% in excess (1.2 times) of maximum BkW of the pump.	The motor shall be rated from 10% to 20%	Follow the BOQ. at least 20%		
34	TPM - Tower ABCE - MEP Works Tender Docuement 20241122	PUMPS – General criteria for all types of pumps	Page No - 1119 of 4237	Pumps	Sound pressure Level shall not exceed 75 dBA @ 1.5 m from source.	Sound level shall be 75 dBA @ 3 m from source	Follow the BOQ and Specification		
35	TPM - Tower ABCE - MEP Works Tender Docuement 20241122	Double Skin Air Handling Units (AHU)	Page No - 1146 of 4237	AHU	Outer sheet of panels shall be made of galvanized pre- plasticized sheet of 0.8mm – 1.00mm thick	Outer sheet of panels shall be made of galvanized pre-coated sheet of 0.8mm thick	Follow the BOQ and Specification		

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36	TPM - Tower ABCE - MEP Works Tender Docuement 20241122	Treated Fresh Air Unit (TFA)	Page No - 1166 of 4237	TFA	The panels shall be of double skin construction with both inner and outer steel sheets being minimum 1.0mm thick pre coated & plasticized.	Outer & Inner sheet of panels shall be made of galvanized pre-coated sheet of 1.0 mm thick	Follow the BOQ and Specification		
37	TPM - Tower ABCE - MEP Works Tender Docuement 20241122	Double Skin Air Handling Units (AHU)	Page No - 1147 of 4237	Cooling Coil	Coils shall have anticorrosive coating equivalent to blygold.	Coils shall have coating as per OEM Standard	Follow the BOQ and Specification		
38	TPM - Tower ABCE - MEP Works Tender Docuement 20241122	UV-C Lamps for Heat recovery / treated fresh air units	Page No - 1148 of 4237		Emitter life should not be less than 20,000 hours of continuous operation.	UVC emitter life shall be 8000 Hours	Follow the BOQ and Specification		
39	TPM - Tower ABCE - MEP Works Tender Docuement 20241122	UV-C Lamps for Heat recovery / treated fresh air units	Page No - 1149 of 4237		The air purifier equipment shall Conform to UL935, UL542, UL635, UL817, CSA, CE, EU and TUV certified.	Codes shall be as per OEM	Follow the BOQ and Specification		
40	BOQ_MEP_T OWER_ABCE TPM - Tower ABCE - MEP Works Tender Docuement 20241122	B.TFA & C.AHU Air Handling units with EC fans	Page No - 1152 of 4237	AHU/TFA	BOQ calls for "The motor selected shall be energy efficient having IE-3 efficiency rating at full load with motor rating as per specification." however Specs calls "EC motor with at least IE5 efficiency and maintenance-free ball bearings that are permanently lubricated"	Please let us confirm efficiency of Motor to be consider	Follow the BOQ and Specification & kindly recheck the BOQ.		
41	Make Lists - HVAC	Make Lists - HVAC				We request you to accept following additional list of make :- 1)Insulated Valve Kit for FCU - Flowcon/Seimens/ATS 2) Butterfly Valve/Balancing Valve/NRV/Motorized Butterfly Valve - Advance 3) BTU Meter - Adept 4)Duct - Waves/Ductofab	Please follow the tender make list		
	PLUMBING								

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1	BOQ_MEP_TC	A.Sanitary Fittings &	A		SANITARY FITTINGS & ACCESSORIES (ONLY INSTALLATION, TESTING & COMMISSIONING)	We assume that Sanitary & CP Fittings including consumables will be made available to us. Also unloading, collecting, storing, shifting to respective floors shall be under client scope of work.	Only supply is in client's scope, unloading, storing, shifting to respective floors will be under plumbing contractor's scope.		
2	BOQ_MEP_TC	B. Water Supply	B.1		Domestic & Flushing Water Supply Inside Toilets CPVC PIPES	We request you to please consider chasing, concealing & cementing of concealed pipes under civil scope of work.	Follow the BOQ and Specification, It could be under respective contractor's scope.		
3	BOQ_MEP_TC	B. Water Supply	B.2		Domestic Water, Flushing Water piping, valves and accessories before entering Toilets	We request you to please consider chasing, concealing & cementing of concealed pipes under civil scope of work.	Follow the BOQ and Specification, It could be under respective contractor's scope.		
4	BOQ_MEP_TC	B.3 Water Supply Valves & Accessories	6		Liasoning Work for water mains connection	We will submit all necessary drawing and documents. Official fees will be bear by the Client	In contractor's scope, official fees will be paid by client		
5	BOQ_MEP_TC	C DRAINAGE	C.1, C.2, C.4		Drainage Piping Inside Office and Toilet (Single Stack System)	BOQ State PE 63 PN6 whereas Technical Specs (Pg 36) says PN10. Please clarify the pressure rating	Follow the Specification		
6	BOQ_MEP_TC	C DRAINAGE	C.4		Liasoning Work for Sewage	We will submit all necessary drawing and documents. Official fees will be bear by the Client	In contractor's scope, official fees will be paid by client		
7	BOQ_MEP_TC	D CAR PARK DRAINAGE & EXTERNAL STORM WATER DRAINAGE	3		Liasoning Work for Storm water drain	We will submit all necessary drawing and documents. Official fees will be bear by the Client	In contractor's scope, official fees will be paid by client		
8	BOQ_MEP_TC	E PUMP ROOM EQU	7		Double flanged flexicon rubber connector	Flexicon rubber connector whether single arch or double arch please confirm the same	Double arch type		
9	BOQ_MEP_TC	G WATER TREATMENT PLANT			WATER TREATMENT PLANT	Please share water analysis report			
10	TPM - Tower	LIST OF APPROVED	G		TRANSFER, SUMP PUMP	Please also consider Xlyelm, Wilo, HBD Pumps	Please follow the tender make list		
11	TPM - Tower	LIST OF APPROVED	G		FIRE SEALANT	Please also consider Promat	Please follow the tender make list		
12	TPM - Tower ABCE - MEP Works Tender Document 20241122	SEISMIC CONSIDERATION AND SEISMIC SUPPORTS	1.19		SEISMIC SUPPORT	Please provide separate line item for Seismic & Modular Supports	Follow the BOQ and Specification		
13	TPM - Tower ABCE - MEP Works Tender Document 20241122	FIRESTOPPING AND SMOKE SEALS	1		FIRESTOPPING AND SMOKE SEALS	Please provide separate line item for Fire Stop Material (Fire Sealant)	Follow the BOQ		

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14	BOQ_MEP_TC	G. Water Treatment	G-Boq WTP		Raw water Feed Pump to the WTP Plant 21.5 m3/hr @30 m Head Quantity- 2nos	Capacity of Pump ,Head and quantity need to require	Follow the BOQ & equipment schedule.		
			Boq WTP		All Battery Limits inlet to outlet of WTP	Need to add	Follow the specification.		
			Boq WTP		Liaisoning work	It should be in the scope of clients /Contractor not in WTP vendor	Client /PMC to be confirmed.	Refer the responsibility Matrix for the same where responsibilities are clearly defined	
			Boq WTP		Total Area available for WTP plant	Need to add Total Area available for WTP	Follow the drawings		
			Boq WTP		Inlet parameter	Inlet parameter dissolved iron content is mentioned. You had asked for outlet guarantee of Nil. In the offered scheme of DMF-ACF-Softener, Iron removal can not guaranteed.	Follow the specification.		
			Boq WTP		FRP vessel	FRP vessels are available upto 1200 mm dia only. You had asked for 1500 mm dia. You will have to go for MS vessel for the same	Follow the BOQ		
			Boq WTP		Softener MOC	Please confirm softener MOC	BH will revert.		
			Boq WTP		Sr no 5 is Dosing system	Sr no 5.1 is piping and Sr no 5.2 is control panel. Please confirm if Sr no 5.1 and 5.2 are for entire system of only for dosing system (Sr no 2 to 4)?	For Dosing system.		
			Boq WTP		Feed pumps	Feed pumps are not mentioned in below BOQ. Hope you are ordering them. So please arrange for panel of pump from your side.	Vendor have to propose		
			Boq WTP		Sr no 2, 3, 4 is system	for 21.5 m3/hr flow. Same is repeated for Sr no 6, 7, 8. Is it additional requirement or its by mistake?	Vendor have to propose		
			Boq WTP		Sr no 5, 5.1 and 5.2	are same as 9, 9.1, 9.2 Please check. Is it additional requirement or its by mistake?	Vendor have to propose		
			Boq WTP		Hardness	Please confirm hardness for Sr no 10.			
			Boq WTP		Inlet parameter	1. Inlet water has Iron content but no Iron Removal filter mentioned in BOQ. Also Inlet Iron content need to be specified exactly , Iron content can be up to 3ppm at inlet and at outlet <0.3 PPM	Follow the specification.		
			Boq WTP		FRP Pressure sand filter (Multi Grade filter) -Max flow rate = 21.5 cum/hr,at 10 hrs cycle	2. You have mentioned 21.5 m3/hr capacity DMF,ACF,Softner , And then you have asked for 21.5m3/hr capacity DMF,ACF, Softner again , is plant capacity same for both plant with different operating hrs?	BH will revert.		

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			Boq WTP -10		Softener (suitable for 3.5 Kg/Sqcm working pressure) CT makeup water treatment plant of 56.8 m³/hr Max flow rate = 56.8 cum/hr,at 10 hrs cycle	You have asked for 56.8m3/hr softner for 10hr cycle, Please mention hardness for this.	Follow the specification.		
			Boq WTP		Outlet water paramater Confirm the Total Hardness 5 or 50 PPM	Need to confirm Outlet water Total Hardness	Follow the specification.		
			Boq WTP		Confirm the MOC Of Vessel FRP/MSEP	If MOC of Vessel is MSEP then thickness of Shell and disc need to require	Follow the specification.		
			Boq WTP		Confirm the Warranty /DLP period of material	Need to mention material Warranty /DLP period of material	BH will revert.		
					The earthing probes shall be connected and wired to the building earth systems of the building. The level switch set shall operate with a stepped down voltage at 24V maximum. Stepped down transformers shall be provided for each set of control probes and shall be installed inside centralized control cubicles inside pump room.	It should be in the scope of Electrical vendor	Follow the specification.		
					Core cut - MEP services	pls keep the same is in civil scope or client.	Client/PMC to confirm	Refer the responsibility Matrix for the same where responsibilities are clearly defined	
			Pumps			VFD is for each pump or set.	Follow the BOQ & equipment schedule.		
1	IV-BILL OF QUANTITY	SECTION A - SPRINKLER SYSTEM	A-1 For Horizontal piping	73 of 101	Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239/IS-3589 (heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall /ceiling with clamps, hangers (using anchor fasteners) as per specification.(For horizontal piping - Grooved flexible & rigid coupling type shall be used for all pipe sizes.) Fittings to be UL & FM Approved.	1-Tender calls for both welded & grooved Fittings for Piping , Kindly confirm which type of pipe fittings to be considered. 2- As per IS: 1239, the maximum available thickness for a 150 mm GI/MS pipe is 5.4 mm. Please verify the BOQ and confirm the required thickness.	All grooved type fitting to be considered		

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2	IV-BILL OF QUANTITY	SECTION A - SPRINKLER SYSTEM	A-2 For Risers piping	73 of 101	Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239/ IS-3589(heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) inside the risers (grooved couplings shall be used along with grooved fittings on the pipes) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall /ceiling with clamps, hangers (using anchor fasteners) as per specification. For pressure rating refer specification and schematics. Grooved flexible & rigid coupling type shall be used for all pipe sizes.Fittings to be UL & FM Approved. Supplying, installing, testing and commissioning of Seismic Supports for vertical riser piping to take care of wind , building settlement and seismic effect (Refer schematic for seismic support detail) The contractor shall bring on board the seismic contractor & submit all the required calculations for consultant approval.	1-Tender calls for both welded & grooved Fittings for Piping , Kindly confirm which type of pipe fittings to be considered. 2-Tender calls for Seismic support to be considered for verticals pipe, what about for horizontal pipes, kindly confirm.	All grooved type fitting to be considered		
3	IV-BILL OF QUANTITY	SECTION A - SPRINKLER SYSTEM	A-3 For Drainage		Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239/IS-3589 (heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall /ceiling with clamps, hangers (using anchor fasteners) as per specification.(For horizontal piping - Grooved flexible & rigid coupling type shall be used for all pipe sizes.) Fittings to be UL & FM Approved.	Tender calls for both welded & grooved Fittings for Piping , Kindly confirm which type of pipe fittings to be considered.	All grooved type fitting to be considered		

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4	IV-BILL OF QUANTITY	SECTION A - SPRINKLER SYSTEM	A-4 Sprinkler		Supplying, installing, testing and commissioning of 15mm CP Brass quartzoid bulb type screw end connected Sprinkler Head, UL & FM Approved with approved chrome finish including joints to steel pipe work. The sprinkler shall be either pendent or upright type/side wall as specified on drawings and shall not be universal style. All concealed pendent sprinkler will be provided with rosette plate. Sprinkler shall be quick response type for all areas except car park and mechanical areas (i.e. Refer technical specs for sprinkler discharge coefficient table. (Sprinkler K-Value = K-80 Contractor shall supply 2% spare sprinkler heads and one spanner neatly installed in a wooden cabinet with glass shutters as approved by the Employer. This is to be included in the cost. Fittings to be UL & FM Approved. Technical Specification Ref : 3.4 Sprinkler Heads	Please note that Tender calls for all type of sprinklers required are UL listed & FM approved & makes for the same are mentioned in tender document as Tyco, Newge, HD & Safex, However Newage, HD & Safex does not have FM approval for all type of sprinklers, they are having only UL listing. In this connection can we go for only UL listed sprinklers.	Follow the BOQ and Specification		
5	IV-BILL OF QUANTITY	SECTION A - SPRINKLER SYSTEM	A-4 Sprinkler		Quick Response type sprinkler (Light hazard NFPA-13-2019 Clause 9.4.3.1)	Quantity of sprinklers should be in round not in digit	Follow the BOQ and Specification		
6	IV-BILL OF QUANTITY	SECTION A - SPRINKLER SYSTEM	A-6 ZCV		Zone Control Valve Supplying, installing, testing and commissioning of Zone Control Valve consisting of Butterfly valve with flange joint. Butterfly valves shall be slim seal, short wafer type with standard finish. Valves shall be suitable for mounting between flanges drilled to ANSI 125, Vane type flow switch, Supervisory switch / tamper switch. Accessories, complete with tap off socket arrangement as required Potential free contact with 2 Nos. NONC contacts all complete strictly as per drawing and schematics. Tamper switch quantity is equal to no. of ZCV. (Control cabling shall be considered in Electrical section) Pressure Rating - PN 16. Installation to be inline with technical specifications. Color No. 536 of IS:5 for valves. Technical Specification Ref : 3.11 Zone Control Valve & Inspection and Test Valve Assembly	Please note that under this item we are considering price of : 1-Flow Switch 2- Butterfly valve with Tamper switch 3-Pressure gauge (MS body 50 mm dial size)	Follow schematic & standard details		

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7	IV-BILL OF QUANTITY	SECTION A - SPRINKLER SYSTEM	A-10 Flow Switch		Water Flow switch (Non ZCV) Supplying, installing, testing and commissioning of electrically operated Water flow switches (Vane type) and accessories, complete with tap off socket arrangement as required, with potential free contact with 2 Nos. Water flow switches - UL /FM.Cabling till the Monitor module shall be in the scope of the fire fighting contractor. Cable to be quoted separately as mentioned in the electrical cable items. Technical Specification Ref : 3.11 Zone Control Valve & Inspection and Test Valve Assembly	Kindly clarify the scope of cabling for flow switch	It will be in fire contractor's scope.		
8	IV-BILL OF QUANTITY	SECTION B - Fire hydrant (Internal) & Wet Riser system	B-1 Hydrant Pipeing		Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239/IS-3589 (heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall /ceiling with clamps, hangers (using anchor fasteners) as per specification.(For horizontal piping - Grooved flexible & rigid coupling type shall be used for all pipe sizes.) Fittings to be UL & FM Approved.	Tender calls for both welded & grooved Fittings for Piping , Kindly confirm which type of pipe fittings to be considered.	All grooved type fitting to be considered		
9	IV-BILL OF QUANTITY	SECTION B - Fire hydrant (Internal) & Wet Riser system	B-1 Hydrant Riser Pipeing		Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239/ IS-3589(heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) inside the risers (grooved couplings shall be used along with grooved fittings on the pipes) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall /ceiling with clamps, hangers (using anchor fasteners) as per specification. For pressure rating refer specification and schematics. Grooved flexible & rigid coupling type shall be used for all pipe sizes.Fittings to be UL & FM Approved. Supplying, installing, testing and commissioning of Seismic Supports for vertical riser piping to take care of wind , building settlement and seismic effect (Refer schematic for seismic support detail) The contractor shall bring on board the seismic contractor & submit all the required calculations for consultant approval. Technical Specification Ref : 2.1 Pipes & Fittings	1-Tender calls for both welded & grooved Fittings for Piping , Kindly confirm which type of pipe fittings to be considered. 2-Tender calls for Seismic support to be considered for verticals pipe, what about for horizontal pipes, kindly confirm.	All grooved type fitting to be considered		

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	Volume No.	Main Section No. Sub-Section or Appendix No.	Clause No.						
10	IV-BILL OF QUANTITY	SECTION B - Fire hydrant (Internal) & Wet Riser system	B-4 Hose Cabinet/ Door		Supplying, installing, testing and commissioning of indoor type Lockable Fire Hose Cabinet (Concealed Type), The entire cabinet will be masonry construction except for the door which shall be two door and fabricated from 16 gauge MS sheet with half front glass door and locking arrangement, suitable to accommodate the below listed fittings/accessories/equipment's, Technical Specification Ref : 2.9 Fire Hose Cabinets (FHC) Concealed type	1-Approved venders like Tyco & HD Fire are not manufacturing Hose cabinet & Doors, Kindly provide some alternate makes like Arrow , Seerat etc. 2-Kindly provide size of Fire Door as same is not given in BOQ as well as in technical specification.	Follow the tender makes.		
11	IV-BILL OF QUANTITY	SECTION B - Fire hydrant (Internal) & Wet Riser system	B-4.01 Hydrant Valve		UL Listed Landing valves S.S-304 fire hydrant Double head landing valve with 100 mm N.B. flanged inlet, brass spindle controlled 63 mm dia female instantaneous outlet type. SS coupling, blank cap, chain, twist release type lug & all accessories Conforming to IS:5290-1993 including fixing and flanged tapping from wet riser. Technical Specification Ref : 2.5 Internal Hydrants	1-BOQ calls for UL listed hydrant valve however Technical specification calls for as per IS Standard, Please clarify. 2-Makes given for Hydrant valves are Tyco, HD fire , Newage & Safex , out of those makes only Newage & safex are manufacturing Hydrant valves. Newage has IS standard & UL listed hydrant valves & safex having only IS standard Hydrant valves . In this regard kindly clarify whether UL listed Hydrant or IS Standard Hydrant valves to be quoted. 3- Kindly provided some more makes.	1. Follow the BOQ 2. Follow the tender makes.		
12	IV-BILL OF QUANTITY	SECTION B - Fire hydrant (Internal) & Wet Riser system	B-4.21 Hose Pipe		UL Listed Non-percolating,(as Shown in Standard Installation drawing, BH-XX-XX-DR-xxxx) rubber reinforced lined canvas Fire Hose Pipe (as per IS : 636-1988) of 63 mm dia. The hose shall be rated for burst pressure of 37.5 Kg/sqcm. Hose shall be complete with ISI marked Stainless Steel male & female coupling (IS:903-1993) bound & riveted to hose pipe with copper rivets & 1.5 mm copper wire. (Type-B). Scope includes necessary supporting, bracket arrangement to hang the hose pipe inside FHC/Fire Shaft. 15 m length X 2 Nos - 1 set Technical Specification Ref : 2.10 Fire Hose Pipe (FHP)	1-BOQ calls for UL listed Hose Pipe however Technical specification calls for as per IS Standard, Please clarify. 2-Makes given for Hose pipe are Tyco, HD fire , Newage & Safex , out of those makes only Newage & safex are manufacturing Hose Pipe. Newage has IS standard & UL listed & safex having only IS standard Hose Pipe . In this regard kindly clarify whether UL listed Hydrant or IS Standard Hose Pipe to be quoted. 3- Kindly provided some more makes.	1. Follow the BOQ 2. Follow the tender makes.		

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13	IV-BILL OF QUANTITY	SECTION B - Fire hydrant (Internal) & Wet Riser system	B-1 Butterfly Valve		Butterfly Valve (Hydrant system-Non Plant room area) Providing ,fixing, testing and commissioning of Cast Iron Butterfly valve, (Body :Cast Iron, Shaft : SS, Disc : C.I. Liner : HT - EPDM). Including rubber gasket, flanges, nuts, bolts, washers & painting with necessary mating flanges to suit the specified piping material and complete as required and In accordance to the technical specification. Pressure rating shall be as per schematic and specification. Color No. 536 of IS:5 for valves Technical Specification Ref : 2.3.3 Butterfly Valves	Kindly specify the pressure rating for butterfly valve	PN rating shall be as per Pump haed.		
14	IV-BILL OF QUANTITY	SECTION C - DRENCHER SYSTEM FOR BASEMENT CAR PARK & FAÇADE PROTECTION IN OFFICE FLOORS	C-1 Deluge Valve		Supplying, installing, testing and commissioning of 180° pattern type UL Approved Cast iron Deluge Valve assembly consisting of all related required controls ,trim assembly necessary by-pass piping.	Tender calls for CI body Deluge valve however most of the approved venders provides DI body Deluge valve, can we go ahead with the same, Please confirm	Follow the BOQ,specification & tender makes.		
15	IV-BILL OF QUANTITY	SECTION C - DRENCHER SYSTEM FOR BASEMENT CAR PARK & FAÇADE PROTECTION IN OFFICE FLOORS	C-4 Horizontal Piping		Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239/IS-3589 (heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall /ceiling with clamps, hangers (using anchor fasteners) as per specification.(For horizontal piping - Grooved flexible & rigid coupling type shall be used for all pipe sizes.) Fittings to be UL & FM Approved.	Tender calls for both welded & grooved Fittings for Piping , Kindly confirm which type of pipe fittings to be considered.	All grooved type fitting to be considered		
16	IV-BILL OF QUANTITY	SECTION D - Fire Hydrant (External)	D-1 Piping		Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) conforming to IS- 1239/ IS-3589 (heavy class) with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall /ceiling with clamps, hangers (using anchor fasteners) as per specification (For horizontal piping - Grooved flexible & rigid coupling type shall be used upto dia 65mm and butt welded joints for dia 80mm and above). Fittings to be UL & FM Approved. For pressure rating refer specification and schematics.	Tender calls for both welded & grooved Fittings for Piping , Kindly confirm which type of pipe fittings to be considered.	All grooved type fitting to be considered		

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17	IV-BILL OF QUANTITY	SECTION D - Fire Hydrant (External)	D-1 Hose Chose Pipe		Supply ,Installation,testing and commissioning of non-percolating, rubber reinforced lined Fire Hose Pipe (as per IS : 636) of 63 mm dia. and 2 nos.of 15 m length confirming to IS:8423. The hose shall be rated for burst pressure of 37.5 bar. Hose shall be complete with ISI marked brass male & female coupling (IS:903-1993) bound & rivetted to hose pipe with copper rivets & 1.5 mm copper wire. These hose reels shall be placed along side of each external hydrant, there shall be a weather proof lockable cabinet to accommodate two hose reels as described above. (Type-B) Technical Specification Ref : 2.10 Fire Hose Pipe (FHP)	As all Hydrant valve & accessories are SS MOC, can we consider SS MOC for Hose Pipe coupling instead of Brass (GM)	Follow the BOQ and Specification		
18	IV-BILL OF QUANTITY	SECTION D - Fire Hydrant (External)	D-1 Hose Cabinet External		Supply, installation, testing and commissioning of Hose Cabinet shall be of glass fronted with hinged door & lock. The cabinet shall be made of 16 gauge thick SS 304 sheet having central opening and 4 mm thick glazed glass doors(2nos.)suitably marked on the outside with the letters' FIRE HOSE' MS sheet and spray painted to shade No. 536 of IS:5. sized to accommodate single landing valve(s), 30m(2x15m) long hoses, branch pipe nozzle. size of cabinet not less than 0.9 x 0.6 x 0.45 mtr FIRE HOSE' including necessary locking and supporting arrangement along with SS Branch pipes of 63 mm dia with nozzle of 20 mm dia, as per IS 903-1993.	1-Technical specification calls for MS 16 SWG while BOQ says SS-304 MOC for External Hose cabinet, Please confirm MOC. 2- Approved venders like Tyco & HD Fire are not manufacturing Hose cabinet & Doors, Kindly provide some alternate makes like Arrow , Seerat etc.	1. Follow the BOQ 2. Follow the tender makes.		
19	IV-BILL OF QUANTITY	SECTION E - Pump Room Equipment	E-1 Fire Pumps		Supply, installation, testing and commissioning of electrically driven centrifugal Fire Hydrant and Sprinkler Pump suitable for automatic operation as per the CFO NOC. Refer Equipment schedule PEPL_MUM_PLA_FP_SCH_0001 for pump technical parameters & details.	Pump Schedule document for tower A calls for UL listed / FM approved Fire Pumps but Pump schedule for tower C calls for IS Standard/ TAC approved, Please confirm which Standard / Approval to be considered.	Non UL listed / FM approved fire pumps		
20	IV-BILL OF QUANTITY	SECTION E - Pump Room Equipment	E-2 Jockey Pump		Supply, installation, testing and commissioning of Jockey Pump (pressurisation pump) along with motor, suitable for automatic operation as per the CFO NOC. Refer Equipment schedule PEPL_MUM_PLA_FP_SCH_0001 for for pump technical parameters & details.	Jockey Fire Pumps are available as per IS standard only , Kindly confirm	ok		
21	IV-BILL OF QUANTITY	SECTION E - Pump Room Equipment	E-3 Diesel Engine Pump		Supply, installation, testing and commissioning of Diesel driven Standby Pump , Horizontal split case centrifugal pump with mechanical gland packing, suitable for automatic operation consisting of Diesel Engine,standard accessories etc. as indicated in Specifications (MS Medium grade exhaust pipe for diesel engine with mineral wool insulation and 22 gauge aluminium sheet cladding including fittings, fixings and brackets)	Pump Schedule document for tower A calls for UL listed / FM approved Fire Pumps but Pump schedule for tower C calls for IS Standard/ TAC approved, Please confirm which Standard / Approval to be considered.	Non UL listed / FM approved fire pumps		

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22	IV-BILL OF QUANTITY	SECTION E - Pump Room Equipment	E-15 Level Indicator		Providing ,fixing, testing and commissioning electronic type Water Level Indicator with LED panel mounted as required by the Project Manager complete with probes, pressure switch & all control wiring etc.	Kindly specify the scope, size & length of cabling	All cable which mentioned in Fire BOQ shall be in Fire vendor's scope.		
23	IV-BILL OF QUANTITY	SECTION E - Pump Room Equipment	E-20 Relief Valve		Providing & fixing heavy forged brass tested ISI marked 3200 Lpm Flow Pressure Relief valves of approved quality with all necessary Piping connections till the Bypass valve to Water Tank. Technical Specification Ref : 2.3.8 Pressure Relief Valve	Kindly provide the size of relief valve	Follow the schematic & typical details.		
24	IV-BILL OF QUANTITY	SECTION E - Pump Room Equipment	E-21 Gate Valve		Gate valve Supply ,Installation,testing and commissioning of Brass Gate valve with suitable flanges, nuts, bolts, gaskets, etc. pressure rating as per specification and schematics - (PN-16) Technical Specification Ref : 2.3.1 Gate Valve	We will quote for screw end Gate valve as it is getting difficult in Flanged end	Follow the BOQ and Specification		
25	IV-BILL OF QUANTITY	SECTION H - ELECTRICAL INSTALLATION	H-2 FS Cable		Design, Manufacture, Supply, Unloading, Handling & Laying of 1.1kV grade, 2 hour fire rated Cu. conductor, FS, XLPE insulated, inner and outer PVC sheathed, galvanised steel wire armoured Power cables on cable trays or within cable risers, Complete as per technical specifications, drawings,SLDs, relevant standards & site conditions. Price to be complete with all hardware such as clamps, saddles etc., or for clamping along cable trays or along walls etc., as the case may be. The rates shall be inclusive of cable markers / cable tag / cable tie etc.	BOQ calls for FS cable for Fire Pump room however FS cable is not available in below 500 RMT quantity by approved vender, Please suggest	Follow the BOQ and Specification		
26	IV-BILL OF QUANTITY	SECTION H - ELECTRICAL INSTALLATION	H-5 Earthing		Earthing. Allow for supply, installation, terminating and testing of two no separate earthing cables (1c 10mmsq Cu G/Y LSF cable each) for bonding between the casing of each motor and its respective switchboard earthing bar. Allow 40m runs for each bonding cable.	Request to specify the size of Strip or wire	Alredy specify in BOQ pl refer.		
27	IV-BILL OF QUANTITY	SECTION H - ELECTRICAL INSTALLATION	H-6 PUSH BUTTON STATION AND CONTROL WIRING		Supply, Installation, Testing & Commissioning of Mushroom head Push Button Station provided in glass fronted box. The actuators shall be mushroom head type and colored RED. The background immediately around the device actuator should be colored YELLOW complete in all respect with its control wiring.	Kindly provide separate line item for cable for push button	No need for extra line item. Cost included in same line item.		
28	IV-BILL OF QUANTITY	SECTION G -LIST OF APPROVED MAKE	G-25 Hydrant Valve & Accessories		HDFire/ Safex/ Tyco/ Newage	HD Fire & Tyco does not manufacturing Hydrant valve & Hose cabinet, Please provide some other alternate make i.e Safeguard,Winco, Exflame, omex fire for the same	Follow the tender makes.		

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	HVAC										
1	TPM - Tower ABCE - MEP Works Tender Document 20241122	4. Equipment & Product Specification	4.1.1 i	40	Pressure vessels shall be designed, constructed, tested, stamped and complete with devices in accordance with ANSI/ASHRAE 15-1989 Safety Code and ASME Code.	Please confirm whether to considered ASME stamping for Chillers.	Only Air vessels will be ASME stamped.				
2	BOQ_MEP_TOWER_ABCE	Section A Chiller Plant & High Side Equipment	A. I. 2		Chiller plant optimizer system	Please provide I/O summary for mentioned item.	Refer the BMS tedner for I/O summary.				
3	TPM - Tower ABCE - MEP Works Tender Document 20241122 / BOQ_MEP_TOWER_ABCE	Section A Chiller Plant & High Side Equipment & 4. Equipment & Product Specification	A.I.3 & 4.24	141	Cooling Towers shall be suitable for outdoor use. Tower shall be vertical, induced draft, counter/cross flow type.	As per BOQ Cross Flow , but as per Specs cross / counter flow , please clarify.	Please follow the BOQ.				
4	BOQ_MEP_TOWER_ABCE	Section A Chiller Plant & High Side Equipment	A.I.4		Complete assembly including pump, motor and drive shall be designed to limit THDI to less than 5%	Kindly review & confirm the requirement of then THDI less than 5% for the pump's VFDs, as same will need provision of harmonic filters.	Inbuilt harmonic filter shall be considered, refer the electrical section. Active filters to be considered				
5	BOQ_MEP_TOWER_ABCE	Section A Chiller Plant & High Side Equipment	A.I.5		Expansion Tank shall have third party certificate (TUV / Iloyds registered).Vessel 100-1000 Liters :in accordance with EN13831/ 1200-8000 liters : in accordance with AD2000.	Expansion Tank shall have third party certificate (TUV / Iloyds registered). We shall consider only certification from OEMs for the said approvals. please confirm.	Yes,Only certification from OEMs.				
6	TPM - Tower ABCE - MEP Works Tender Document 20241122 / BOQ_MEP_TOWER_ABCE	Section A Chiller Plant & High Side Equipment & Section G - List of approved makes	A.I.5	1 of 8	Expansion tank as per the manufacturer without any ASME stamping.	As per make list provided all imported manufacturer provides Expansion tank only with ASME stamping as their standard offerings. Please suggest.	Yes, ASME stamping required.				
7	BOQ_MEP_TOWER_ABCE	Section A1. Chw Pumps & PHEX	A1 1		Complete assembly including pump, motor and drive shall be designed to limit THDI to less than 5%	Kindly review & confirm the requirement of then THDI less than 5% for the pump's VFDs, as same will need provision of harmonic filters.	Inbuilt harmonic filter shall be considered,refer the electrical certtinn Active filters to be considered				
8	BOQ_MEP_TOWER_ABCE	Section A1. Chw Pumps & PHEX	A1 3		Expansion Tank shall have third party certificate (TUV / Iloyds registered).Vessel 100-1000 Liters :in accordance with EN13831/ 1200-8000 Liters : in accordance with AD2000.	Expansion Tank shall have third party certificate (TUV / Iloyds registered). We shall consider only certification from OEMs for the said approvals. please confirm.	Yes,Only certification from OEMs.				
9	BOQ_MEP_TOWER_ABCE	Section A1. Chw Pumps & PHEX	A1 3	1 of 8	Expansion tank as per the manufacturer without any ASME stamping.	As per make list provided all imported manufacturer provides Expansion tank only with ASME stamping as their standard offerings. Please suggest.	Yes, ASME stamping required.				
10	BOQ_MEP_TOWER_ABCE	Section B TFA	B 1		The unit shall comprise of filter section, fresh air inlet / exhaust air outlet, fan section with EC plug type fan with motor and drive package including the variable frequency drive housed in the control panel. / TFAs to be with control panel which will house the VFD. VFDs shall be designed to limit THDI to less than 5%.	BOQ description calls for both EC fans & VFD provision, request you please confirm the consideration.	No additional VFD required for EC plug Fan. Inbuilt harmonic filter shall be considered,refer the electrical section.Active filters to be considered				
11	BOQ_MEP_TOWER_ABCE	Section B TFA	B 1		TFAs to be with control panel which will house the VFD. VFDs shall be designed to limit THDI to less than 5%. Cost to include cable from panel to Unit.	Kindly review & confirm the requirement of then THDI less than 5% for the TFA's VFDs, as same will need provision of harmonic filters.	Inbuilt harmonic filter shall be considered,refer the electrical section.Active filters to be considered				
12	BOQ_MEP_TOWER_ABCE	Section C AHU	C 1		The fan shall be selected for minimum 70% efficiency and shall be EC plug type. / For units with VFD, complete equipment shall be designed to limit THDI to less than 5%.	BOQ description calls for both EC fans & VFD provision, request you please confirm the consideration.	No additional VFD required for EC plug Fan.				
13	BOQ_MEP_TOWER_ABCE	Section C AHU	C 1		For units with VFD, complete equipment shall be designed to limit THDI to less than 5%.	Kindly review & confirm the requirement of the THDI less than 5% for the AHU's VFDs, as same will need provision of harmonic filters.	No additional VFD required for EC plug Fan. Inbuilt harmonic filter shall be considered, refer the electrical				
14	BOQ_MEP_TOWER_ABCE	Section F PIPING	F 1		Supplying, installing, testing and commissioning of Seismic Supports for vertical riser piping to take care of wind , building settlement and seismic effect (Refer schematic for seismic support detail)	Seismic supports need to be considered for the shaft pipes however quantities for the shaft pipes have not been segregated separately. Request you to please provide shaft risers quantities separately for the consideration of the Seismic supports.	Please refer the Schematic for shaft risers.				
	ELECTRICAL										
1	Volume No. III	Technical Specification ,2.1 HT PANEL AND SWITCHGEARS / RMU AND TRANSFORMER	2.1.1 TECHNICAL DATA B. Metal-Clad, Circuit-Breaker Switchgear	Page No. 1588	Circuit Breakers: Vacuum Circuit Breakers, Three-pole, single-throw, electrically operated, draw out-mounting units, with SF6 insulated modules	As per SLD, HT Panel is with VCB breakers whereas as per Specification it shall be with SF6 insulated VCB, please confirm the requirement.	Please follow the SLD				
2	Volume No. III	Technical Specification ,2.1 HT PANEL AND SWITCHGEARS / RMU AND TRANSFORMER	2.1.2 CONSTRUCTION REQUIREMENTS J. COMPONENTS	Page No. 1589	a. Main Bus: Copper for full length of switchgear.	As per technical specification, Bus bar of HT Panel Should be Copper. however in the Schematic drawings of HT Panel, Bus bar is of Aluminium. Request you to specify the material of the Bus bar.	Please follow the SLD				
3	Volume No. III	Technical Specification ,2.1 HT PANEL AND SWITCHGEARS / RMU AND TRANSFORMER	2.1.2 CONSTRUCTION REQUIREMENTS U. Cable Termination	Page No. 1591	Cable Termination : Each Cable compartment shall be provided with three bushings of adequate sizes to terminate the incoming outgoing 11kV 3C XLPE (E)cables sizes as per SLD, with Bottom Cable entry	As per BOQ, cable entry is top whereas as per Specification it shall be a bottom entry, please confirm the cable entry required for HT Panels.	Please follow BOQ				
	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet :- A. HT & XMER A HT PANEL	Page No. 4052	The panel shall be prewired, ready for External Cable termination. (Cable entry/Exit-Top).						
4	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet :- A. HT & XMER B HT CABLES & TERMINATIONS	Page No. 4052	Include for Supply, laying, dressing, testing and commissioning of 11kV (Unearthed) grade cables strictly as per specifications and as per IS and of following sizes of XLPE insulated, flat armoured cable with Al. conductor, with GI strip /wire armoured to be laid on wall / ceiling / tray cable rack with proper clamping, end terminations, jointing kits, cable identification tags etc.	1) BOQ asks for HT Cables with flat strip / round wire armouring, please clarify the armouring requirement. 2) Kindly share the technical specification for HT cable and Termination.	1) HT cable is with flat armoured 2) Will be shared in addendum				
5	Volume No. III	Technical Specification ,2.1 HT PANEL AND SWITCHGEARS / RMU AND TRANSFORMER	2.1.7 BATTERY, BATTERY CHARGER & DC DISTRIBUTION Data Sheet of Battery:	Page No. 1599	2.0 Quantity 1 No	In the data sheet of battery, qty is 1 number. But as per BOQ total quantity is 7. Therefore we have understand that the battery qty will be 7 as per BOQ. Kindly confirm.	Battery qty will be as per BOQ				
	Volume No. III	Technical Specification ,2.1 HT PANEL AND SWITCHGEARS / RMU AND TRANSFORMER	2.1.7.2 110 V BATTERY CHARGER: Loads on Charger:	Page No. 1602	3.0 A.C. input system : 3.1 Voltage 415 V + 10%, 3 phase		battery charger suitable for 415V				

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6	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet :- A. HT & XMER A HT PANEL 1.2.2 battery Charger	Page No. 4052	Battery Charger :- Supply, Installation, Testing & Commissioning of Battery Charger unit with provisions of Boost Charging and Float Charging. Suitable for 240V single phase +/- 10% AC, 50 Hz power supply as per specifications. Refer Technical Specification Ref: 2 1 7 2 Battery Charger	As per specification, Battery charger should be suitable for 415 volts 3 phase . As per BOQ, Battery Charger should be suitable for 240 volts 1 phase . Request you to confirm the discrepancies between the charger rating.					
7	Volume No. III	Technical Specification, 2.1 HT PANEL AND SWITCHGEARS / RMU AND TRANSFORMER	2.1.6.16 Data Sheet for transfor	Page No. 1597	Class of Insulation of HV & LV windings - F/ H	3) Technical Specification asks for Class F/ H Insulation, please clarify the class of Insulation required.	Insulation class will be H				
8	Volume No. III	Technical Specification, 2.1 HT PANEL AND SWITCHGEARS / RMU AND TRANSFORMER	2.1.6.16 Data Sheet for transfor	Page No. 1596	Transformer Quantity - 8 Nos	2) As per BOQ, Transformer quantity is 16 no's, whereas as per Technical data sheet mentions 8 nos. transformers , please confirm.	Please follow the boq				
9	Volume No. III	Technical Specification, 2.1 HT PANEL AND SWITCHGEARS / RMU AND TRANSFORMER	2.1.6.16 Data Sheet for transfor	Page No. 1597	19. No Load Losses@50% Watts:- To be confirmed by Vendor 20 Load Losses@50% Watt:- 7500 21. Load Losses (@ 100%load) at 75 °C:- 2000	The losses Specified in the technical data sheet is for 2000KVa capacity transformer. We will consider the transformer losses as per ECBC 2017 latest for 2500 KVA capacity Transformer. Please confirm	Please consider losses as per ECBC 2017 for 2500KVA transformer				
10	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet :- A. HT & XMER C Transformer_2	Page No. 4053	Supply, Installation, Testing and Commissioning of below listed Isolators/ACB in IP 21 RAL 7032 metal enclosure & all outdoor isolator shall be IP 55 with all necessary accessories, supporting arrangement, gland plate etc; complete in all respect to the satisfaction of client and consultant and concern statutory authority. Please refer SLD sheet: PEPL_MUM_PL_C_EL_D_7001 & PEPL_MUM_PL_C_EL_D_7007	Request you to specify the thickness of metal Enclosure.	1.6mm thick				
11	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet :- B. TENANT BOQ G ATS (Auto Transformer Switch)	Page No. 4055	Supply, Installation, Testing and Commissioning of below listed ATS equivalent to 7000 Series of Asco, in RAL7032 IP 21 metal enclosure & with all necessary accessories, supporting arrangement, gland plate etc; . Fortson cable entry from top ,and second is busbar and exit is busbar .All entry/exit from top. Please refer Slashed: PEPL_MUM_PL_C_EL_D_7001 to PEPL_MUM_PL_C_EL_D_7010 Refer Technical Specification Ref : 3 19 Auto Transformer Switch (ATS)	Request you to specify the thickness of metal Enclosure. Request you to share the technical specification for ATS	Thickness of metal enclosure is 1.6mm . The ATS specifications are in section 6A Technical Specifications , clause 2.2.13.. Specific model number are mentioned in the Schematic as per application				
12	Volume No. III	Technical Specification, 2.3 L.V. BUSDUCT, BUSBARS & ACCESSORIES:	2.3.3 TECHNICAL : C	Page No. 1644	The complete bus bar enclosure housing must be of from minimum 2.5 mm thermal processed extruded aluminium housing from all four side aluminium RAL 7038 (or such other shade) electrostatic painted, or equivalent Treated and finished sheet steel Enclosure of minimum 1.6 MM thick, epoxy powder coated.	As per the technical specification Bus bar enclosure should be extruded Aluminium . But as per BOQ, Enclosure should be extruded GI . Request you to kindly confirm the enclosure material.	GI enclosure				
	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet B. TENANT BOQ I BUSDUCT AND RISING MAINS AND TAP OFF UNITS	Page No. 4055	Supply, Installation, testing, Transport, Loading, unloading on proper place at site as directed by Engineer In Charge with Supervision & Commissioning of compact sandwiched Type, Aluminium Busduct / Rising Mains with Class B insulation in GI extruded enclosure .						
13	Volume No. III	Technical Specification, 2.3 L.V. BUSDUCT, BUSBARS & ACCESSORIES:	2.3.5 CONSTRUCTION REQUIRE	Page No. 1646	F. Ratings - Three phase with Neutral Bus Bars shall be designed to carry the rated full load current under normal operating site conditions without exceeding a 55 degree C rise above a 40 degree C ambient or total hot spot temperature of 95o C (for Class B Insulated Bus risers) and 115 o C Centigrade with Max hot Spot temp of 155 dig for Class F Insulated busbars . There shall be no derating of busbar up to 55 dig.	Specification is asking for both class B and Class F insulation. But as per BOQ description, we have understand that the only Class B insulation is required.	Please adhere to boq				
	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet B. TENANT BOQ I BUSDUCT AND RISING MAINS AND TAP OFF UNITS	Page No. 4055	Supply, Installation, testing, Transport, Loading, unloading on proper place at site as directed by Engineer In Charge with Supervision & Commissioning of compact sandwiched Type, Aluminium Busduct / Rising Mains with Class B insulation in GI extruded enclosure .						
14	Volume No. III	Technical Specification, 2.4 L.V. POWER & CONTROL CABLES, IT/ TELCOM & TV CABLES :	2.4.3 CABLE TYPES : 2.4.3.1 Power Cabling	Page No. 1652	Armoured Normal Service Cable (LSZH/SWA/LSZH/XLPE)		Cables will be of FRLSZH and flat armour				
	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet B. TENANT BOQ J CABLES AND TERMINATIONS	Page No. 4056	Supplying, unloading at site, shifting into the store, Laying, Testing & Commissioning of 1100V grade, Fire-retardant Low Smoke Halogen (FRLSH) , XLPE Insulated, Flat Steel Wire Armour, Aluminium conductor Cables to be laid with necessary supports on cable trays/ on ceiling / wall complete in all respect as per site requirement, technical specifications, and as per approval from client / consultant.	There is a discrepancies between the type cable in Specification and BOQ. Request you to confirm the type of cable along with the type of armour for the cable (round wire or flat Strips)					
15	VOLUME – IV	PRESTIGE LIBERTY TOWERS - SALES TOWER TOWER A B C E BOQ (ELECTRICAL)	Sheet :- C. Common Area N LV ELECTRICAL PANEL BOARDS, DISTRIBUTION BOARDS AND ISOLATORS	Page No. 4058	Panels	We have understand that the Main LT Panel (MLTP) will be type tested (TTA) as per IEC 61439 and other panels(SMDBs) will be Partial type test (PTTA) as per IEC 60439	MLTPS and SMDBs to be as per IEC61439 .TTA and PTT terms are Not applicable . Please provide IEC 61439 report for each panel type .Also discrimination report of similar scale project for reference				
16	Volume No. III	G. List of Approve Make	Cubicle LV Panels Main Distribution Boards (MDB), Sub-Main Distribution Boards (SMDB), Distribution Boards (DB)	Page No. 1362	Panels	3) As per Approved Make List, Panel vendors are Lauritz Knudsen (earlier L & T), ABB, Siemens & Schneider. Kindly confirm whether to consider System Integrated Panels of Lauritz Knudsen (earlier L & T), ABB, Siemens & Schneider, please confirm & amend the make list with panel vendors details.	OEM approved panel vendors are ok .				

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1	BOQ - Sprinklers	Section - A	Item No. 1 & 3		Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239/IS-3589 (heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings up to 150mm dia) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall / ceiling with clamps, hangers (using anchor fasteners) as per specification.(For horizontal piping - Grooved flexible & rigid coupling type shall be used for all pipe sizes.) Fittings to be UL & FM Approved.	BOQ description calls for Grooved fittings for the all the sizes for the horizontal pipes. Kindly confirm the same.												
2	BOQ - Hydrant	Section - B	Item No 1				Follow the BOQ and Specification											
3	BOQ- Drencher	Section - C	Item No 4															
4	Volume - II	Section B- Technical Specification / 4. Fire Pumps and Ancillaries	4.2 General Requirements / Point No. 1	3457	All materials & equipment used shall be UL listed.	Kindly confirm whether pumps shall be UL/FM approved as BOQ calls for Non-UL/FM approved pumps.	Non UL/FM pumps.											
5	Volume - II	Section B- Technical Specification / 4. Fire Pumps and Ancillaries	4.1 Scope / Point 1	3457	Electrically operated Multi-stage-Multi Outlet/Horizontal End suction centrifugal type/Horizontal split case centrifugal main fire pumps for Hydrants, Sprinkler and car park Drencher system	Discrepancy between BOQ & Tech Specs. Kindly confirm.	Follow the BOQ.											
	Volume - IV	SCHEDULE OF QUANTITIES FOR FIRE PROTECTION WORKS TOWER A, B, C & E	E. Pump Room Equipment / Point No. 1.a	4097	Multi-stage Multi-outlet Vertical Centrifugal Pump , the installation shall be complete with flexible coupling and coupling guard as required.													
6	Volume - III	Fire Protection Pump Schedule - Tower A	Sr. No 1,2,3,4,7,8,9,10	3666	<table><tr><th>Equipment description</th><th>Pump Type</th></tr><tr><td>MAIN HYDRANT PUMP</td><td>VERTICAL INLINE MSMO PUMP</td></tr><tr><td>JOCKEY PUMP (HYDRANT)</td><td>VERTICAL INLINE MSMO PUMP</td></tr><tr><td>MAIN SPRINKLER PUMP</td><td>VERTICAL INLINE MSMO PUMP</td></tr><tr><td>JOCKEY PUMP (SPRINKLER)</td><td>VERTICAL INLINE MSMO PUMP</td></tr></table>	Equipment description	Pump Type	MAIN HYDRANT PUMP	VERTICAL INLINE MSMO PUMP	JOCKEY PUMP (HYDRANT)	VERTICAL INLINE MSMO PUMP	MAIN SPRINKLER PUMP	VERTICAL INLINE MSMO PUMP	JOCKEY PUMP (SPRINKLER)	VERTICAL INLINE MSMO PUMP	As per OEMs, MSMO pumps are not available in Vertical INLINE type. MSMO pumps are either available in Horizontal Split Case/Vertical Centrifugal. Kindly Confirm	Follow the BOQ for Pump type.	
Equipment description	Pump Type																	
MAIN HYDRANT PUMP	VERTICAL INLINE MSMO PUMP																	
JOCKEY PUMP (HYDRANT)	VERTICAL INLINE MSMO PUMP																	
MAIN SPRINKLER PUMP	VERTICAL INLINE MSMO PUMP																	
JOCKEY PUMP (SPRINKLER)	VERTICAL INLINE MSMO PUMP																	
7	Volume - IV	SCHEDULE OF QUANTITIES FOR FIRE PROTECTION WORKS TOWER A, B, C & E	E. Pump Room Equipment / Point No. 2.a	4097	Supply, installation, testing and commissioning of Jockey Pump (pressurisation pump) along with motor, suitable for automatic operation as per the CFO NOC. Refer Equipment schedule PEPL_MUM_PLA_FP_SCH_0001 for pump technical parameters & details. a) Multi-Stage Multi-outlet Vertical Centrifugal Pump, Jockey pump shall be of stainless steel construction. The pump head and base shall be CI GG20 construction and bronze impellers, diffusers, shaft sleeve, seal bush & pump shaft shall be in SS 304 material. The pump shall be provided with Gland packed seal (non asbestos). The system shall be complete with necessary pressure gauge with gun metal shut off cock on delivery side (The pump should meet the condition and shall be gotten approved by the Local fire Authority).	Discrepancy in MOC of Jockey Pump. Kindly provide clarity.	Follow the BOQ for MOC.											
	Volume - II	Section B- Technical Specification / 4. Fire Pumps and Ancillaries	4.3 Fire Pumping Sets / Point No. 11	3459	Casing - Stainless steel for Vertical inline Jockey pumps. Impeller - Stainless steel for Vertical inline Jockey pumps. Shaft - Stainless steel													
8	Volume - IV	SCHEDULE OF QUANTITIES FOR FIRE PROTECTION WORKS TOWER A, B, C & E	A. Sprinklers / Point No. 4	4091	Contractor shall supply 2% spare sprinkler heads and one spanner neatly installed in a wooden cabinet with glass shutters as approved by the Employer. This is to be included in the cost.	Kindly amend the BOQ with separate item for the spare sprinkler in wooden cabinets providing quantities of cabinets as per the sprinkler zones.	The cost of the spare sprinkler heads, the spanner, and the wooden cabinet should be included in the overall price. There should be no additional charges for these items.											
10	Volume - IV	SCHEDULE OF QUANTITIES FOR FIRE PROTECTION WORKS TOWER A, B, C & E	D. Hydrant (External) / Point 3	4096	Excavate trench not exceeding 1.5m deep for laying pipes up to 300mm dia. including forming bottom surface to required level, refilling the trenches with selected excavated earth around the pipes in layers of 150mm thick, watering, consolidating, disposing off surplus earth out side the site to a dump yard acceptable to relevant authorities or as directed	Please check & confirm the quantity for the excavation is RO however pipe quantities needs to be considered with anti-corrosive wrapping & coating.	Please provide the rate only.											
11	Volume - IV	SCHEDULE OF QUANTITIES FOR FIRE PROTECTION WORKS TOWER A, B, C & E	B. Hydrant / Point No. 4.02	4091	UL Listed Non-percolating,(as Shown in Standard Installation drawing, BH-XX-XX-DR-xxxx) rubber reinforced lined canvas Fire Hose Pipe (as per IS : 636-1988) of 63 mm dia. The hose shall be rated for burst pressure of 37.5 Kg/sqcm. Hose shall be complete with ISI marked Stainless Steel male & female coupling (IS:903-1993) bound & riveted to hose pipe with copper rivets & 1.5 mm copper wire. (Type-B). Scope includes necessary supporting, bracket arrangement to hang the hose pipe inside FHC/Fire Shaft.	Request you to consider Hydrant valves & accessories as per IS standards for the internal hydrants & not UL listed. Please note as per BOQ, External hydrant shall be non-UL listed.	Follow the BOQ and Specification											

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1	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C - Section 1	01:01:49	102	Mobilization: Advance for each Package	10% of the contract value shall be paid in instalments as shown below against Bank Guarantee as per Engineer recommendation. A. Part I – 5% – Upon submission of Bank guarantee and Site mobilization after completing the requirements and documentation as per the respective clause. B. Part II – 5% – Upon submission of Bank guarantee and within 1 month after successful commencement of the work. The Employer shall recover the Mobilization Advance by deducting an amount on pro rata basis of the gross Running Bill value from each Running Bill raised by the Contractor starting from the first Running Bill. Provided however that, the entire Mobilization Advance should be recovered on prorata basis prior to payment of 80% (Eighty percent) of the Contract Price, Package wise.	We propose, Advance Bank Guarantee (ABG) is issued on a reducing balance basis, it means that the value of the guarantee decreases over time as the contractor repays or adjusts the advance payment provided by the employer	Contractor to follow the tender BG format.	
2	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C - Section 5	1	103	Obligations of Contractor	- Punitive Damages on non- appointment of Authorized representative within stipulated period -0.25% of the Contract Price per week subject to a maximum of 2% of the Contract Price.	Not acceptable and contractor shall be given sufficient time to replace / appoint the required resource.	Tender shall prevail.	
3	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C Section 13	1.1	104	Defects Liability period	The Defects Liability Period shall be Twenty Four Months (24) months from the date of issue of Completion Certificate by the Engineer.....	Please note, in case phase wise completion & handover to client, DLP period for the equipment / system installed for the handing over area shall commence thereafter & will not be linked overall completion of the project.	Tender shall prevail. Employer reserves the right to takeover overall functional system or phasewise. Contractor to request the same if such condition arises and get written confirmation from Employer / PMC for such DLP commencement and Employer at their sole decretion can accept or reject the request based on the type of system and situation.	
4	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C - Section 1	3	105	Basis of Contract - Electricity and Water	The electricity shall be provided by the Employer chargeable basis at one point on site....	Request to consider supply of Electricity for construction on free of charge basis on alternate floors.	Tender shall prevail. Contractor to quote charges considered for the same separately in the summary sheet of the BOQ.	
5	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C - Section 14	1.8	106	Payment Mechanism - Secured Advance Against Specified Material	The Contractor may request for secured advance against delivery of materials up to 75% of its verified tax invoice value. This advance is recoverable from the subsequent interim payment certificates. Recovery of the same shall be completed within 3 months from the date of payment of advance.	Please provide breakup for the installation portion.	This is advance is for materials only. Installation shall be considered in the item rates.	
6	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C - Section 14	5.1	106	Payment Mechanism - Submission and Verification of Final Bill	Final Bill to be certified based on joint measurement of installed work, submission all required back up documents such as marked up drawings, measurement books, quality and safety certificates and other compliances as per tender documents and bill check list process shall be submitted within 30 days after completion of the Project. The submitted Final bill by the contractor, shall be certified by the PM within 90 days of submission.	Request to consider - the submitted final bill by the Contractor shall be certified by the Project Manager within 60 days of submission	Tender shall prevail.	
7	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C - Section 14	3.1 / 3.2.1	107	Retention Money	From all running accounts bills and final bill payments, Deposit against Defects Liability Period services/retention money at 5% of the gross certified bill amount of the work performed shall be withheld. Such withhold amount shall be released 30 (Thirty) days after completion of Defect Liability Period.	Request you to consider submission of retention BG equivalent 5% of the contract value with validity till DLP period to be submitted before first RA bill in lieu of cash retention.	Tender shall prevail.	
8	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C - Section 14	2.1	107	Valuation of Variation	In case of Varied Work, the Employer shall pay 12% of the sum of Cost of materials, labour and Plant and machinery for that work, towards contractor's establishment, water and power charges, all other cost and over heads and profit.	We propose, In case of Varied Work, the Employer shall pay 20% of the sum of Cost of materials, labour and Plant and machinery for that work, towards contractor's establishment, water and power charges, all other cost and over heads and profit.	Tender shall prevail.	
9	SECTION – G – SPECIAL CONDITIONS OF CONTRACT	G.C.C – Section 16	1	108	Base Prices and Cost Escalation / De-Escalation	All the rates quoted by the contractor shall remain unchanged till satisfactory completion of the project. No escalation shall be payable on rates quoted by the contractor except for Basic rate variation for specified materials pursuant to Appendix – 2 and Appendix – 4.	Please note, our prices shall remain fixed & firm for the original completion period of the tender. For any extension for the reasons not attributing to the contractor, prices to be revised with mutual discussion for the extended period.	Tender shall prevail.	
10	SECTION – B – NOTICE INVITING TENDER		1.13	10		The Tender shall remain valid for acceptance for a period of 120 days from the last date of receipt of the Tender or revised quotes /Final offer (if any), whichever is later.	We propose offer validity of maximum 90 days from the date of offer submission.	Tender shall prevail.	

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11	SECTION – B – NOTICE INVITING TENDER		1.1.0	10		All quoted rates in the BOQ will be firm till the satisfactory completion of the project and extension, if any	We propose, All quoted rates in the BOQ will be firm till the satisfactory original completion of the project and extension, if any .	Tender shall prevail.	
12	SECTION – B – NOTICE INVITING TENDER		01:03:03	12		The rates quoted against each item in the BOQ shall be deemed to cater for all minor constructional details which are not specifically mentioned, but are fairly and obviously intended and are essential for the full & final completion of the Works and it is the responsibility of the Bidder to point out to the Employer / PM about any major constructional details missing or gaps/ ambiguities or incorrect information related to the Works, in the Drawings, Specifications and or Bill of Quantities, before providing his offer on or before the Tender Submission Date .The Successful Bidder is not entitled to make any extra claim on this account.	Please note, we shall be quoting as per the tender documents received, any new or additional scope / service / ancillary works required will be charged extra as applicable.	Tender terms and conditions shall prevail.	
13	SECTION F – GENERAL CONDITIONS OF CONTRACT		2.6	31		(a) In the event of any discrepancy between the details and / or description given in the Contract, then the Item of the Works / Specified Materials shall be deemed to have been priced in accordance with the details and / or description confirming to the most superior provisions contained in any of the following: (i) Contract Agreement/ Work Order (ii) Special Conditions of Contract; (iii) General Conditions of Contract; (iv) Technical Specifications; (v) Tender Drawings and related notes/ instructions etc.; (vi) Priced bid (vii) Any other ancillary documents entered between the parties	We propose to add (viii) Pre-Bid Queries, All Amendments, MOM's MOU's	To be discussed during final commercial meeting.	
14	SECTION F – GENERAL CONDITIONS OF CONTRACT		3.1	32	Basis of Contract	This is an Item rate contract as mentioned and shall be subject to remeasurements during the process of works and billings except for items having unit of measurement as "Lumpsum". It is clearly understood and agreed that the quantities mentioned in the Bill of Quantities are approximate estimated quantities and can therefore vary to an extent and the Contractor shall not have any claim for loss or profit whatsoever for such variation to subject to Section 15 of the General Conditions of the Contract. The Contractor shall be paid at the unit rates (as mentioned in the Bill of Quantities) for the actual quantities of work executed	Request you to consider ceiling of +/- 20% for the quantity variation/scope changes. If contract value varies +/- 20% due to change in quantity or scope modifications then contractor shall be allowed to revise the rates of the concerned items.	Tender terms and conditions shall prevail.	
15	SECTION F – GENERAL CONDITIONS OF CONTRACT		1	35	SECTION 4 – OBLIGATIONS AND POWERS OF THE EMPLOYER AND PROJECT MANAGER	The Employer shall have the absolute authority to enforce compliance with the Contract Documents, on all questions relating to quantities, the acceptability of materials, equipment, or Works, the adequacy of the performance of the Work, and the interpretation of the Drawings and specifications. The decision of the Employer will be final and binding and shall proceed to any payment under the contract agreement unless otherwise provided in the contract documents.	Priority of the documents shall be referred for any technical ambiguities and decision to be taken mutually by client & contractor.	Tender terms and conditions shall prevail.	
16	SECTION F – GENERAL CONDITIONS OF CONTRACT		3	35	SECTION 4 – OBLIGATIONS AND POWERS OF THE EMPLOYER AND PROJECT MANAGER	The Employer shall have the right to appoint its representative namely, Project Manager (PM) for general supervision of Works and to get the Works carried out in strict confirmation of the Contract documents and shall have authority to put into effect the following:-.....	Please note, PM appointed by the client would have no direct authority / powers over the contractor's staff and any information required by PM shall be placed appropriately to contractor's site in charge giving him sufficient time to respond.	Tender terms and conditions shall prevail.	
17	SECTION F – GENERAL CONDITIONS OF CONTRACT		4	36	SECTION 4 – OBLIGATIONS AND POWERS OF THE EMPLOYER AND PROJECT MANAGER	Any instructions or approval given by the Employer/ PM to the Contractor verbally / writing in connection with the Works shall bind the Contractor as though it had been given by the Employer, provided failure of the Employer/ PM to disapprove any work or materials shall not prejudice the Employer/ PM /Consultants to disapprove such works or materials and to order the pulling down, removal or breaking up thereof at the cost of the Contractor	Instruction has to be in recordable form i.e. in writing and verbal instructions needs to be validate in writing.	Tender terms and conditions shall prevail.	

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18	SECTION F – GENERAL CONDITIONS OF CONTRACT		9	36	SECTION 4 – OBLIGATIONS AND POWERS OF THE EMPLOYER AND PROJECT MANAGER	In the event, the performance of this Contract is likely to interfere with simultaneous execution of some other separate contract or contracts, the Employer/PM shall decide which of such Contractors shall precede the others. The Employer shall not be responsible for any damages suffered or extra costs incurred by the Contractor resulting directly or indirectly from any decision or omission of the Employer/PM with regard to the order of precedence in the performance of the separate contracts awarded for completion of the project. Any costs caused by ill-timed Work shall be borne by the Contractor responsible thereof.	The contractor needs to be compensated in case there is idle period of more than 48 hrs due to employer’s decision to give precedent to other contractor.	Tender terms and conditions shall prevail.	
19	SECTION F – GENERAL CONDITIONS OF CONTRACT		1.2 F	37	SECTION 5 – RESPONSIBILITY/ OBLIGATIONS OF CONTRACTOR	f. Performance Guarantee for Waterproofing works: -	Mentioned works does not fall in our scope of works	Noted	
20	SECTION F – GENERAL CONDITIONS OF CONTRACT		6	40	Inspection of Site	No subsequent claims shall be raised by the Contractor, in respect of the condition of the Site, for any delays in the execution of the Work as per the agreed milestones/timelines under this Contract as far as those conditions which are within the control of Contractor.	I). Employer agrees to provide for equitable adjustments to the differing site conditions affecting performance / cost of the work, in terms of the following: (1) the contractor as indicated in the tender document made the site visit (2) the actual conditions were reasonably unforeseeable based upon all the information available to the contractor at the base date; (3) the contractor incurred additional costs as a result of the difference between the expected conditions at the base date and the actual conditions. II).The Employer agrees to disclose all vital information affecting the project’s performance, costs, or duration that is not available to the Contractor. Additionally, the Company indemnifies the Contractor concerning any ambiguity in the contract specifications that did not sufficiently alert the Contractor to seek further clarifications.	Tender terms and conditions shall prevail.	
21	SECTION F – GENERAL CONDITIONS OF CONTRACT		20	46	SECTION 5 – RESPONSIBILITY/ OBLIGATIONS OF CONTRACTOR	Life Saving and Safety Appliances; Ambulance - The Contractor shall at his own cost provide and maintain upon the works sufficient, proper and efficient lifesaving and first aid appliances including ambulance to the satisfaction of the Employer/ PM and in compliance with statutory or other regulations.	Please note, we shall be having tie ups with local hospitals in vicinity for the site’s medical emergencies so we shall not consider provision of keeping Ambulance at site.	Tender terms and conditions shall prevail.	
22	SECTION F – GENERAL CONDITIONS OF CONTRACT		20	47	SECTION 5 – RESPONSIBILITY/ OBLIGATIONS OF CONTRACTOR	22. Liability for Damage of Property - The Contractor shall be solely responsible for any damage to any adjoining structures caused by its workmen/ machines deployed. The O w n e r / PM bears no liability whatsoever on this account. If the Employer/ PM is called upon to pay for any such damage, then the sum involved will be recovered from the amounts payable to the Contractor.	Please note, contractor’s overall liability shall be 100% of the contract value and contractor shall not be responsible for any consequential damages.	Tender terms and conditions shall prevail.	
23	SECTION F – GENERAL CONDITIONS OF CONTRACT		31	49	Programme and Progress Report	The program, suitably amended after discussions with the Project Manager shall become binding on the Contractor and shall serve as Baseline Schedule. However, during the execution of the project, should it become necessary, in the opinion of the Project Manager to reschedule some of the activities, alter the sequence, duration of any activities; the Contractor shall do so at no extra cost and/or without any other claim.	The contractor needs to be compensated in case there is idle period of more than 48 hrs due to PM’s decision to change in schedule.	Tender terms and conditions shall prevail.	
24	SECTION F – GENERAL CONDITIONS OF CONTRACT		13.5	52	Compliance with Applicable Laws	Cost of labour cess shall be borne by the Employer so labour cess shall be excluded from the quoted rates by the Contractor.	Kindly confirm, if any deduction from RA bills for the same.	Contractor can mention the BOCW cess separately in the summary sheet of the Bid submitted by them additionally.	

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25	SECTION F – GENERAL CONDITIONS OF CONTRACT		7	58	SECTION 5 – RESPONSIBILITY/ OBLIGATIONS OF CONTRACTOR	Lien - The Employer/ PM shall have a lien over all the materials brought to the Site for Works by the Contractor/ Sub Contractors and all or any monies that may become due and payable to the Contractor under the conditions of Contract and also over the retention money against the Defects Liability Period held by the Employer/ PM under the Contract and becoming repayable to the Contractor.	We request for deletion of the this clause as the Contractor has submitted Performance BG and Retention Money also is withhold till issuance of Taking over certificate, thus we request for deletion of this provision as already sufficient monies are kept for security against performance of contract.	Tender terms and conditions shall prevail.	
26	SECTION F – GENERAL CONDITIONS OF CONTRACT		3	59	3. Testing required by Employer/PM	- The Contractor shall provide such assistance, labour, apparatus and instruments as are normally required for examining, measuring and testing any Specified Materials or Contractor's Equipment and shall supply samples of Specified Materials, before incorporation in the Works, for testing as may be selected and required by the Employer / PM from time to time. The costs of any tests conducted by the Contractor or sub-contracted to any outside agency, including but not limited to the costs of samples, transportation and testing fee, shall be solely borne by the Contractor.	Please convey if any provisions to be considered for the third party testing.	Tender terms and conditions shall prevail. Contractor shall be responsible for all type of tests and associated costs including third party testing.	
27	SECTION F – GENERAL CONDITIONS OF CONTRACT			59	4. Test Timings, Place and Attendance by Employer/PM	The Employer/PM shall endeavour to give the Contractor a minimum of 24 (Twenty Four) hours prior notice of its intention to carry out the inspection or to conduct / attend the tests. The Contractor shall agree with the Employer/PM on the time and place for the inspection or testing of any Specified Materials, Contractor's Equipment or Works. In the event the Contractor is of the view that the presence of the Employer/PM or their representative is required for inspection and testing, the Contractor shall intimate the same to the Employer/PM.	Please confirm whether expenses towards travel , logging & boarding of client / consultant's persons needs to be considered by contractor .	Consider the same if tests or inspections to be conducted outstation.	
28	SECTION F – GENERAL CONDITIONS OF CONTRACT		4	62	Handing over of floors	Contractor shall start handing over of floors to Employer / PM after completion of Contractors scope of Works for start of other trade activities. However DLP will start from final handing over of the project / Tower / Cluster.	Please note, in case phase wise completion & handover to client, DLP period for the equipment / system installed for the handing over area shall commence thereafter & will not be linked overall completion of the project.	Tender terms and conditions shall prevail.	
29	SECTION F – GENERAL CONDITIONS OF CONTRACT		3	67	Deductions and Retention Money	Retention money shall be deducted @ 5% from each R.A. Bill. The Contractor shall be entitled to claim refund of Retention money of 2.5% (two and half percent) after issue of Completion Certificate of Work and acceptance of final bill and measurements.	Request you to consider submission of retention BG equivalent 5% of the contract value with validity till DLP period to be submitted before first RA bill in lieu of cash retention.	Tender terms and conditions shall prevail.	
30	SECTION F – GENERAL CONDITIONS OF CONTRACT	SECTION 15 - VARIATION IN THE SCOPE OF THE WORKS	1.1	71	Variations	The Contractor acknowledges and agrees that the PM/Employer shall have the option of making any variation in the form, quality or quantity of the Works or any part thereof that may, in his/their opinion, be necessary or advisable during the progress of the work, and for that purpose, or if for any other reason it shall, in his/their opinion, be desirable, he/they shall have the power to order the Contractor to do and the Contractor shall do any or all of the following ("Variations"): (a) increase or decrease of areas (b) Omit any Works; (c) Change the character or quality or kind of any such Works; (d) Execute additional Works of any kind necessary for the completion of the Works; and no such Variation shall in any way vitiate or invalidate the Contract, but the value, if any, of all such variations shall be taken into account in ascertaining the amount of the Contract Price	Request you to consider ceiling of +/- 20% for the quantity variation/scope changes. If contract value varies +/- 20% due to change in quantity or scope modifications then contractor shall be allowed to revise the rates of the concerned items.	Tender terms and conditions shall prevail and no limit for the quantity variation.	
31	SECTION F – GENERAL CONDITIONS OF CONTRACT	SECTION 21 - SUSPENSION OF CONTRACTOR'S RIGHTS/ WORKS	1	82		b. exercise such rights itself and perform the obligations hereunder or authorize any other Person to exercise or perform the same on its behalf during such suspension ("Suspension"). Suspension hereunder shall be effective forthwith upon issue of notice of Suspension by the Employer / PM to the Contractor and may extend up to a period not exceeding cumulative 90 (Ninety) days and 60 days at a stretch from the date of issue of such notice	b. exercise such rights itself and perform the obligations hereunder or authorize any other Person to exercise or perform the same on its behalf during such suspension ("Suspension"). Suspension hereunder shall be effective forthwith upon issue of notice of Suspension by the Employer / PM to the Contractor and may extend up to a period not exceeding cumulative 90 (Ninety) 30 days and 60 days at a stretch from the date of issue of such notice	Tender terms and conditions shall prevail.	

Prebid Queries

Name of Contractor : Vendor 5					Date : 29:11:24				
Sr. No. / Query No.	Bidding Document Reference(s) [Volume No / Section Number / Clause Number]			Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required		Response to the Query [JLL / PEPL]	Remarks
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32	SECTION F – GENERAL CONDITIONS OF CONTRACT	SECTION 21 - SUSPENSION OF CONTRACTOR'S RIGHTS/ WORKS	4	82		In the event that Suspension is not revoked within cumulative 90(Ninety) days or 60 days at a stretch from the date of Suspension hereunder or within the extended period	In the event that Suspension is not revoked within cumulative 90(Ninety) days or 60 30 days at a stretch from the date of Suspension hereunder or within the extended period	Tender terms and conditions shall prevail.	
33	SECTION F – GENERAL CONDITIONS OF CONTRACT		10.3	88	Extension of Construction Period	In the event that a Material Breach by either Party causes delay in achieving Completion Date or leads to Suspension, as the case may be, the Employer may, in its sole discretion, extend the Construction Period, such extension being equal in duration to the period by which Completion Date was delayed.	If the delay is not attributed to contractor then contractor shall be compensated for the expenses incurred towards PMC & Site overheads for the such extended periods.	Tender terms and conditions shall prevail and without any compensation to the contractor.	
34	SECTION F – GENERAL CONDITIONS OF CONTRACT		4	90	3.1 Conciliation and Arbitration	4. Governing Law This Contract shall be governed and construed in accordance with the laws of India and subject to the provisions of arbitration as set out above in Clause 3.0, the courts at Bangalore shall have exclusive jurisdiction.	Please consider jurisdiction place Mumbai	Tender terms and conditions shall prevail.	
35	SECTION F – GENERAL CONDITIONS OF CONTRACT	SECTION 25 - LIABILITY AND INDEMNITY	2.2	91		Without limiting the generality of Clause 1.1, the Contractor shall fully indemnify, hold harmless and defend, without demur or protest, the Employer/PM and the Indemnified Persons from and against any claims and/or all loss and/or damages and/or expenses and/or liabilities and/or penalties arising out of or with respect to:	Please note, contractor's overall liability shall be 100% of the contract value and contractor shall not be responsible for any consequential damages.	Tender terms and conditions shall prevail.	
36	SECTION F – GENERAL CONDITIONS OF CONTRACT		20	99	Penalty / Penalties :	If a contractor failed to meet their contractual obligations will receive financial penalties. The value of the penalty is as mentioned below and in the Special conditions of contract. If the contractor failing to comply to the intermediate milestones, quality and safety standards or the rectifications thereof, this clause becomes operative, and Contractor will have to pay the stipulated sum of money as imposed. Total amount of penalties will be deducted from the balance payment of the Contractor. The penalties shall be applicable as below.....	Request you to delete said clause as LD clause is there to cover the delays.	Tender terms and conditions shall prevail.	
37	MASTER SCOPE RESPONSIBILITY MATRIX		5	117	Protected Stores of Trade Contractors at site	Limited space available at site. Offsite storage is necessary for the complete material. Main contractor to provide available space to the MEP Contractor. Vendor must adopt a just in time procurement and no stock holding at site.	Please convey the tentative area & location for the office & store to plan our material deliveries accordingly.	Very limited space shall be available at site and contractor has to make the arrangement for offsite storage. Space can't be allocated at this stage and shall be only provided to the contractor during the actual construction stage as per availability.	

Prebid Queries

Name of Contractor : Vendor 6 - Query 01

Date :

29-11-2024

Sr. No. / Query No.	Bidding Document Reference(s) [Volume No / Section Number / Clause Number]			Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.						
FAPA									
1	MEP works for Tower ABCE	D_Voice Annunciation System_D10	D_Voice Annunciation System_D10		Supply, installing, testing and commissioning of 20W, Surface Mounted, horn speaker with required accessories. The Units shall be UL listed and FM approved. To include for terminating circuit cabling into device and backbox.	20W Speakers are not available with all of the OEM's as per Approved Make list. Kindly suggest.	15W non UL speakers		
2	MEP works for Tower ABCE	D_Voice Annunciation System_D11	D_Voice Annunciation System_D11		Supply, installing, testing and commissioning of 20W, Surface Mounted, horn with Strobe & required accessories. The Units shall be UL listed and FM approved. To include for terminating circuit cabling into device and backbox.	20W Hooter with Strobes are not available with all of the OEM's as per Approved Make list. Kindly suggest.	Standard hooter strobe to be considered .Strobe shall have minimum 110cd output .It shall UL listed .The hooter output shall be 75db at 10 feet		
3	MEP works for Tower ABCE	D_Voice Annunciation System_D12	D_Voice Annunciation System_D12		Supply, installing, testing and commissioning of 20W, Weatherproof, Surface mounted horn with Strobe & required accessories. The Units shall be UL listed and FM approved. To include for terminating circuit cabling into device and backbox.	20W Hooter with Strobes are not available with all of the OEM's as per Approved Make list. Kindly suggest.	Standard weatherproof hooter strobe to be considered .Strobe shall have minimum 110cd output .It shall UL listed .The hooter output shall be 75db at 10 feet		
BMS									
1	MEP works for Tower ABCE	2_CONTROLLERS & SYSTEM INTERFACES_2.1	2_CONTROLLERS & SYSTEM INTERFACES_2.1		ASC/DDC for TPM (The mentioned quantities are tentative and they have to be verified as per the vendors requirement) Refer to BHE-PL-ZZ-SH-BM-9000-01 to 10 for I/O point schedule (Refer to the BMS schematic)	BOQ & IO Summary is not matching. We are quoting as per BOQ qty.	IO summary is given in BMS tender		
2	MEP works for Tower ABCE	2_CONTROLLERS & SYSTEM INTERFACES_2.3	2_CONTROLLERS & SYSTEM INTERFACES_2.3		Repeater panels as per OEM / system requirement	Qty is in Lot. Kindly confirm the number qty required.	Follow the BOQ & refer layouts		
3	MEP works for Tower ABCE	2_CONTROLLERS & SYSTEM INTERFACES_2.4	2_CONTROLLERS & SYSTEM INTERFACES_2.4		Hand held portable operator terminals(POT). These shall be required if the Controllers and Integration Units do not have a built in keypad on the facia. BACnet/IP shall have .Display with adjustable contrast	Qty is in Lot. Kindly confirm the number qty required.	Follow the BOQ & refer layouts		
4	MEP works for Tower ABCE	2_CONTROLLERS & SYSTEM INTERFACES_2.6	2_CONTROLLERS & SYSTEM INTERFACES_2.6		Third Party Integration	Qty is in Lot. Kindly share the IO Summary for 3rd party integrations.	IO summary is given in BMS tender		
Fire Protection System (FPS)									
1	MEP works for Tower ABCE	A_For Horizontal piping_SI No_01	A_For Horizontal piping_SI No_01		Providing, laying, jointing, testing and commissioning of following sizes of GI ERW - C (Heavy class) pipes conforming to IS-1239(S-3589 (heavy class) cut to required lengths with all accessories like all fittings (schedule 40 seamless fittings upto 150mm dia) including tees, elbows, reducers, union, flanges, rubber gaskets, GI nuts bolts, washer including supporting/fixing the pipe on floor / wall /ceiling with clamps, hangers (using anchor fasteners) as per specification.(For horizontal piping - Grooved flexible & rigid coupling type shall be used for all pipe sizes.) Fittings to be UL & FM Approved. G.I. pipe sleeve of suitable higher size shall be provided wherever the pipes are crossing the walls/floors and sealing the sleeves with glass wool in between & fire sealant compound at either end all as per Project Manager's / Consultants requirements including cutting holes and chases in brick, R.C.C work and making good the same to original conditions complete in all respects. Allow for builders work/sleeves associated with fire protection installation work. All hangers, clamps, brackets etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns as per the approved makes unless specified otherwise and then supply, installation of the same shall also be included for rates under this head. Welding of any kind on the galvanized support / hanger shall not be permitted. For pipe pressure rating refer specification and schematics. Pipes shall be coated with the two coats of Zinc chromate (yellow colour) and painted with two coats of synthetic enamel paint of approved make and shade No. 536 of IS:5 as per specifications and final approval of consultant.Allow for all protective and decorative painting and cleaning of pipes, fittings and equipment's.	Kindly Confirm the type of Fitting. 25mm to 50mm dia & Above 65mm dia.	Grooved fitting to be used		
2	MEP works for Tower ABCE	A_For Horizontal piping_SI No_4.18	A_For Horizontal piping_SI No_4.18		Supply installation testing and commissioning of (UL Listed) Flexible steel braided connection to sprinkler head consisting of sprinkler drop assembly comprised of a branchline adapter nipple, stainless steel flexible hose with two slip nuts, sprinkler reducer, and lightweight ceiling bracket components. The quantity indicated is tentative as this is a provisional item. Contractor to use the same only as per site requirement. 4.180.0.0 as per drawing	Can we consider 1000mm as per the OEM standards, Kindly confirm.	OK, 1m is RO		
3	MEP works for Tower ABCE	A_For Horizontal piping_SI No_5	A_For Horizontal piping_SI No_5		Multiport Valve (Test & Drain Valve) consisting of sight glass sectional drain valve, union with pressure gauge all complete and strictly as per drawing and schematics). The assembly will be factory fabricated. The necessary pipes for this assembly considered separately in the Pipes section.	32mm dia is not standard as per the OEM, Kindly confirm the std size.	32mm size is RO		
4	MEP works for Tower ABCE	A_For Horizontal piping_SI No_8	A_For Horizontal piping_SI No_8		Automatic Air release (Sprinkler system) Supply,installation,testing and commissioning of Brass body Automatic Air release valve with stop cock, screwed inlet connection and drain arrangement with 15mm dia valve, pressure gauge with gun metal stop cock complete with all accessories as required and conforming to IS 4736-1968.	Can we consider the Ball valve make as same as to Air release valve, Kindly confirm.	Follow makelist		
5	MEP works for Tower ABCE	C_DRENCHER SYSTEM FOR BASEMENT CAR PARK & FAÇADE PROTECTION IN OFFICE FLOORS_SI No_1	C_DRENCHER SYSTEM FOR BASEMENT CAR PARK & FAÇADE PROTECTION IN OFFICE FLOORS_SI No_1		Supplying, installing, testing and commissioning of 180° pattern type UL Approved Cast iron Deluge Valve assembly consisting of all related required controls ,trim assembly necessary bypass piping. 1.001.65 mm dia	As per the OEM Sds 65mm Dia is not available, Kindly confirm.	65mm dia is in RO		
6	MEP works for Tower ABCE	Valves	Valves		Non-Return valve.	Non return valve PN-20 is not available as per the approved make list, Can we consider Lehry make, Kindly confirm. If any deviations in values we not remarks along with our commercial offer	PN 20 is RO, follow makelist		
PHE									
1	MEP works for Tower ABCE	A_SANITARY FITTINGS & ACCESSORIES (ONLY INSTALLATION, TESTING & COMMISSIONING)	A_SANITARY FITTINGS & ACCESSORIES (ONLY INSTALLATION, TESTING & COMMISSIONING)		SANITARY FITTINGS & ACCESSORIES (ONLY INSTALLATION, TESTING & COMMISSIONING)	As per the BOQ, We understood that supply of all sanitary fixtures, CP Fittings, and SS Rack bolts / chair bracket are under Client scope. Kindly confirm the same.	Yes		

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1	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-A. CHILLER PLANT HIGHSIDE EQUIP-SL NO 1	COOLING TOWERS INDUCED DRAFT CROSS FLOW		Supply, installing, testing and commissioning of WATER COOLED CENTRIFUGAL CHILLERS test complete with 1000 TR - 5 Nos Working and 1 No Standby actual capacity single compressor with motor, VFD starter panel, water cooled condenser, insulated evaporator. Refer to PEPL MUM_PLE_ME_SCH_0001 for the chiller equipment schedule & Performance Characteristics. The chillers shall be configured to work on Constant Primary & Variable Secondary Flow Chilled Water System. Motor shall be suitable for 415±10% V, 50 cycles, 3 phase AC supply with efficiency of class IE3 and motor cable terminal box shall be suitable to connect Aluminium Cabling. Supply of adaptor box suitable for the chiller and suitable copper cabling & termination from adaptor box to the chiller. CHILLER PLANT OPTIMIZER SYSTEM Supply, installation, testing & commissioning of chiller plant Optimizer as per chillers requirement. The plant optimizer shall be suitable to control all the chillers, pumps & cooling towers being proposed and all suitable modules (I/O) shall have to be provided so individual Micro Processor controlled panels of machines can be synchronised for chillers and pumps operation and to enable sequential operation to achieve equal operating time and take care of load patterns 1. To control and operate the Primary only Variable speed chilled water system, Chillers and Cooling Towers. 2. To optimise the run time hours of all the pumps, Chillers and Cooling Towers. 3. To control and monitor the temperature of the chilled water and condenser water entering and leaving the chillers. 4. To control and operate the cooling towers and associated pumps 5. Chilled water energy usage for the different zones. 6. Interface with BMS system	1. Requesting you to kindly confirm whether the SITC of chillers is in Client or contractor scope. 2. Towards Chiller COP, IPLV, kW/TR and noise level shall we consider as per OEM standard. Requesting you to kindly confirm. 3. Kindly confirm that whether we need to consider the ASME stamped on vessels or not? 4. We presume that civil related works such as RCC foundation & supports shall come under the scope of civil contractor, hence we are not considering the same in our proposal.	1. Chiller SITC is in contractor's scope 2. Follow BOQ 3. Vessels will be ASME stamped 4. Client to confirm		
2	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-A. CHILLER PLANT HIGHSIDE EQUIP-SL NO 1	Chilled Water Pumps Vertical Inline		Supply, installation, testing and commissioning of COOLING TOWERS INDUCED DRAFT CROSS FLOW CTI approved for air-conditioning system. Each tower shall be complete with basin (with connection for drain, makeup water overflow, quick fill/float valve etc), casing, distribution system, filling, louvers, ladder, fan and motor. Motor shall be suitable for 415±10% volts, 50 cycles, 3 phase power supply with efficiency of class IE3, Isolator panel with MCCB with suitable enclosure for cooling tower and earthing, shall also be included. Cooling Tower for air-conditioning system shall be selected on the basis of leaving water temperature 31°C & entering water temperature 38° C, ambient wet bulb 28° C and nominal capacity shall be as per Equipment Schedule Ref: PEPL_MUM_PLE_ME_SCH_0006. Cold water basin shall be a deep sump and made out of 5 mm thick FRP construction and UV stabilized on which cooling tower super structure shall be supported. Fillings shall be made of corrosion proof and rigid PVC film in honeycomb design and arranged in square / rectangular form. The cooling tower shall be quiet operating, and shall produce an overall sound no higher than 80 dB at a distance of 1.5 m.	1. Requesting you to kindly confirm whether the SITC of CPM is in Client or contractor scope. 2. We presume that cabling for CPM and field devices shall be under the scope of BMS agency. Please confirm.	1. SITC of CPM is in contractor's scope 2. CPM wiring will be in HVAC contractor's scope, connecting with BMS will be in BMS scope		
3	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-A. CHILLER PLANT HIGHSIDE EQUIP-SL NO 3	CHILLED WATER PUMPS & HEAT EXCHANGERS		Supply, installation, testing and commissioning of COOLING TOWERS INDUCED DRAFT CROSS FLOW CTI approved for air-conditioning system. Each tower shall be complete with basin (with connection for drain, makeup water overflow, quick fill/float valve etc), casing, distribution system, filling, louvers, ladder, fan and motor. Motor shall be suitable for 415±10% volts, 50 cycles, 3 phase power supply with efficiency of class IE3, Isolator panel with MCCB with suitable enclosure for cooling tower and earthing, shall also be included. Cooling Tower for air-conditioning system shall be selected on the basis of leaving water temperature 31°C & entering water temperature 38° C, ambient wet bulb 28° C and nominal capacity shall be as per Equipment Schedule Ref: PEPL_MUM_PLE_ME_SCH_0006. Cold water basin shall be a deep sump and made out of 5 mm thick FRP construction and UV stabilized on which cooling tower super structure shall be supported. Fillings shall be made of corrosion proof and rigid PVC film in honeycomb design and arranged in square / rectangular form. The cooling tower shall be quiet operating, and shall produce an overall sound no higher than 80 dB at a distance of 1.5 m.	1. Requesting you to kindly confirm whether we need to consider single cell or multi cell cooling tower. 2. Kindly confirm any height constrain in terrace for cooling tower. 3. BOQ calls for cross flow cooling tower whereas Technical specification calls for counter/cross flow cooling tower. Requesting you to kindly confirm. 4. Shall we consider NON CTI certified cooling tower. Requesting you to kindly confirm. 4. We presume that foundation, cooling tower pedestal and any structural support for cooling tower shall come under scope of civil contractor, hence we are not considering the same in this line item. 5. Shall we consider cooling tower with deflection cowl in order to match the sound level. Requesting you to kindly confirm. 6. Since panel is quantified separately in Electrical BOQ section , hence we are not considering the same in this line item.	1. No. of cells shall be based on the selection. No height constraint 2. Consider cross flow type cooling tower. 3. Follow BOQ 4. Client to confirm 5. OK 6. OK	4. Please refer the responsibility matrix for the same.	
4	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-A1. CHW PUMPS & PHEX-SL NO 1,2	Pressurized Expansion Tank, Pressurization Unit		Supplying, installing, testing and commissioning of Chilled Water Pumps Vertical Inline Centrifugal Pumps consisting of adjustable frequency drive (VFD) for each pump, one dedicated microprocessor based pump controller per zone housed within the enclosure of one of the VFD with parallel pumping software duly down loaded and multiple differential pressure sensor / transmitters along with complete control cabling from transmitter to VFD Panel as per the site requirement. The entire system along with chilled water pumps as described above must be sourced from single manufacturer only, to ensure unit responsibility. Necessary factory test certificates for entire system shall be submitted along with the equipment. Sensor / Transmitters as described in specification. Complete assembly including pump, motor and drive shall be designed to limit THDI to less than 5%. System shall be BMS compatible with open protocol communication port, so that complete data can be transported to remote BMS console. The pump motor shall be suitable for 415 ± 10% volts, 50 cycles, 3 phase power supply. The motors of secondary chilled water pumps shall be suitable for operation on adjustable frequency drive. The motor selected shall be energy efficient having IE3 efficiency class rating. (The quoted price shall include cost of thermal insulation and cladding.) Supply Installation Testing and commissioning of Imported portion of VFD, Pump Controller & Sensor Transmitters. Refer to the chilled water schematic for the Location and number of Differential pressure sensor. Inspection and Factory Test Prior to Shipment, Pumps shall be subjected to witness Factory Test by owners representative (4 Persons) to ascertain the design requirements. Load testing and performance testing (Air Flow, Static, Power and efficiency) of the AHU shall be done as per standard procedure. The Load test report & performance test report to be submitted to owners representative during inspection. All expenses inclusive of airfare, transportation, accommodation, food etc. related to witness test will be borne by the contractor	1. As per BOQ and Equipment schedule, there is a discrepancy in water flow rate & HEAD, working configuration of pump (Tag No CDWP-TE-802-01 to CDWP-TE-802-06). Requesting you to kindly confirm. 2. We are not considering any factory witness test in our proposal. 3. Technicals specification calls for primary , condensor water pumps are constant flow where as secondary pumps are with variable speed so we are considering VFD for secondary pumps only. 4. If Pump mounted VFD is required, shall we consider pump with sensorless technology. Requesting you to kindly confirm. 5. If Panel mounted VFD is required, we presume that panel is quantified separately in Electrical BOQ Section, hence we are not considering pump electrical starter panel in this line item. 6. Towards THDI level shall we consider less than 35% for current and less than 5% for voltage. Kindly confirm. 7. Towards Pump RPM, Efficiency and Noise level shall we consider as per OEM standard. Requesting you to kindly confirm. 8. Since BOQ & Technical specification is silent on pump pressure rating, requesting you to kindly confirm. 9. We presume that civil related works such as RCC foundation & supports shall come under the scope of civil contractor, hence we are not considering the same in our proposal. 10. Technical spec & Equipment schedule calls for Inertia block with isolators since it is vertical inline pumps so we are not considering any type of inertia block with isolators in our	1. Follow BOQ 2. Consider for 1 pump 3. Secondary pumps will be variable speed 4. Follow BOQ 5. OK, refer the electrical section 6. Follow BOQ & specs 7. Follow BOQ & specs 8. Follow specification pg.no. 84 9. Client to confirm 10. OK	9. Please refer the responsibility matrix for the same.	
5	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-A1. CHW PUMPS & PHEX-SL NO 1,2	Chemical dosing unit		Supplying, installing, testing and commissioning of Variable Tertiary Chilled Water Pumps Vertical Inline Centrifugal Pumps consisting of adjustable frequency drive (VFD) for each pump, one dedicated microprocessor based pump controller per zone housed within the enclosure of one of the VFD with parallel pumping software duly down loaded and multiple differential pressure sensor / transmitters along with complete control cabling from transmitter to VFD Panel as per the site requirement	1. Towards Variable Tertiary Chilled Water Pumps BOQ calls for Vertical inline pumps whereas equipment schedule calls for Horizontal end suction pumps kindly confirm on the same.	Vertical inline pumps will be there		

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6	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-A1A. CHILLER PLANT HIGHSIDE EQUIP-SL NO 1,2	Side stream filtration unit		Supply, installation, testing & commissioning of Pressurized Expansion Tank, Pressurization Unit with vacuum cylinder c/w break tank as well as additive tank for chemical dosing as detailed in specification. High and low pressure alarm setting shall be selectable by the user. Expansion and pressurization of the chilled water system to be provided by an integrated pump with pressurized expansion tank(s) for Approximate system volume 50,000 liter at Max ambient temp 40 Deg C and total Static height above the expansion tank 100 M. The total Volume to be checked by vendor. The complete system shall be suitable for 16kg/cm2 as maximum operating pressure. The system shall consist of following : 1) Pressurization Unit with Vacuum Degasser & Additive Tank c/w two Pump ,vacuum cylinder, Break tank, additive tank for anti-corrosive chemicals, float valve, over flow connection, pressure sensor for system and for vacuum cylinder, solenoid valve, isolation Valve, NRV, Drain Valve, flood protection in the event of a serious leak, Pipe fittings c/w Controller & display, full unit duly tested, housed in a mild steel cabinet. 2) Pressurized Expansion Tank 1200 Liter made from top quality steel and finished with a gleaming red epoxy-powder coating c/w high quality butyl rubber removable bladder, Nitrogen filling for longer maintenance of pre pressurization. Expansion Tank shall have third party certificate (TUV / Iloyds registered) Vessel 100-1000 liters in accordance with EN13831/ 1200-8000 liters : in accordance with AD2000. 3) Controller : Controller for pressurization as well as vacuum degassing with IP X4 rating, bright LED display showing scrolling messages and status of the main operating components, top up function, flood limiter/shut down function in event of serious leak. in case of any fault, controller shall display fault code and generate alarm. Expansion tank as per the manufacturer without any ASME stamping. Refer Technical Specification Ref - 4 & 8 Pressurised expansion tank with inbuilt vacuum based	1.As per OEM selection, shall we consider pressure rating of PN 10 instead of PN 16.Requesting you to kindly confirm. 2.Shall we consider Expansion tank as per IS standard.Requesting you to kindly confirm. 3.We seek your approval for below proposed additional make. Expansion Tank & Air Separator - Emerald,Anergy.	1.Follow BOQ 2.Follow BOQ & specification 3.Follow makelist		
7	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-A1A. CHILLER PLANT HIGHSIDE EQUIP-SL NO 7	TREATED FRESH AIR UNITS		Supplying, installation, testing and commissioning of Chemical dosing unit as shown in schematic with necessary arrangement as per specification for Chilled water & condenser water circuit.	1.Towards the chemical dosing system ,we are considering 1st dose of chemicals in our proposal , kindly confirm whether 1 year chemicals required for both chilled water & Condenser water side. 2.To wards the chemical dosing system , the chemicals used for dosing are as per OEM selected chemicals only.	1.1 year dosing to be considered 2.Follow Specification		
8	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-A1A. CHILLER PLANT HIGHSIDE EQUIP-SL NO 6	AIR HANDLING UNITS		Supplying, installation, testing and commissioning of Side stream filtration unit as shown in schematic with necessary arrangement as per specification for Chilled water & condenser water circuit.	1.We seek your approval for below proposed additional make. Expansion Tank & Air Separator - Emerald,Anergy. 2.Shall we consider Side stream filtration unit as per IS standard.Requesting you to kindly confirm.	1.Follow makelist 2.Follow specs		
9	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-B.TFA-SL NO 1	FAN COIL UNIT		Supplying, installing, testing and commissioning of TREATED FRESH AIR UNIT WITH Heat Recovery Wheel (HRW) with 50mm thick double skin construction with Thermal Break Profile as per specification. The unit shall comprise of filter section, fresh air inlet / exhaust air outlet, fan section with EC plug type fan with motor and drive package including the variable frequency drive housed in the control panel. The unit shall be have MERV-15 + PCO + UV LAMP. Prefilter should be MERV-8.Refer to the equipment schedule PEPL_MUM_PLA_ME_SCH_0004 and specification for air filtration requirements. Min 6 Row Deep water coil of copper tube & aluminium fins construction and base frame, condensate drain connections with U trap. Coil size shall be selected for a maximum face velocity of 500 feet/minute. Double skinned panels shall be made of galvanized steel, pressure injected with foam insulation 50 mm thickness having density 40 kg/m3	1.As per Technical specification panels shall be of double skin construction with both inner and outer steel sheets being minimum 1.0mm thick pre coated & plasticized instead shall we consider outer skin of 0.8mm thick pre coated GI and inner skin of 0.6mm thick plain GI.Requesting you to kindly confirm. 2.As per BOQ TFA with MERV 8 & MERV 15 filter+PCO+UV lamp but whereas technical specification calls for MERV 8 & MERV 13 /UV Lamps as mentioned in schematic drawing for TFA in Page no 136 kindly confirm which one we need to refer. 3.Towards noise level shall we consider upto 70±5 dB @ 1m.Requesting you to kindly confirm. 4.Towards fan outlet velocity we are considering upto 10m/s in our proposal. 5.Towards cooling coil shall we consider coil with MS header and aluminium fins without hydraulic coating.Kindly confirm. 6.Kindly confirm whether pre commissioning filter & spare filters i.e additional MERV 8 & MERV 15 are required. 8.To words BOQ calls Motor shall be EC plug type fan whereas equipment schedule & technical specification calls for Backcurved/Aeroflo/Plug fans kindly confirm on the same which one we need to select. 8.As OEM recommendation, UV lamp without PCO can be considered.Requesting you to kindly confirm. 9.Since TFA control panel is quantified separately in Electrical BOQ Section L SI No 1, hence we are not considering the same in this line item. 10.If the selection of motor with EC motorwith plug fan then VFD is not required kindly confirm on the same. 11.Towards harmonic level shall we consider less than 35% for current and less than 5% for voltage.Kindly confirm. 12.We are not considering any factory witness test in our proposal. 13.Shall we consider drain pan of 22G SS304 with 19mm thick nitrile rubber insulation.Kindly confirm. 13.We seek your approval for below proposed additional make. TFA - Luftek	1.Follow BOQ 2.Follow BOQ 3.Follow specs 3.Follow specs 5.Follow BOQ & specs 6.Pre commissioning pre filter will be considered 7.Follow BOQ 8.Follow BOQ 9.OK 10.Yes 11.Follow BOQ 12.Consider for 1 equipment 13.Follow BOQ & Specs 14.Follow makelist 15.Follow makelist		
10	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-B.AHU NO 1	BTU Meters		Supplying, installing, testing and commissioning of double skin construction draw thru type AIR HANDLING UNITS (CHV AHUs) with Thermal Break Profile as per specification, each complete with motor and drive package. The unit shall be have MERV - 8 + PCO + UV LAMP. Refer to the equipment schedule and specification for air filtration requirements. Double skinned panels shall be made of galvanized steel, pressure injected with foam insulation 50 mm thickness having density 40 kg/m3. Min 6 Row Deep water coil of copper tube & aluminium fins construction and base frame, condensate drain connections with U trap. coil size shall be selected for a maximum face velocity of 500 feet/minute. The fan shall be selected for minimum 70% efficiency and shall be EC plug type. The motor selected shall be energy efficient having IE-3 efficiency rating at full load with motor rating as per specification. Motor shall be suitable for 415±10% volts, 50 cycles, 3 phase AC supply. Fan motor assembly (as whole) shall be statically and dynamically balanced to grade G 6.3 as per ISO 1940/AMCA 204-3. Duct flexible connector (fire proof) shall be provided at outlets along with volume control dampers. For units with VFD, complete equipment shall be designed to limit THDI to less than 5%, Cost to include cable from panel to Unit. The Fan shall be mounted on Spring Isolators as per the specification The unit shall have terminal box outside the AHU with protective guard near the AHU door.	1.We are considering MERV 8 +PCO+UV Lamp only in our proposal. 2.Towards noise level shall we consider upto 70±5 dB @ 1m.Requesting you to kindly confirm. 3.Towards fan outlet velocity we are considering upto 10m/s in our proposal. 4.Towards cooling coil shall we consider coil with MS header and aluminium fins without hydraulic coating.Kindly confirm. 5.Kindly confirm whether pre commissioning filter & spare filters i.e additional are required. 6.As OEM recommendation, UV lamp without PCO can be considered.Requesting you to kindly confirm. 7.We are not considering any factory witness test in our proposal. 8.To words BOQ calls Motor shall be EC plug type fan whereas equipment schedule & technical specification calls for Backcurved/Aeroflo/Plug fans kindly confirm on the same which one we need to select. 9.Shall we consider drain pan of 22G SS304 with 19mm thick nitrile rubber insulation.Kindly confirm. 10.Since AHU control panel is quantified separately in Electrical BOQ Section L SI No 1, hence we are not considering the same in this line item. 11.If the selection of motor with EC motor with plug fan then VFD is not required kindly confirm on the same. 12.Towards THDI level shall we consider less than 35% for current and less than 5% for voltage.Kindly confirm. 13.We seek your approval for below proposed additional make. AHU,CSU - Luftek 14.Since NIT is silent on approved make for UV lamp, we seek your approval for below proposed make.	1.Follow BOQ 2.Follow BOQ & specs 3.Follow BOQ & specs 4.Follow BOQ & specs 5.Pre commissioning pre filter will be considered 6.Follow BOQ 7.Consider for 1 equipment 8.Follow BOQ 9.Follow BOQ & specs 10.OK 11.Yes. 12.Follow BOQ & specs 13.Follow makelist 14.BH to revert		

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	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.							
11	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-D.FCU sl no-1	Chilled Water Pipes		Supply, installing, testing and commissioning of horizontal FAN COIL UNITS blow thru type, each complete with 3/4 rows deep Chilled Water Coil One or more centrifugal fans Cleanable fabric filters Main condensate drain and extended condensate SS drain pans, condensate drain connections with U trap. Fan Coil Units to be provided with condensate drain pump.	1.Requesting you to kindly confirm whether we need to consider AHRI or Eurovent certified FCU. 2.We are considering FCU of single skin construction in our proposal. 3.Towards FCU valve package, we presume that all the valves are quantified under valves section, hence we are not considering the same in our proposal. 4.If the valve package to be considered in the same line item, shall we consider as per OEM standard.Kindly confirm. 5.We presume that the supply of thermostat shall be under BMS scope, hence we are not considering the same in our proposal. Kindly confirm. 6. Towards noise level shall we consider as per OEM standard, requesting you to kindly confirm. 7.As per equipment schedule kindly confirm whether we need to consider 10% extra duty on sensible & total cooling and 10% extra on dehumidified air quantities. 8.Equipment schedule calls for single phase motor with IE 3 efficiency, as per OEM recommendation we are considering FCU with standard motor efficiency in lieu of IE 3 motor. 9.We seek your approval for below proposed additional make.	1.Eurovent 2.Follow specification 3.Follow BOQ, FCU Valves are considered in FCU line item 4.Follow specs & makelist 5.Supply of thermostat will be in HVAC scope 6.Follow specs 7.Yes. 8.Follow schedule 9.Follow makelist			
12	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-E.BTU METER & THERMOSTAT sl no-2	Condenser Water Pipes		Supply, installation, testing and commissioning of BTU Meters with necessary temperature measuring Instruments (2 per BTU Meter) (No need to quantify these temp sensors separately), test ports controls (2 per BTU Meter) (No need to quantify these temp sensors separately) and arrangement as shown in the schematic. The Cooling/BTU meter shall be MID approved and comply with EN 1434 standard. The MID approval must be displayed on the meter. (MID - Measuring Instruments Directive)Minimum Pressure Rating: 16 bar Shall be Ultra sonic - Side Wall mounted, with RTD Sensor and consisting of M-bus protocol which would support BMS. Selection The flow meter shall be selected such that the expected flow shall be below the meter's Nominal flow rate. The Minimum flow (Q) measurable shall be 1/100 of the meter's Nominal flow (Qp). The flow meter shall be selected such that the below criteria are fulfilled; Threaded meters: The outer connection on the meter shall be of the same size as the pipe size. In no condition meter shall be selected that outer connection on meter is one size lower than the pipe size. This is because in threaded meter actual measuring is done in one size lower chamber inside the meter. This will ensure longer life of meters as no cavitations happen in the meter. Flanged meters: The meter shall be of same size as pipe size or maximum one size lower than pipe size. In no condition meter shall be selected two sizes lower than pipe size. This will ensure longer life of meters as no cavitations happen in the meter. The Minimum flow measurable by the meter shall be 1% of the Nominal flow of the meter. Q/Op = 1:100 (Q - Minimum flow, Qp - Maximum flow)	1.We presume that integrator shall come under the scope of BMS, hence we are not considering the same in our proposal. 2.Towards BTU meter we are considering Inline type for upto 150mm dia and clamp on type for above 150mm dia in our proposal. 3.We seek your approval for below proposed additional make BTU Meter - Engelmann,Shenitech,Omicron	1.BH to revert 2.Follow Specification 3.Follow makelist			
13	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-F.PIPING sl no-1	CONDENSATE DRAIN PIPING		CHILLED WATER PIPING Supplying, installation, testing and commissioning of the following MS ERW class C pipes with specified wall thickness as per IS3589, cut to required lengths and supporting arrangements with all welded joints, providing and fixing in position necessary fittings like bends, elbows, tees, reducers, mating flanges & sockets for BAS, fabricated to size and shape in workshop or brought out ready made where available of acceptable quality.Shoe fittings in pipe header to be factory fabricated. The rate shall be inclusive of Rubber support inserts having steel reinforcement, with overall material density of 1400 Kg/m³ for the pipes where the Supports would have contact with the chilled water pipes. All hangers, clamps, brackets etc. shall be modular hot dip galvanized having galvanizing thickness 75 microns as per the approved makes unless specified otherwise and then supply, installation of the same shall also be included for rates under this head. Supplying, installing, testing and commissioning of Seismic Supports for vertical riser piping take care of wind , budding settlement and seismic effect. The contractor shall bring on board the seismic contractor & submit all the required calculations & CONDENSER WATER PIPING Supplying, installation, testing and commissioning of MS ERW Class C pipes with specified wall thickness as per IS3589, cut to required lengths and supporting arrangements with all welded joints, providing and fixing in position necessary fittings like bends, elbows, tees, reducers, mating flanges & sockets for BAS, fabricated to size and shape in workshop or brought out ready made where available of acceptable quality.	1.Towards pipe supports for pipe upto 80mm dia we are considering U clamp support and MS supports for above 80mm dia. 2.Towards pipe to pipe joints, upto 50mm dia shall we are considering socket welding and above 65mm dia shall we consider butt welding.Requesting you to kindly confirm. 3.Technical specification calls for grooved fittings instead we are considering welded joints for pipes. 4.Towards pipe to valves, shall we consider Table E flanges.Requesting you to kindly confirm. 5.We presume that wall opening,clipping and any civil related works shall come under the scope of civil contractor, hence we are not considering the same in this line item. 6.We presume that SITC of fire sealant wherever pipes pass through a fire rated wall / slab openings shall come under the scope of fire agency, hence we are not considering the same in this line item. 7. If fire sealant is in our scope , requesting you to please share us a separate line item for same. 8.Please note that we are not considering the painting for chilled water since all chilled water pipes are insulated. Kindly confirm.	1.Follow BOQ 2.Follow BOQ & specs 3.Follow BOQ 4.BH to revert 5.Client to confirm 6.Under respective vendor scope 7.Sealant for service crossing in wall to be in respective service line item 8.Follow BOQ & specs	5. Please refer the responsibility matrix for the same.		
14	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-F.PIPING sl no-1,2,3	Electronic Pressure Independent Balancing & Control Valve (PIBCV)		CONDENSATE DRAIN PIPING Providing and fixing in position the following UPVC Class-3 (IS 4985) pipes cut to required lengths with 10mm of Nitrile rubber insulation, support to be with pipe clamp & rubber gasket and providing and fixing in position the necessary elbows, tees and reducers as per specifications for AHU / ECL drain.	1.We are considering UPVC pipe of density 6 kgf/cm2 in our proposal. 2.As per OEM standard, we are considering 9mm thick nitrile rubber insulation instead of 10mm thick.	1.Follow BOQ & specs 2.Follow BOQ			
15	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-F.PIPING sl no-4	Butterfly Valves		Supply, installation, testing and commissioning of Electronic Pressure Independent Balancing & Control Valve (PIBCV) PN16 with independent type dynamic balancing valve for AHUs / TFA with integrated 2 way modulating control valves in a single body. The actuator shall be capable of accepting 2-10 VDC / 4-20 mA electric signal and shall provide similar transduced feedback output signal to control system for flow measurement. Contractor shall show actual flow measurement v/s flow indicated by valve for atleast 5% of the valves.	1.Shall we consider PIBCV with Non spring return type actuator.Kindly confirm. 2.We presume that supply of thermostat shall come under scope of BMS, hence we are not considering the same in our proposal. 3.Can we consider PIBCV with Non Delta P controller. 3.We seek your approval for below proposed additional make. PIBCV - Honeywell,Flowcon	1.Follow BOQ & specs 2.Thermostat will be supplied by HVAC vendor, follow BOQ 3.Follow BOQ & specs 4.Follow makelist			
16	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-G.VALVES 1	Manual Balancing Valves		Supply, installation, testing commissioning of Butterfly valve (Body : Cast Iron, Shaft: SS, Disc: D.I Liner : HT - EPDM) Including rubber gasket, flanges, nuts, bolts, washers & painting with necessary mating flanges to suit the specified piping material and complete as required and in accordance to the technical specification. To suit pressure rating as specified on schematics and specification. (Operating Pressure for all Valves & Accessories in Express Risers(transfer Lines)). Refer Technical Specification Ref : 2.9.3 Butterfly Valves	1.We seek your approval for below proposed additional make. Butterfly Valves - Advance,Honeywell	1.Follow makelist			
17	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-G.VALVES 2	Ball Valve		Supply, installation, testing commissioning of Butterfly valve (Body : Cast Iron, Shaft: SS, Disc: D.I Liner : HT - EPDM) Including rubber gasket, flanges, nuts, bolts, washers & painting with necessary mating flanges to suit the specified piping material and complete as required and in accordance to the technical specification. To suit pressure rating as specified on schematics and specification. (Operating Pressure for all Valves & Accessories in Express Risers(transfer Lines)). Refer Technical Specification Ref : 2.9.3 Butterfly Valves	1.We seek your approval for below proposed additional make. Butterfly Valves - Advance,Honeywell	1.Follow makelist			
18	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-G.VALVES 3	Check Valves		Supplying, installation, testing and commissioning in position the following double regulating Manual Balancing Valves with counter flanges, nuts & bolts and gasket to suit pressure rating as per specification. The valves shall be suitable for following pipe sizes with necessary fittings:	1.Since BOQ & Technical specification is silent on MOC, shall we consider Gun metal body for dia upto 50mm and Cast iron body with SS disc for dia above 50mm.Kindly confirm. 2.We seek your approval for below proposed additional make. Balancing Valves - Advance,Honeywell	1.BH to revert 2.Follow makelist			
19	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-G.VALVES 5	"Y" strainers		Supply, installation, testing commissioning of full way lever operated forged brass Ball valve of Brass body with SS ball with threaded / flanged joints complete with nuts, bolts, gaskets, washers, necessary mating flanges, to suit the specified piping material etc. as specified in the specification. To suit Pressure rating as specified in schematics and specification. Refer Technical Specification Ref : 2.8.2 Ball Valves	1.We seek your approval for below proposed additional make. Ball Valves - RB,ITAP	1.Follow makelist			
20	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-G.VALVES 6	Flexible Connections		Supply, installation, testing commissioning of Cast Iron Non-Return valve , seat metal to metal, Dual plate SS 316 rings pins, threaded/flanged, fixed screwed with flange accessories and with necessary mating flanges to suit the specified piping material and in accordance to the technical specification. To suit pressure rating as specified on schematics and specification. Brass up to 40mm and CI above 50mm dia can be considered. (Operating Pressure for all Valves & Accessories in Express Risers (transfer Lines)). Refer Technical Specification Ref : 2.8.2 Ball Valves	1.Shall we consider check valve of PN 16 rating.Kindly confirm. 2.Towards check valve disc shall we consider SS disc instead of CI.Kindly confirm. 3.We seek your approval for below proposed additional make. Check Valves -Honeywell	1.Follow BOQ 2.Follow BOQ 3.Follow makelist			

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21	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-G.VALVES 7	Motorised Globe valve		Supply, installation, testing and commissioning of brass (up to 40mm dia)/Cast iron (above 40mm dia) "Y" strainers conforming to IS fixed screwed / flanged with removable type SS screen with 3.175 mm perforations with flaring accessories, mating flanges to suit the piping material specified and in accordance with specification. To suit the pressure rating as specified on schematics and specification. Strainer to be with drain valve for sizes 65 mm and above dia. Refer Technical Specification Ref - 2.9.6 V-Strainers	1.Towards Y strainer MOC, shall we consider MS body instead of CI body. Kindly confirm. 2.We seek your approval for below proposed additional makes. Y Strainer - Emerald, OS Eng	1.Follow BOQ 2.Follow makelist		
22	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-G.VALVES 9	Auto air vent		Supply, installation, testing and commissioning of Flexible Connections (multi Arch type) of suitable pressure rating to be installed at pump suction, discharge and at pipes crossing building expansion joints with flanges and necessary fittings as per specification. The flexible connectors shall be reinforced using high tensile synthetic fibre	1.Shall we consider single arch rubber bellows. Requesting you to kindly confirm.	Follow BOQ		
23	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-G.VALVES 11	Flow switch		Supply, installation, testing and commissioning of Motorised Globe valve with Linear fast acting type actuator for the Bypass line in the Primary chilled water pumping system (PN20). The actuator shall be capable of accepting 2-10 VDC / 4-20 mA electric signal and shall provide similar transduced feedback output signal to control system for flow measurement. Contractor shall show actual flow measurement v/s flow indicated by valve for atleast 5% of the valves. The valves shall be suitable for following pipe sizes with necessary fittings.	1.Since NIT is silent on approved make for Motorised Globe valve, requesting you to kindly share the same.	1.BH to revert		
24	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-H.FIELD DEVICES 1	Temperature sensor		Supply, installation, testing and commissioning of Auto Air vent with brass body and non ferrous float for supply risers, with isolation valve, and accessories with necessary mating flanges to suit the specified piping material and in accordance with specification. Refer Technical Specification Ref - 5.3.12 Automatic Air Vents	1.We seek your approval for below proposed additional make. Auto Air Vent - Anergy	Follow makelist		
25	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-H.FIELD DEVICES 2	Thermowell		Supply installation testing & commissioning of Flow switch for chilled water & condenser water system compatible to connect with the chiller and chiller plant manager. Refer Technical Specification Ref - 6.13 Flow Switch for Water	1.We seek your approval for below proposed additional make. Flow Switch - Johnson	Follow makelist		
26	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-H.FIELD DEVICES 3	Tube axial fan/Smoke spill axial fan		Providing and fixing in position Dial type thermometers with thermowells . Thermometers shall be dial type 100 mm dia or V form industrial type. Body shall be aluminum alloy, anodized gold-coloured surface. The casing shall be adjustable sideways for reading from the front. The glass capillary shall be triangular in shape with blue mercury filled in glass for better visibility. Scale of reading shall be of the range 0°C to 60°C & +32°F to 150°F. Graduation of scale shall be 1° in both readings. Range of scales shall be 30-120 degree F (0-50 degree C) for air conditioning applications of cooling only. Quality has been included in Schedule of Quantities. Thermometer shall be suitable for 15 mm connections. Thermometer for chilled water shall be with long stem, so that thermometer is removable without damaging the insulation. M.S socket to be welded on pipes shall be provided with thermometer. Thermometers shall be installed on chilled water supply and return at each air handling unit, supply and return at each chiller and condenser as shown on the Drawings and included in Schedule of Quantities. Refer Technical Specification Ref - 6 Automatic Controls & Instruments under "Instruments"	1.Shall we consider thermometer dial MOC of SS construction. Requesting you to kindly confirm.	Follow BOQ		
27	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-H.FIELD DEVICES 6	DUAL SPEED JET FANS Impulse type		Providing and fixing of Testpoints (thermowells) for BMS devices in coordination with BMS vendor. Refer Technical Specification Ref - 6.2 Immersion Temperature Sensors	1.We seek your approval for below proposed additional make. Test Point - Anergy	Follow makelist		
28	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-I.FANS-SI no 1,2	CO sensors		Supply, installation, testing and commissioning of TUBE AXIAL FANS . The motor shall be suitable for 415 V + 10% volts three phase 50cycle. The fan shall be equipped with mounting bracket inlet cone & outlet cone required at discharge. Fans shall be selected for minimum 65% efficiency. Fan shall be with flexible fire proof connection at inlet and outlet. The motor selected shall be energy efficient having IE3 efficiency class rating.	1.Since technical specification tube axial fan casing construction is 14 G steel sheet, shall we consider axial fan of MS powder coated casing. Requesting you to kindly confirm. 2.Towards fan outlet velocity, we are considering up to 20m/s in our proposal. 3.BOQ calls for sound level shall not exceed 75dB @ 3m whereas equipment schedule calls for sound level shall not exceed 55dB @ 1m. Kindly confirm. 4.Towards pressurization fans shall we consider with class F motor whereas Basement ventilation shall we consider Class H in our proposal kindly confirm. 5.We presume that control panel for Axial fans is quantified separately in Electrical BOQ section L SI No 1, hence we are not considering the same in this line item. 6.We seek your approval for below proposed additional makes. Axial Fans - Air Flow, Elta, Nicotra	1.Follow BOQ & specs 2.Follow specs 3.Follow BOQ 4.OK 5.OK 6.Follow makelist		
29	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-I.FANS-SI no 3	Cabinet Fans with single skin /INLINE CENTRIFUGAL DUCT FANS		Supply, installation, testing and commissioning of DUAL SPEED JET FANS impulse type including motor, drive, starter panels, Sound attenuators at inlet and outlet, vibration isolators etc. casing shall be suitable for smoke exhaust application. Motor shall be mounted inside and shall have fire rating of 250°C for 2 hours. Fans shall be suitable for rating as per ISO-1940/AMCA 204-3. All motors shall be IE-3 class or better suitable for 415 V +/- 10% volts three phase 50cycle, AC supply as per specifications. Fan and motor shall be fire rated as specified in the spec (Exact quantities for jet fans and their locations shall be decided based on CFD Analysis by Vendor)	1.Since BOQ & technical specification is silent on casing construction, requesting you to kindly confirm. 2.We are considering Jet fan of standard motor efficiency instead of IE 3 motor in our proposal. 3.Towards fan outlet velocity, Thrust shall we consider as per OEM selection. Requesting you to kindly confirm. 4.Equipment schedule calls for noise level shall not exceed 55 dB @ 1m instead shall we consider as per OEM standard. Kindly confirm. 5.We seek your approval for below proposed additional make. Jet Fans - Air Flow, Elta, Nicotra	1.BH to revert 2.BH to revert 3.Follow specs 4.BH to revert 5.Follow makelist		
30	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-I.FANS-SI no 4.1	Kitchen Exhaust Air Fan with Scrubber unit		CO Sensors currently shown in the drawing for the Car parking are indicative, exact quantity to be confirmed based on the CFD analysis	1.We are considering CO sensor without display. 2.Since NIT is silent on approved make, we seek your approval for below proposed make. CO Sensor - Greystone, Omicron, Honeywell	1.OK 2.BH to revert		
31	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-I.FANS-SI no 6,7	General		Supplying, installing, testing and commissioning of Cabinet Fans with single skin construction as per specifications, each complete with squirrel cage induction motor (mounted internally), DIDW centrifugal fan, belt drive and vibrations isolators. Motors shall have IE-3 efficiency rating shall be suitable for 415 + 10% volts, 50 cycles, 3 phase AC supply. Fan static pressure shall be as indicated. Fan-motor assembly (as whole) shall be statically & dynamically balanced to grade G6.3 as per ISO-1940/AMCA 204-3. The motor shall have IE3 class efficiency rating. Fan shall be with flexible fire proof connection at inlet and outlet.	1.BOQ calls for sound level shall not exceed 75dB @ 3m whereas equipment schedule calls for sound level shall not exceed 55dB @ 1m. Kindly confirm.	Follow BOQ		
32	BOQ MEP TOWER ABCE	BOQ MEP TOWER ABCE-I.FANS-SI no 8	General		Kitchen Exhaust Air Fan with Scrubber unit Supply, installation, testing and commissioning of Kitchen Exhaust air fan with Scrubber Unit (Electro static precipitator & Activated carbon filter bank units or Electro static precipitator unit only) for the treatment of the exhaust from the kitchen as per specifications and equipment schedule.	1.BOQ calls for sound level shall not exceed 75dB @ 3m whereas equipment schedule calls for sound level shall not exceed 55dB @ 1m. Kindly confirm.	Follow BOQ		
33	BOQ MEP TOWER ABCE		General		Drawings	1.Kindly share the autocad drawings	CAD layouts are already shared in tender	Files have already been shared on 07/12/2024	
34	BOQ MEP TOWER ABCE		General		BOQ	1.Kindly recheck on BOQ Quantity since some BOQ line items are Reference (Formula error)	Follow BOQ QTY	The revised BOQ with corrections will be included in the addendum	
35	BOQ MEP TOWER ABCE		General		Equipment Schedule	1.Kindly share the Tower B Equipment Schedule.	Will be shared	To be added in the addendum	
36	BOQ MEP TOWER ABCE		General						
37	BOQ MEP TOWER ABCE		General		Minimum Billing for Grilles, Damper and Fire damper	1. Please note that as per OEM standard minimum billing of Grilles/Diffuser shall be 0.2 Sqmtr and Fire damper shall be 0.1 Sqmtr.	Follow Specs		
38	BOQ MEP TOWER ABCE		General		Labour Cess/ BOCW	1. We presume that BOCW/Labour Cess Shall be under the scope of Client, hence we have not considered the same in our proposal.	Client to confirm	Mention it separately in the summary sheet. Do not include in the autobid rates.	
39	BOQ MEP TOWER ABCE		General		Civil Works	1. Please note that we have not considered any kind of civil works such as wall chipping, wall chasing, wall opening, wall closing etc in our proposal	Client to confirm	Vendors Scope at their own cost.	

Sr. No. / Query No.	Bidding Document Reference(s) [Volume No / Section Number / Clause Number]			Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.						
40	BOQ MEP TOWER ABCE		General		Power and Water	1. We presume that Power and water shall be provided free of cost at one point	Client to confirm	Vendors Scope. Mention the electricity cost in summary sheet.	
41	BOQ MEP TOWER ABCE		General		Third party Testing & commissioning	3. Please confirm the requirement of Third party or can we consider our inhouse team for the same if third party required please provide the desired make.	Client to confirm	It will be vendor's scope and cost if such case and requirement will be there.	

Sr. No. / Query No.	Bidding Document Reference(s) [Volume No / Section Number / Clause Number]	Main Section No. / Sub Section or Appendix number	Clause No.	Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
					As per BOQ	As per Specification			
	ELECTRICAL: TECHNICAL SPECIFICATIONS : TOWER A, B, C AND E								
1	HV/LV Electrical Tender Package		Under HT Works - SI No- 1.1 to 1.7 - 11kV Switchgear SI No - 1.1 & 1.16 - Transformer Under Electrical Package - SI No - 1.1 to 1.3, 1.1.1 to 1.1.2, 1.2.1 to 1.4.7, 1.1.1 to 1.1.3.16 - LT Switchgear			We understand that our quoted rate for Installation Cost for HT Switchgears, Transformers and LT Switchgears and does not include the cost of Civil Foundations. The same needs to be got done by your Civil Vendor. Kindly Confirm.	The civil foundation for HT switchgears, transformers and LT switchgears is not part f Electrical contractor scope	Refer the responsibility Matrix for further details.	
2	HV/LV Electrical Tender Package	Elec BOQ - Under HT & XMER BOQ, Sec A	Item No - 1 - [1.1 to 1.7]	Page 33 of 154	11kV /22kV / 33kV HV Panel	a) We are considering "ELINS SWITCHBOARDS PVT. LTD" System house of ABB Make, which is our own concern to achieve better time schedules, manufacturing Quality Switchboards. 8) The drawing no. specified in the BOQ line item is not matching with the drawings provided. b) Kindly Provide the SLD and AutoCad Drawings.	a) Please adhere to th approved makes only. B) Pleas specify the mismatch for us to address the query	Has been shared.	
					HT Panel 1.3 HTP-TB-L01-01 AIR INSULATED (VCB) 11 kV SWITCHGEAR 2IC ZOG 1.4 HT-DG-TB-PWR-B01-01 AIR INSULATED (VCB) 11 kV SWITCHGEAR 1IC ZOG 1.5 HTP-TC-L00-01 AIR INSULATED (VCB) 11 kV SWITCHGEAR 2IC 3OG 1.7 HTP-TE-L01-01 AIR INSULATED (VCB) 11 kV SWITCHGEAR 1IC 3OG	SLD Not available, we request you to provide us the same, which will help us to quote.	SLD has been shared by BH .Please check in Tower C . Tower C panels and xmers are located in Tower B ,Hence the nomeclature . SLD has been shared by BH Pleas check in Tower B HTP TB L00 01 AIR INSULATED (VCB) 11kV SWITCHGEAR IC 3OG . Please check in Tower C SLD has been shared by BH Pleas check in Tower E		
					General Queries Make	Shall we consider following makes, 1. LT Panels (MDB, SMD& DB) -"ELINS SWITCHBOARDS PVT. LTD" 2. Fire suppression system - Fire Trex/Supremex, 3. Protective Relays - CSPC. Kindly confirm.	Please adhere to the makes list		
3	HV/LV Electrical Tender Package	Elec BOQ - Under HT & XMER BOQ, Dec G & Sec N	Under Electrical Package - SI No - 1.1 to 1.3, 1.1.1 to 1.1.2, 1.2.1 to 1.4.7, 1.1.1 to 1.1.3.16 - LT Switchgear		LT Panels	Shall we consider Elins make (Eube) panels which is latest type tested as per IEC 61439 - 1&2. And Internal Arc type tested as per IEC 61641 , 50KA @ 0.3 sec.	Please adhere to the makes list . Submit the IEC 61439 certificate for al the types of MLTPs and SMDBs in the project . Also share Discrimination report of similar scale project as part of Technical submission		
				Page 36 of 154	Constructional Details SI.No. BB) All the feeders shall be withdrawable type. ACB Shall be withdrawable type, however MCCB shall be fixed type.	Shall we consider Fixed type execution panels with drawtype ACBs & fixed type MCCBs, Please confirm.	Please refer SLD and the legends		
				Page 38 of 154	Current Density SI.No: G) Maximum current density of the busbar shall be 0.8A/Sq mm for Aluminium and 1.4 A/Sq mm for Copper.	As per SLD Notes, SI.No 10 . Current density for copper busbar is 1.2A/Sq mm. As per BOQ ; Shall we consider 1.6A/Sq mm Current density for copper busbar, Please confirm.	Please follow the specifications		
				Page 35 of 154	Constructional Details SI No. K) The structure, including doors and panels, shall be suitable of withstanding the stress due to seismic conditions as specified in a for mentioned site conditions.	Elins Make E Cube LT panels are design to withstand the Sismic conditions, But not type tested for sismic ty pe test. Please confirm.	Please adhere to the makes list . Type test certificate ifor seismic withstand is a must .		
				Page 37 of 154	BMS Integration. CC. Each compartment shall have the following features SI No. g) Smart Link to be provided for BMS integration at one point in LT panels.	Shall we consider ACB/MCCB's Potential free contacts only ON/OFF/trip status) for BMS integration and we are not considering Smartlinks. Please confirm.	Potential free contacts are ok.		
				Page 37 of 154	Thermal monitoring system SI No. a) All Main LT panel (MLTP)/main Distribution Boards both life safety and Normal power shall be provided with Temperature sensors and panel Mounted temperature Monitoring. SI No. c) Thermal monitor shall be multi line LCD/Digital Display and with necessary selector switch to change function and scan the particular temp sensor. SI No. f) The contractor shall provide a gateway device to building BACnet System. SI. No. g) The Temperature Sensor shall be provided at each Breaker connection.	1. We are not considering Thermal monitoring system in our offer . 2. If required we will consider Thermal monitoring system only in Main LT panel ACB feeders. Please confirm.	1. Please adhere to the specifaions 2.. Thermal monitoring systems are required in all MLTPs and SMDBs		

			Page 48 of 154	ACB	2.2.8 : Air Circuit Breaker: SI No. G) Protections The breaker should be equipped with Micro-controller based, communicable type release with RS485 port for communication to offer accurate and versatile protection with complete flexibility .	Shall we consider ACB with RS485 Port wherever Communication ports are specified in SLD, Please confirm.	Pease consider RS485 for all incomers ACB and MCCB except Tenant metering panels and control panels		
			Page 48 of 154	ACB	2.2.8 : Air Circuit Breaker: SI No. F) All Incomer ACBs for LT panel shall have the following additional protections other than mentioned above. A. Under and over voltage B. Under and Over frequency. SI No. G) The release of incoming breakers should provide comprehensive metering with the following parameters. i) Maximum demand (KVA & KW) J. Total Harmonics Distortion (From Fundamental to 50th Harmonic)	1. Shall we consider Schneider make NW 6E releases for Incoming ACBs & NW 6A releases for Outgoing ACBs in all the panels. 2. Shall we consider Incomer ACB with Metering to measure the 50th Harmonic in MLTP Panel and we are not considering breaker with Inbuilt 50th Harmonic. Please confirm.	1. Please adhereto the specifications and provide 2. please adhere to the specifications		
			Page 50 of 154	MCCB	2.2.9.1 : Protection functions: D. The LSI feature shall be used for all panels except Main LT Panel & Main DG set Panel.	Shall we consider only LI releases wherever LSI releases are mentioned in SLD.	Please adhere to the SLD		
				MCCB	General	(i) Shall we consider TM based MCCB with O/C, S/C releases for ratings upto & inclusive of 250A & MP based MCCB with O/C, S/c releases (LI Releases) for the rating above 250A. Please confirm. ii) Shall we consider E/F releases wherever specified in SLD. Please confirm.	i) Please adhere to the specifications ii) EF will be in incomers of MLTPs and SMDBs		
			Page 50 of 154	MCCB	2.2.9.1 : Protection functions: The breaker should be equipped with micro-controller based, communicable type release with RS485 port for communication to offer accurate and versatile protection with complete flexibility and shall offer complete over current protection to the electrical system	Shall we consider RS485 port wherever mentioned in SLD, Please confirm.	already answered above		
				Spare Capacity		As per SLD Notes, All the Panels to be provided with 20% spare capacity. As per BOQ: we are considering spare feeders as per SLD only and we are not considering any additional spare feeders apart from SLD. Please confirm.	Please consider the spare feeders as specified in SLD .No additional spares to e considered		
			Page 50 of 154	APFC Panel	2.2.11.2: Reactor: Copper foil windings is mentioned.	a) Shall we consider 480V MPP type capacitor. b) Shall we consider 7% detuned Aluminium Reactor, Please confirm	a) 480V MPP type capacitor is ok b) Please adhere to the specifications		
				Fire Suppression system		We shall consider fire suppression system wherever specified in BOQ, Please confirm.	yes		
				ATS		As per BOQ Notes, ASCO 7000 Series is mentioned, Shall we consider ASCO Transferpact/WOTPC Series ATS instead of ASCO 7000 Series, Please confirm.	Plase adhere to the boq description		
				1.1 Main Distribution Boards (MDB/MLTP) (As per Technical Specification, Schedules & SLD)					
				Tower B					
				1.1.9 MLTP-TB-PWR-L00-01 (FIRE SUPPRESSION SYSTEM) 50KA fault level				Please check the latest tender repository shared by BH	
				1.1.10 MLTP-TB-PWR-LS-L00-02 (FIRE SUPPRESSION SYSTEM) 50KA fault level		SLD Not available, we request you to provide us the same, which will help us to quote.			
				1.1.11 MLTP-TB-PWR-L02-01 (FIRE SUPPRESSION SYSTEM) 50KA fault level					
				1.1.12 MDB-TA-EVSE-B1-01 (FIRE SUPPRESSION SYSTEM) 50KA fault level					
				Tower E					
				1.1.25 BB-TE-PWR-L01-01 (FIRE SUPPRESSION SYSTEM) 65KA fault level		SLD Not available, we request you to provide us the same, which will help us to quote.		Please check the latest tender repository shared by BH	
				1.2.1 Normal SMDb					
				Tower B					
				1.2.1.4 SMDb-TB-EVSE-B1-01		SLD Not available, we request you to provide us the same, which will help us to quote.		Please check the latest tender repository shared by BH	
				1.2.2 DG Backed SMDb (Essential Supply)					
				Tower A					
				1.2.2.8 SMDb-TA-VENT-B4-01					
				1.2.2.9 SMDb-FCU-TA-B1-01		SLD Not available, we request you to provide us the same, which will help us to quote.		Please check the latest tender repository shared by BH	
				1.2.2.10 SMDb-TA-VENT-B1-02					
				Tower B					
				1.2.2.12 SMDb-PL-TB-L00-01					
				1.2.2.13 SMDb-TB-LIFT-L00-01					
				1.2.2.14 SMDb-TB-LIFT-L00-02					
				1.2.2.15 SMDb-TB-PWR-L+P-L00-01					
				1.2.2.16 SMDb-TB-VENT-N-L00-01					
				1.2.2.17 SMDb-TB-EVSE-B1-01		SLD Not available, we request you to provide us the same, which will help us to quote.			
				1.2.2.18 SMDb-TB-PARK-B3-01					
				1.2.2.19 SMDb-TB-F&L-L00-01					
				1.2.2.20 SMDb-TB-ST-B4-01					
				1.2.2.21 SMDb-TB-ST-B3-01					

						1.2.2.22 SMD8-TB-ST-B1-01						
						1.2.3 Life Safety SMDB (DG Backed) (FIRE SUPPRESSION SYSTEM)						
						Tower B						
						1.2.3.6 SMD8-TB-VENT-LS-L00-01					Please check the latest tender repository shared by BH	
						1.2.3.7 SMD8-TB-PWR-LS-B1-01		SLD Not available, we request you to provide us the same, which will help us to quote.				
						PHE Panel						
						1.5 MCC-TA-PL-LS-B4-02		SLD Not available, we request you to provide us the same, which will help us to quote.			Please check the latest tender repository shared by BH	
						FPS Panel						
						1.03 MCC-TA-FF-LS-L51-01						
						1.05 MCC-TC-FF-LS-L46-01					Please check the latest tender repository shared by BH	
						1.06 MCC-TC-FF-LS-L48-01		SLD Not available, we request you to provide us the same, which will help us to quote.				
						1.07 MCC-TC-FF-LS-L70-01						
4	HV/LV Package	Electrical Tender	Under Elec BOQ - Sec A	Item No. 1.2 [1.2.1 to 1.2.3] - Battery & Battery Charger	Page 27 of 154	As per BOQ : 110 V DC, 200AH Battery Charger					Please check the latest tender repository shared by BH	
5	HV/LV Package	Electrical Tender	Under Ele BOQ - Sec C	Item No - 1 [1.1 - 1.16] - Transformer	Page 22 of 154	As per BOQ : 2500 & 1600 KVA Dry type off load tap changer stepdown transformer.						
6	HV/LV Package	Electrical Tender	Under Ele BOQ - Sec E	Item No - 1 [1.1 - 1.1.1 to 1.1.5 - 1.2.1 to 1.2.4] - Cable Containment, Trunkings and Raceways	Page 94 of 154	Cable Containment, Trunkings and Raceways		a) We request you to kindly approve "ELUNS SHEET METAL INDUSTRIES" Make for Cable trays, Raceways and Junction Box.	Please adhere to the approved makes list			
								b) Whether Cable Containment is required with pre Galvanised or Hot Dip Galvanised as there will be huge cost difference between Pregalvanised and Hot Dip Galvanised. Kindly Confirm.	Please adhere to the specifications			
						Ladder type cable trays shall be made from 2mm thick sheet formed in 'C' section of 75mm height and inward flanges of 15mm as side runners and 30mm wide x 10mm high rungs ('C' shaped) from a 1.5mm thick sheet. Perforations as mentioned above shall be provided in the width of the rungs. Pitch of the rungs shall not exceed 250 mm centre to centre. Rungs shall be tack welded to the side members. Perforated trays shall be of the formed channel shape. The trays will be joined with fish plates,nuts,bolts and NOT welded.	For all cable trays upto and Including 300mm , Thickness of Material to be 1.6mm and Trays above 300mm Thickness of material to be 2mm	c) Kindly Confirm , whether we have to Consider thickness for cable trays upto and Including 300mm , Thickness of Material to be 1.6mm and Trays above 300mm Thickness of material to be 2mm.	Confirm			
7	HV/LV Package	Electrical Tender	Under Ele BOQ - Sec I	Item No - 1.1 - [1.1.1 to 1.2.6] - Busduct and Rising Mains and Tap Off Units.	Page 73 of 154	Busduct and Rising Mains and Tap Off Units.		a) We request you to kindly approve "ELUNS" Make for Busducts b) Kindly provide separate line item for Terminal Flange and Bends.	Please adhere to the approved makes list			
								b) Kindly provide quantity for steel, we will not be able to arrive at the quantity required at this stage, provide quantity in MT, since it is ITEM RATE Tender.	Please adhere to the boq description			
						Supply, Installation, testing, Transport, Loading, unloading on proper place at site as directed by Engineer In Charge with Supervision & Commissioning of compact sandwiched Type, Aluminium Busduct / Rising Mains with Class B insulation in GI extruded enclosure with 2 Nos Integral GI strip earth strip to run along with rising mains with all accessories like bends, supports,copper flexibles, phase changeovers,flange end, end piece, necessary supporting arrangements, hangers etc. as below complete in all respects as per approval of Client / Consultant / PMCs. The Rising Mains shall be suitable for IP54 class of protection.		Kindly Confirm whether the Enclosure of Busduct / Rising Main shall be MS Powder Coated or GI Extruded or Al Extrusion, Since GI Extruded is not Standard as per Manufacturers standards.	Please adhere to the boq description			
				Item No. : 1.2. Tap off box :1.2.1		Tap off Box -100A, TP, 25KA, TMD release with class-2 Surge Protection		As per Elins 'Standard we Cannot Considered Class-2 Surge Protection.	Please adhere to the boq description			
8	HV/LV Package	Electrical Tender	Elec BOQ - Under Ele BOQ - Sec N	Item No - 1.3 - [1.3.1 to 1.4.34] - DISTRIBUTION BOARDS :	Page 73 of 154	Supplying and fixing metal double door Horizontal TPN MCB, PPI Distribution board with IP-42 protection and IK 09, concealed in wall after cutting the wall & mending good the damages to original finish incl. Inter connection with suitable size of copper wire and neutral link & provision for earthing attachment.	C. Each unit shall be of the 500 volt pattern as a minimum requirement and be complete with a pressed or fabricated steel case manufactured from sheet steel having a minimum thickness of 1.6 swg. double door configuration, painted with primer of 2 coats of low VOC paint of approved colour.	a) We are considering Metal Double Door with 1.2mm thickness IP43 for all Distribution Boards as per manufacturer standards as mentioned in BOQ. Since ,1.6mm thick sheet steel is mentioned in Specifications. Kindly Confirm. b) Kindly Confirm whether we have to Consider Timer Switch & Contactor for Common Area DB's. c). Subincommer Ampere rating is not Specified.	a) Please adhere to specifications b) yes c) Please refer the DB schedule, subincomer rating is same as main incomer			
9	HV/LV Package	Electrical Tender	Elec BOQ - Under Ele BOQ - Sec R	Item No - 3-[3.1 to 3.10] & 6-[6.1 to 6.10] - CABLE AND TERMINATIONS	Page 81 of 154	Supplying, unloading at site, shifting into the store, Laying, Testing & Commissioning of 1100V grade, Fire Survival (FS/PHI20), compliant to IS17505 as per section B of Specification no 101-P1-BHE-ZZ-ZZ-SP-E-1000, Steel Wire Armour, Copper conductor Cables		Kindly Confirm , whether we have to Consider Aluminium FS Cable for Cable Sizes for Item No - 3-[3.1 to 3.10] & 6-[6.1 to 6.10] from 3.5C x 25 Sqmm to 3.5C x 300 Sqmm & for Item No - 3-[3.11 to 3.20] & 6-[6.11 to 6.22]Copper FS Cable from 4C x 4 Sqmm to 4C x 25 Sqmm & 1 Core Cables.	Fire survival cables have to be Copper only			

10	HV/LV Electrical Tender Package	Under Ele BOQ - Sec U	Item No - 1 - [1.1 to 1.8] - AVIATION LIGHTING	Page 116 of 154	Aviation Lights - High-intensity obstruction light Type A, 360 degree Conforming to regulations from CAR, ICAO, FAA. It should be flashing white light for Omni-Directional usage having SMD High Intensity LEDs for white light output operations in Day, Twilight and Night. Intensity of light should be Day Time: 2,00,000 Cd, Twilight Time: 20,000 Cd & Night Time: 2,000 Cd (White)	The system shall comprise Low intensity obstacle lights , external photocell, gps unit, wiring system and power supply units serving the obstacle lights. All obstruction lights shall be provided with an interface point to allow a central controller to monitor the status of each light fitting. This shall be provided at a single point on the floor where the lights are located to allow connection into the interface by the contractor.	Kindly Confirm, whether we have to Consider Low Intensity or High Intensity as per BOQ . Since BOQ calls for both High Intensity having SMD High Intensity LEDs.	High Intensity light to be considered		
11	HV/LV Electrical Tender Package	Under Ele BOQ - Sec X	Item No - 1 to 7- [1.1 to 1.18, 2.1 to 2.20, 3.2 & 3.2, 4.1 to 4.4, 5.1 to 5.4, 6.1 & 6.2 & 7.1-7.2] - POINT WIRING	Page 89 of 154	Point Wiring		a) We Understand that the Wire type is HFFR PVC Insulated Copper Conductor, multi-stranded Wire and PVC Conduit as per the BOQ for Point Wiring. Kindly Confirm the Wire and Conduit type/Size (GI/MS/PVC), since the same is not mentioned in Layout drawings / BOQ or we have to follow the size as specified in Point Wiring Scheme. Kindly Confirm.	For wire size please refer the DB schedule. The conduits will be prelaidd in the slab .		
12	HV/LV Electrical Tender Package	Under Ele BOQ - Sec V	Item No - 5 - [5.1] - EV CHARGING		Wiring from DB to EV charging point including 16A switch socket . Cabling to be FRZHLS PVC insulated copper conductor single core wire in GI conduit. SITC of cable and earthing the point with FRZHLS PVC insulated copper conductor singlecore cable etc. as required. Wiring to be P: 4 N: 4 E: 2.5sqmm,		We Understand that the for EV Charging we are providing only 16A Domestic Switch & Socket. Kindly Confirm.	confirmed		
13	HV/LV Electrical Tender Package	Under Ele BOQ - Sec Y & Z	Item No - [1.1 to 1.16 & 1.1 to 1.8] - LIGHT FIXTURES		LIGHT FIXTURES		a) We understand we should quote our rates only for Installation Cost for Items Covered under Sec Y & Z - Item No - [1.1 to 1.16 & 1.1 to 1.8] i.e., Lighting and Residential Units. Kindly Confirm.	This query is not of this project		
							b) We understand that we should quote only Installation Cost for Light Fixtures . Since the supply of the same is not in our scope, we request you to consider Handling charges @ 15% of the Supply Cost for these free issue items. The handling charges are basically required for assembling, providing hardware's and accessories and to meet our Engineering / Man Power charges for putting into operation of the materials supplied by you.			
14	HV/LV Electrical Tender Package	General			Temporary Lighting Works		a) We understand the Temporary Works to be carried out by our End. Request you to kindly furnish us the separate line items for the Temporary Lighting Works and Temporary DG along with capacity and Working hours. Kindly Confirm.		Task light and its allied infra for the use of MEP contractor is in the scope of MEP contractor at their own cost. However, As discussed during the meeting vendor can quote below the BOQ by adding the separate line item for the power provision to other contractors. Contractor to fulfill all the requirement and provisions for such temporary power required by all the vendors. No separate cost shall be payable for the same.	
							b) We presume EB power is available in this premises and the same can be utilised for construction purpose. Kindly confirm.		Please refer the special conditions of contract and responsibility matrix point no. 17 and 19 carefully for detailed responsibility of the MEP contractor. The cost of creating site Infra for own use and other contractors use, employer will ask MEP contractor to provide power and will be in scope of MEP contractor. All the meter and sub meter provisions are in the scope of MEP contractor. MEP contractor to include all such costs in their quoted rates.	
15	HV/LV Electrical Tender Package	Under Ele BOQ - Sec F	Item No - 1-[1.9] - MISCELLANEOUS ITEMS		Charges for submission and obtaining approval before start of work. Survey & approval of Installation location. Inspection and approval of the work under scope of this tender (H.T. & L.T.) Co-ordination with Supply Company & Electrical Inspector for commissioning of the installation. Any other activity / approval related to the scope of this tender. Submission of required Test Reports to E. I. / Supply Authority. (This includes the statutory fees of liaison, incidental expenses and any other charges paid by the contractor).		We understand that we should quote only for coordination with Supply Company and Load Sanction is not a part of Scope. However if the Load Sanction needs to be done please provide kVA required. Kindly Confirm.		Tender terms or BOQ item shall prevail. Discuss in first round technical meeting for further clarity.	
16	HV/LV Electrical Tender Package	Under Ele BOQ - Sec M	Item No - [1] - MISCELLANEOUS ITEMS		Charges for submission and obtaining approval before start of work. Survey & approval of Installation location. Inspection and approval of the work under scope of this tender L.T. meters approval by PWD Co-ordination with Supply Company & Electrical Inspector for commissioning of the installation. of LT meters for offices .Any other activity / approval related to the scope of this tender. Submission of required Test Reports to E. I. / Supply Authority. (This includes the statutory fees of liaison, incidental expenses and any other charges paid by the contractor).		Kindly confirm whether all Official fees should be paid to Supply Companies as per the Line Item by Contractor or the Developer will pay the Official fees. Kindly Confirm.		Tender terms or BOQ item shall prevail. Discuss in first round technical meeting for further clarity.	

17	HV/LV Electrical Tender Package	General			Temporary Lighting Works		a) We understand the Temporary Works to be carried out by our End. Request you to kindly furnish us the separate line items for the Temporary Lighting Works and Temporary DG along with capacity and Working hours. Kindly Confirm.		Task light and its allied infra for the use of MEP contractor is in the scope of MEP contractor at their own cost. However, As discussed during the meeting vendor can quote below the BOQ by adding the separate line item for the power provision to other contractors. Contractor to fulfill all the requirement and provisions for such temporary power required by all the vendors. No separate cost shall be payable for the same. Refer the responsibility matrix for further details.	
							b) We presume EB power is available in this premises and the same can be utilised for construction purpose. Kindly confirm.			
18	HV/LV Electrical Tender Package	General			MS Steel		a) Kindly Provide Steel as separate line item in MT for HT, LT, Busduct & Cable Tray Installation.		Tender terms or BOQ item shall prevail. Discuss in first round technical meeting for further clarity.	
19	HV/LV Electrical Tender Package	General			Sealing of Electrical Shafts/Wall/Floor opening with Fire Stop Cement Mortar .		Kindly Provide Fire Sealant line item in Sqmtr for Sealing of Electrical Shafts/Wall/Floor opening with Fire Stop Cement Mortar		Tender terms or BOQ item shall prevail.	
20	General				BOQ Calculation Error		Summation error in the Total Quantity Column. Kindly provide the revised BOQ with Correction in Total Quantity.		The revised BOQ with corrected total quantities will be included in the addendum.	
21	Section-G - List of Approved makes									
a	Cable Trays & Raceway	Elins Sheet Metal Industries					Please confirm whether we can consider the following Makes along with the Makes specified in List of Approved Makes.		Please adhere to the approved makes list given by the consultant.	
b	HT & LT CABLES	AVOCAB								
c	Electrical Wires - FR/ LSZH/ FRLS, PVC Insulated multi-stranded, copper Wires	AVOCAB								

Prebid Queries

Vendor 6 - Query 03

Date :

29-11-2024

Sr. No. / Query No.	Bidding Document Reference(s) [Volume No / Section Number / Clause Number]			Tender PDF File Page Number	Content of Tender Requiring Clarification	Points of Clarification Required	Response to the Query [BH]	Response to the Query [JLL / PEPL]	Remarks
	Volume No.	Main Section No. / Sub Section or Appendix number	Clause No.						
FAPA									
1	MEP works for Tower ABCE	SEC-A	Fire Detection & Alarm System_A1, A2 & A3		The panel shall have 10 loop capacity, with Network card (RJ45) supporting BACnet protocol and Digital Alarm Communicator Transmitter (DACT) .	Digital Alarm Communicator Transmitter (DACT) is not available with any of the OEM's. Instead of DACT we shall be considering Auto Dialler with Securico Make as per the Approved list of Make.	1 number Standard auto dialler to be used.For makes adhere to makes list		
1	MEP works for Tower ABCE	SEC-A	Fire Detection & Alarm System_A1, A2 & A3		Main FAVA panel shall have built-in 2 way fire fighting communication system in built(microphone) Paging system. The panel shall have built-in ethernet/USB port , webserver	There is no Webserver available with OEM & the same has not been Considered	ok		
2	MEP works for Tower ABCE	SEC-A	Fire Detection & Alarm System_A26		Fire alarm red revolving Beacon light with required accessories ,equivalent to model no AVIL HD S <u>List of Make: FA BeaconCooper / Tonar Electronics / Federal Signal Corporation.</u>	We are considering Kheraj Make Beacon light Mode NoI: AVIL HD S ,Kindly Conform the same	kheraj make can be considered		
3	MEP works for Tower ABCE	SEC-D	Voice Annunciation System_D4 to D7		Power Amplifier	240 & 500 Watt Combination of Power Amplifier are available with the OEM ,we are Considering the same Combination to Comply the Wattage requirement	ok ,they should be at least CE approved		
4	MEP works for Tower ABCE	SEC-D	Voice Annunciation System_D8 to D9		Power Amplifier-The units shall be UL listed and FM approved	We are Considering the Amplifier without any Approval ,kindly Confirm the same	It should have at least CE approval		
5	MEP works for Tower ABCE	SEC-D	Voice Annunciation System_D8 to D12		2 Watt false ceiling mounted type speakers,The Units shall be UL listed and FM approved	As per OEM Confirmation there is availability of 2 watt UL listed Speaker ,we are Considering the same	ok		
6	MEP works for Tower ABCE	SEC-D	Voice Annunciation System_D10 to D12		20W, Surface Mounted, horn speaker with required accessories. The Units shall be UL listed and FM approved	As per OEM Confirmation there is no availability of 20 watt Horn Speaker with UL & FM approval We are Considering 15 Watt Horn Speaker without any approvals,Kindly Confirm the same	Please consider 15W non UL		
BMS									
1	MEP works for Tower ABCE	2_CONTROLLERS & SYSTEM INTERFACES_2.1	2_CONTROLLERS & SYSTEM INTERFACES_2.1		ASC/DDC for TPM (The mentioned quantities are tentative and they have to be verified as per the vendors requirement) Refer to BHE-PL-ZZ-SH-BM-9000-01 to 10 for I/O point schedule <u>Refer to the BMS schematic</u>	BOQ & IO Summary is not matching,kindly share the same	Please refer the I/O schedule included in the tender package.		
2	MEP works for Tower ABCE	2_CONTROLLERS & SYSTEM INTERFACES_2.3	2_CONTROLLERS & SYSTEM INTERFACES_2.3		Repeater panels as per OEM / system requirement	Qty is in Lot. Kindly confirm the number qty required & Kindly Share the Detailed requirement	Please refer the BOQ note number 1A, Refer the BMS schematic for more details & proceed with Lot value.		
3	MEP works for Tower ABCE	2_CONTROLLERS & SYSTEM INTERFACES_2.4	2_CONTROLLERS & SYSTEM INTERFACES_2.4		Hand held portable operator terminals.(POT). These shall be required if the Controllers and Integration Units do not have a built in keypad on the face. BACnet/IP shall have Display with adjustable contrast	Qty is in Lot. Kindly confirm the number qty required.	Please refer the BOQ note number 1A, Refer the BMS schematic for more details & proceed with Lot value.		
4	MEP works for Tower ABCE	2_CONTROLLERS & SYSTEM INTERFACES_2.6	2_CONTROLLERS & SYSTEM INTERFACES_2.6		Third Party Integration	Qty is in Lot. Kindly share the IO Summary for 3rd party integrations.	Please refer the I/O schedule included in the tender package.		

Vendor 7 No Queries

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